

2025 CFA®
Exam Prep

SchweserNotes™

Fixed Income, Derivatives, and
Alternative Investments

Level II Book 4

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Book 4: Fixed Income, Derivatives, and Alternative Investments

SchweserNotes™ 2025

Level II CFA®



SCHWESERNOTES™ 2025 Level II CFA® BOOK 4: FIXED INCOME, DERIVATIVES, AND ALTERNATIVE INVESTMENTS

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Learning Outcome Statements (LOS)

23. The Term Structure and Interest Rate Dynamics

The candidate should be able to:

- a. describe relationships among spot rates, forward rates, yield to maturity, expected and realized returns on bonds, and the shape of the yield curve.
- b. describe how zero-coupon rates (spot rates) may be obtained from the par curve by bootstrapping.
- c. describe the assumptions concerning the evolution of spot rates in relation to forward rates implicit in active bond portfolio management.
- d. describe the strategy of rolling down the yield curve.
- e. explain the swap rate curve and why and how market participants use it in valuation.
- f. calculate and interpret the swap spread for a given maturity.
- g. describe short-term interest rate spreads used to gauge economy-wide credit risk and liquidity risk.
- h. explain traditional theories of the term structure of interest rates and describe the implications of each theory for forward rates and the shape of the yield curve.
- i. explain how a bond's exposure to each of the factors driving the yield curve can be measured and how these exposures can be used to manage yield curve risks.
- j. explain the maturity structure of yield volatilities and their effect on price volatility.
- k. explain how key economic factors are used to establish a view on benchmark rates, spreads, and yield curve changes.

24. The Arbitrage-Free Valuation Framework

The candidate should be able to:

- a. explain what is meant by arbitrage-free valuation of a fixed-income instrument.
- b. calculate the arbitrage-free value of an option-free, fixed-rate coupon bond.
- c. describe a binomial interest rate tree framework.
- d. describe the process of calibrating a binomial interest rate tree to match a specific term structure.
- e. describe the backward induction valuation methodology and calculate the value of a fixed-income instrument given its cash flow at each node.
- f. compare pricing using the zero-coupon yield curve with pricing using an arbitrage-free binomial lattice.
- g. describe pathwise valuation in a binomial interest rate framework and calculate the value of a fixed-income instrument given its cash flows along each path.
- h. describe a Monte Carlo forward-rate simulation and its application.
- i. describe term structure models and how they are used.

25. Valuation and Analysis of Bonds with Embedded Options

The candidate should be able to:

- a. describe fixed-income securities with embedded options.

- b. explain the relationships between the values of a callable or putable bond, the underlying option-free (straight) bond, and the embedded option.
- c. describe how the arbitrage-free framework can be used to value a bond with embedded options.
- d. explain how interest rate volatility affects the value of a callable or putable bond.
- e. explain how changes in the level and shape of the yield curve affect the value of a callable or putable bond.
- f. calculate the value of a callable or putable bond from an interest rate tree.
- g. explain the calculation and use of option-adjusted spreads.
- h. explain how interest rate volatility affects option-adjusted spreads.
- i. calculate and interpret effective duration of a callable or putable bond.
- j. compare effective durations of callable, putable, and straight bonds.
- k. describe the use of one-sided durations and key rate durations to evaluate the interest rate sensitivity of bonds with embedded options.
- l. compare effective convexities of callable, putable, and straight bonds.
- m. calculate the value of a capped or floored floating-rate bond.
- n. describe defining features of a convertible bond.
- o. calculate and interpret the components of a convertible bond's value.
- p. describe how a convertible bond is valued in an arbitrage-free framework.
- q. compare the risk–return characteristics of a convertible bond with the risk–return characteristics of a straight bond and of the underlying common stock.

26. Credit Analysis Models

The candidate should be able to:

- a. explain expected exposure, the loss given default, the probability of default, and the credit valuation adjustment.
- b. explain credit scores and credit ratings.
- c. calculate the expected return on a bond given transition in its credit rating.
- d. explain structural and reduced-form models of corporate credit risk, including assumptions, strengths, and weaknesses.
- e. calculate the value of a bond and its credit spread, given assumptions about the credit risk parameters.
- f. interpret changes in a credit spread.
- g. explain the determinants of the term structure of credit spreads and interpret a term structure of credit spreads.
- h. compare the credit analysis required for securitized debt to the credit analysis of corporate debt.

27. Credit Default Swaps

The candidate should be able to:

- a. describe credit default swaps (CDS), single-name and index CDS, and the parameters that define a given CDS product.
- b. describe credit events and settlement protocols with respect to CDS.
- c. Explain the principles underlying and factors that influence the market's pricing of CDS.

- d. describe the use of CDS to manage credit exposures and to express views regarding changes in the shape and/or level of the credit curve.
- e. describe the use of CDS to take advantage of valuation disparities among separate markets, such as bonds, loans, equities, and equity-linked instruments.

28. Pricing and Valuation of Forward Commitments

The candidate should be able to:

- a. describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value.
- b. describe the carry arbitrage model without underlying cashflows and with underlying cashflows.
- c. describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value.
- d. describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value.
- e. describe how interest rate swaps are priced, and calculate and interpret their no-arbitrage value.
- f. describe how currency swaps are priced, and calculate and interpret their no-arbitrage value.
- g. describe how equity swaps are priced, and calculate and interpret their no-arbitrage value.

29. Valuation of Contingent Claims

The candidate should be able to:

- a. describe and interpret the binomial option valuation model and its component terms.
- b. describe how the value of a European option can be analyzed as the present value of the option's expected payoff at expiration.
- c. identify an arbitrage opportunity involving options and describe the related arbitrage.
- d. calculate the no-arbitrage values of European and American options using a two-period binomial model.
- e. calculate and interpret the value of an interest rate option using a two-period binomial model.
- f. identify assumptions of the Black–Scholes–Merton option valuation model.
- g. interpret the components of the Black–Scholes–Merton model as applied to call options in terms of a leveraged position in the underlying.
- h. describe how the Black–Scholes–Merton model is used to value European options on equities and currencies.
- i. describe how the Black model is used to value European options on futures.
- j. describe how the Black model is used to value European interest rate options and European swaptions.
- k. interpret each of the option Greeks.
- l. describe how a delta hedge is executed.
- m. describe the role of gamma risk in options trading.
- n. define implied volatility and explain how it is used in options trading.

30. Introduction to Commodities and Commodity Derivatives

The candidate should be able to:

- a. compare characteristics of commodity sectors.
- b. compare the life cycle of commodity sectors from production through trading or consumption.
- c. contrast the valuation of commodities with the valuation of equities and bonds.
- d. describe types of participants in commodity futures markets.
- e. analyze the relationship between spot prices and futures prices in markets in contango and markets in backwardation.
- f. compare theories of commodity futures returns.
- g. describe, calculate, and interpret the components of total return for a fully collateralized commodity futures contract.
- h. contrast roll return in markets in contango and markets in backwardation.
- i. describe how commodity swaps are used to obtain or modify exposure to commodities.
- j. describe how the construction of commodity indexes affects index returns.

31. Overview of Types of Real Estate Investment

The candidate should be able to:

- a. compare important real estate investment features for valuation purposes.
- b. explain economic value drivers of real estate investments and their role in a portfolio.
- c. discuss the distinctive investment characteristics of commercial property types.
- d. explain the due diligence process and valuation approaches for real estate investments.
- e. discuss real estate investment indexes, including their construction and potential biases.

32. Investments in Real Estate Through Publicly Traded Securities

The candidate should be able to:

- a. discuss types of publicly traded real estate securities.
- b. justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income.
- c. describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation.
- d. calculate and interpret the value of a REIT share using the net asset value, relative value (price-to-FFO and price-to-AFFO), and discounted cash flow approaches.
- e. explain advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles.

33. Hedge Fund Strategies

The candidate should be able to:

- a. discuss how hedge fund strategies may be classified.
- b. discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies.

- c. discuss investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies.
- d. discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies.
- e. discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies.
- f. discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies.
- g. discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies.
- h. describe how factor models may be used to understand hedge fund risk exposures.
- i. evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio.

READING 23

THE TERM STRUCTURE AND INTEREST RATE DYNAMICS

EXAM FOCUS

This topic review discusses the theories and implications of the term structure of interest rates. In addition to understanding the relationships between spot rates, forward rates, yield to maturity, and the shape of the yield curve, be sure you become familiar with concepts like the Z-spread, the TED spread and the MRR-OIS spread. Interpreting the shape of the yield curve in the context of the theories of the term structure of interest rates is always important for the exam. Also pay close attention to the concept of key rate duration.

INTRODUCTION

The financial markets both impact and are controlled by interest rates. Understanding the term structure of interest rates (i.e., the graph of interest rates at different maturities) is one key to understanding the performance of an economy. In this reading, we explain how and why the term structure changes over time.

Spot rates are the annualized market interest rates for a single payment to be received in the future. Generally, we use spot rates for government securities (risk-free) to generate the spot rate curve. Spot rates can be interpreted as the yields on zero-coupon bonds, and for this reason we sometimes refer to spot rates as *zero-coupon rates*. A **forward rate** is an interest rate (agreed to today) for a loan to be made at some future date.



PROFESSOR'S NOTE

While most of the LOS in this topic review have *describe* or *explain* as the command words, we will still delve into numerous calculations, as it is difficult to really understand some of these concepts without getting in to the mathematics behind them.



Video covering
this content is
available online.

MODULE 23.1: SPOT AND FORWARD RATES, PART 1

LOS 23.a: Describe relationships among spot rates, forward rates, yield to maturity, expected and realized returns on bonds, and the shape of the yield curve.

Spot Rates

The price today of \$1 par, zero-coupon bond is known as the discount factor, which we will call P_T . Because it is a zero-coupon bond, the spot interest rate is the yield to maturity of this payment, which we represent as S_T . The relationship between the discount factor P_T and the spot rate S_T for maturity T can be expressed as:

$$P_T = \frac{1}{(1 + S_T)^T}$$

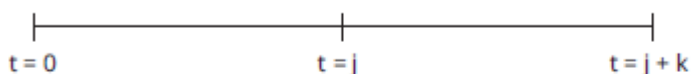
The term structure of spot rates—the graph of the spot rate S_T versus the maturity T —is known as the **spot yield curve** or **spot curve**. The shape and level of the spot curve changes continuously with the market prices of bonds.

Forward Rates

The annualized interest rate on a loan to be initiated at a future period is called the **forward rate** for that period. The term structure of forward rates is called the **forward curve**. (Note that forward curves and spot curves are mathematically related—we can derive one from the other.)

We will use the following notation:

$f(j,k)$ = the annualized interest rate applicable on a k -year loan starting in j years.



$F_{(j,k)}$ = the forward price of a \$1 par zero-coupon bond maturing at time $j+k$ delivered at time j .

$$F_{(j,k)} = \frac{1}{[1 + f(j,k)]^k}$$

Yield to Maturity

As we've discussed, the **yield to maturity (YTM)** or yield of a zero-coupon bond with maturity T is the spot interest rate for a maturity of T . However, for a coupon bond, if the spot rate curve is not flat, the YTM will not be the same as the spot rate.

EXAMPLE: Spot rates and yield for a coupon bond

Compute the price and yield to maturity of a three-year, 4% annual-pay, \$1,000 face value bond given the following spot rate curve: $S_1 = 5\%$, $S_2 = 6\%$, and $S_3 = 7\%$.

Answer:

1. Calculate the price of the bond using the spot rate curve:

$$\text{Price} = \frac{40}{(1.05)} + \frac{40}{(1.06)^2} + \frac{1040}{(1.07)^3} = \$922.64$$

2. Calculate the yield to maturity (y_3):

$$N = 3; PV = -922.64; PMT = 40; FV = 1,000; \text{CPT I/Y} \rightarrow 6.94$$

$$y_3 = 6.94\%$$

Note that the yield on a three-year bond is a weighted average of three spot rates, so in this case we would expect $S_1 < y_3 < S_3$. The yield to maturity y_3 is closest to S_3 because the par value dominates the value of the bond and therefore S_3 has the highest weight.

Expected and Realized Returns on Bonds

Expected return is the ex-ante holding period return that a bond investor expects to earn.

The expected return will be equal to the bond's yield only when *all three* of the following are true:

- The bond is held to maturity.
- All payments (coupon and principal) are made on time and in full.
- All coupons are reinvested at the original YTM.

The second requirement implies that the bond is option-free and there is no default risk.

The last requirement, reinvesting coupons at the YTM, is the least realistic assumption. If the yield curve is not flat, the coupon payments will not be reinvested at the YTM and the expected return will differ from the yield.

Realized return on a bond refers to the actual return that the investor experiences over the investment's holding period. Realized return is based on actual reinvestment rates.

The Forward Pricing Model

The **forward pricing model** values forward contracts based on arbitrage-free pricing.

Consider two investors.

Investor A purchases a \$1 face value, zero-coupon bond maturing in $j+k$ years at a price of $P_{(j+k)}$.

Investor B enters into a j -year forward contract to purchase a \$1 face value, zero-coupon bond maturing in k years at a price of $F_{(j,k)}$. Investor B's cost today is the present value of the cost: $PV[F_{(j,k)}]$ or $P_j F_{(j,k)}$.

Because the \$1 cash flows at $j+k$ are the same, these two investments should have the same price, which leads to the forward pricing model:

$$P_{(j+k)} = P_j F_{(j,k)}$$

Therefore:

$$F_{(j,k)} = \frac{P_{(j+k)}}{P_j}$$

EXAMPLE: Forward pricing

Calculate the forward price two years from now for a \$1 par, zero-coupon, three-year bond given the following spot rates.

The two-year spot rate, $S_2 = 4\%$.

The five-year spot rate, $S_5 = 6\%$.

Answer:

Calculate discount factors P_j and $P_{(j+k)}$.

$$P_j = P_2 = 1 / (1 + 0.04)^2 = 0.9246$$

$$P_{(j+k)} = P_5 = 1 / (1 + 0.06)^5 = 0.7473$$

The forward price of a three-year bond in two years is represented as $F_{(2,3)}$

$$F_{(j,k)} = P_{(j+k)} / P_j$$

$$F_{(2,3)} = 0.7473 / 0.9246 = 0.8082$$

In other words, \$0.8082 is the price agreed to today, to pay in two years, for a three-year bond that will pay \$1 at maturity.



PROFESSOR'S NOTE

In the Derivatives portion of the curriculum, the forward price is computed as the future value (for j periods) of $P_{(j+k)}$. It gives the same result and can be verified using the data in the previous example by computing the future value of P_5 (i.e., compounding for two periods at S_2). $FV = 0.7473(1.04)^2 = \$0.8082$.

The Forward Rate Model

The **forward rate model** relates forward and spot rates as follows:

$$[1 + S_{(j+k)}]^{(j+k)} = (1 + S_j)^j [1 + f(j,k)]^k$$

or

$$[1 + f(j,k)]^k = [1 + S_{(j+k)}]^{(j+k)} / (1 + S_j)^j$$

This model is useful because it illustrates how forward rates and spot rates are interrelated.

This equation suggests that the forward rate $f(2,3)$ should make investors indifferent between buying a five-year zero-coupon bond versus buying a two-year zero-coupon bond and at maturity reinvesting the principal for three additional years.

EXAMPLE: Forward rates

Suppose that the two-year and five-year spot rates are $S_2 = 4\%$ and $S_5 = 6\%$.

Calculate the implied three-year forward rate for a loan starting two years from now [i.e., $f(2,3)$].

Answer:

$$[1 + f(j,k)]^k = [1 + S_{(j+k)}]^{(j+k)} / (1 + S_j)^j$$

$$[1 + f(2,3)]^3 = [1 + 0.06]^5 / [1 + 0.04]^2$$

$$f(2,3) = 7.35\%$$

Note that the forward rate $f(2,3) > S_5$ because the yield curve is upward sloping.

If the yield curve is upward sloping, [i.e., $S_{(j+k)} > S_j$], then the forward rate corresponding to the period from j to k [i.e., $f(j,k)$] will be greater than the spot rate for maturity $j+k$ [i.e., $S_{(j+k)}$]. The opposite is true if the curve is downward sloping.

LOS 23.b: Describe how zero-coupon rates (spot rates) may be obtained from the par curve by bootstrapping.

A **par rate** is the yield to maturity of a bond trading at par. Par rates for bonds with different maturities make up the **par rate curve** or simply the **par curve**. By definition, the par rate will be equal to the coupon rate on the bond. Generally, par curve refers to the par rates for government or benchmark bonds.

By using a process called **bootstrapping**, spot rates or zero-coupon rates can be derived from the par curve. Bootstrapping involves using the output of one step as an input to the next step. We first recognize that (for annual-pay bonds) the one-year spot rate (S_1) is the same as the one-year par rate. We can then compute S_2 using S_1 as one of the inputs. Continuing the process, we can compute the three-year spot rate S_3 using S_1 and S_2 computed earlier. Let's clarify this with an example.

EXAMPLE: Bootstrapping spot rates

Given the following (annual-pay) par curve, compute the corresponding spot rate curve:

Maturity	Par Rate
1	1.00%
2	1.25%
3	1.50%

Answer:

$S_1 = 1.00\%$ (given directly).

If we discount each cash flow of the bond using its yield, we get the market price of the bond. Here, the market price is the par value. Consider the 2-year bond.

$$100 = \frac{1.25}{(1.0125)} + \frac{101.25}{(1.0125)^2}$$

Alternatively, we can also value the 2-year bond using spot rates:

$$100 = \frac{1.25}{(1 + S_1)} + \frac{101.25}{(1 + S_2)^2} = \frac{1.25}{(1.01)} + \frac{101.25}{(1 + S_2)^2}$$

$$100 = 1.2376 + \frac{101.25}{(1 + S_2)^2}$$

$$98.7624 = \frac{101.25}{(1 + S_2)^2}$$

Multiplying both sides by $[(1 + S_2)^2 / 98.7624]$, we get $(1 + S_2)^2 = 1.0252$.

Taking square roots, we get $(1 + S_2) = 1.01252$. $S_2 = 0.01252$ or 1.252%

Similarly, $100 = \frac{1.50}{(1 + S_1)} + \frac{1.50}{(1 + S_2)^2} + \frac{101.50}{(1 + S_3)^3}$

Using the values of S_1 and S_2 computed earlier,

$$100 = \frac{1.50}{(1.01)} + \frac{1.50}{(1.01252)^2} + \frac{101.50}{(1 + S_3)^3}$$

$$100 = 2.9483 + \frac{101.50}{(1 + S_3)^3}$$

$$97.0517 = \frac{101.50}{(1 + S_3)^3}$$

$$(1 + S_3)^3 = 1.0458$$

$$(1 + S_3) = 1.0151 \text{ and hence } S_3 = 1.51\%$$



MODULE QUIZ 23.1

- When the yield curve is downward sloping, the forward curves are *most likely* to lie:
 - above the spot curve.
 - below the spot curve.
 - either above or below the spot curve.
- The model that equates buying a long-maturity zero-coupon bond to entering into a forward contract to buy a zero-coupon bond that matures at the same time is known as the:
 - forward rate model.
 - forward pricing model.
 - forward arbitrage model.



Video covering
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available online.

MODULE 23.2: SPOT AND FORWARD RATES, PART 2

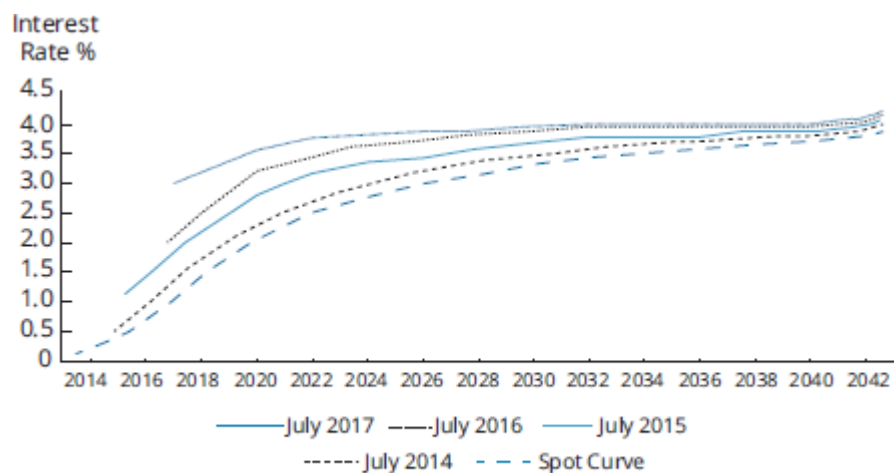
LOS 23.c: Describe the assumptions concerning the evolution of spot rates in relation to forward rates implicit in active bond portfolio management.

Relationships Between Spot and Forward Rates

For an upward-sloping spot curve, the forward rate rises as j increases. (For a downward-sloping yield curve, the forward rate declines as j increases.) For an upward-sloping spot curve, the forward curve will be above the spot curve as shown in Figure 23.1. Conversely, when the spot curve is downward sloping, the forward curve will be below it.

Figure 23.1 shows spot and forward curves as of July 2013. Because the spot yield curve is upward sloping, the forward curves lie above the spot curve.

Figure 23.1: Spot Curve and Forward Curves



Reproduced from Level II CFA Curriculum learning module "The Term Structure and Interest Rate Dynamics," Exhibit 2, with permission from CFA Institute.

From the forward rate model:

$$(1 + S_T)^T = (1 + S_1)[1 + f(1, T - 1)]^{(T-1)}$$

which can be expanded to:

$$(1 + S_T)^T = (1 + S_1) [1 + f(1, 1)] [1 + f(2, 1)] [1 + f(3, 1)] \dots [1 + f(T - 1, 1)]$$

In other words, the spot rate for a long-maturity security will equal the geometric mean of the one period spot rate and a series of one-year forward rates.

Forward Price Evolution

If the future spot rates actually evolve as forecasted by the forward curve, the forward price will remain unchanged. Therefore, a change in the forward price indicates that the future spot rate(s) did not conform to the forward curve. When spot rates turn out to be lower (higher) than implied by the forward curve, the forward price will increase

(decrease). A trader expecting lower future spot rates (than implied by the current forward rates) would purchase the forward contract to profit from its appreciation.

For a bond investor, the return on a bond over a one-year horizon is always equal to the one-year risk-free rate *if the spot rates evolve as predicted by today's forward curve*. If the spot curve one year from today is not the same as that predicted by today's forward curve, the return over the one-year period will differ, with the return depending on the bond's maturity.

An active portfolio manager will try to outperform the overall bond market by predicting how the future spot rates will differ from those predicted by the current forward curve.

EXAMPLE: Spot rate evolution

Jane Dash, CFA, has collected benchmark spot rates as shown here:

Maturity	Spot Rate
1	3.00%
2	4.00%
3	5.00%

The expected spot rates at the end of one year are as follows:

Year	Expected Spot
1	5.01%
2	6.01%

Calculate the one-year holding period return of a:

1. 1-year zero-coupon bond.
2. 2-year zero-coupon bond.
3. 3-year zero-coupon bond.

Answer:

First, note that the expected spot rates provided just happen to be the forward rates implied by the current spot rate curve.

Recall that:

$$[1 + f(j,k)]^k = [1 + S_{(j+k)}]^{(j+k)} / (1 + S_j)^j$$

Hence:

$$[1 + f(1,1)]^1 = \frac{(1 + S_2)^2}{(1 + S_1)} = \frac{(1.04)^2}{(1.03)} \rightarrow f(1,1) = 0.0501 \text{ and}$$

$$[1 + f(1,2)]^2 = \frac{(1 + S_3)^3}{(1 + S_1)} = \frac{(1.05)^3}{(1.03)} \rightarrow f(1,2) = 0.0601$$

1. The price of a one-year zero-coupon bond given the one-year spot rate of 3% is $1 / (1.03)$ or 0.9709.

After one year, the bond is at maturity and pays \$1 regardless of the spot rates.

Hence the holding period return = $\left(\frac{1.00}{0.9709}\right) - 1 = 3\%$

2. The price of a two-year zero-coupon bond given the two-year spot rate of 4%:

$$P_2 = \frac{1}{(1 + S_2)^2} = \frac{1}{(1.04)^2} = 0.9246$$

After one year, the bond will have one year remaining to maturity, and based on a one-year expected spot rate of 5.01%, the bond's price will be $1 / (1.0501) = \$0.9523$

Hence, the holding period return = $\left(\frac{0.9523}{0.9246}\right) - 1 = 3\%$

3. The price of three-year zero-coupon bond given the three-year spot rate of 5%:

$$P_3 = \frac{1}{(1 + S_3)^3} = \frac{1}{(1.05)^3} = 0.8638$$

After one year, the bond will have two years remaining to maturity. Based on a two-year expected spot rate of 6.01%, the bond's price will be $1 / (1.0601)^2 = \$0.8898$

Hence, the holding period return = $\left(\frac{0.8898}{0.8638}\right) - 1 = 3\%$

Hence, regardless of the maturity of the bond, the holding period return will be the one-year spot rate if the spot rates evolve consistent with the forward curve (as it existed when the trade was initiated).

If an investor believes that future spot rates will be lower than corresponding forward rates, then she will purchase bonds (at a presumably attractive price) because the market appears to be discounting future cash flows at “too high” of a discount rate.

LOS 23.d: Describe the strategy of rolling down the yield curve.

“Rolling Down the Yield Curve”

The most straightforward strategy for a bond investor is *maturity matching*—purchasing bonds that have a maturity equal to the investor's investment horizon.

However, with an upward-sloping interest rate term structure, investors seeking superior returns may pursue a strategy called **riding the yield curve** (also known as **rolling down the yield curve**). Under this strategy, an investor will purchase bonds with maturities longer than his investment horizon. In an upward-sloping yield curve, shorter maturity bonds have lower yields than longer maturity bonds. As the bond approaches maturity (i.e., rolls down the yield curve), it is valued using successively lower yields and, therefore, at successively higher prices.

If the yield curve remains unchanged over the investment horizon, riding the yield curve strategy will produce higher returns than a simple maturity matching strategy, increasing the total return of a bond portfolio. The greater the difference between the forward rate and the spot rate, and the longer the maturity of the bond, the higher the total return.

Consider Figure 23.2, which shows a hypothetical upward-sloping yield curve and the price of a 3% annual-pay coupon bond (as a percentage of par).

Figure 23.2: Price of a 3%, Annual Pay Bond

Maturity	Yield	Price
5	3	100
10	3.5	95.84
15	4	88.88
20	4.5	80.49
25	5	71.81
30	5.5	63.67

A bond investor with an investment horizon of five years could purchase a bond maturing in five years and earn the 3% coupon but no capital gains (the bond can be currently purchased at par and will be redeemed at par at maturity). However, assuming no change in the yield curve over the investment horizon, the investor could instead purchase a 30-year bond for \$63.67, hold it for five years, and sell it for \$71.81, earning an additional return beyond the 3% coupon over the same period.

In the aftermath of the financial crisis of 2007–08, central banks kept short-term rates low, giving yield curves a steep upward slope. Many active managers took advantage by borrowing at short-term rates and buying long maturity bonds. The risk of such a leveraged strategy is the possibility of an increase in spot rates.



MODULE QUIZ 23.2

1. If the future spot rates are expected to be lower than the current forward rates for the same maturities, bonds are *most likely* to be:
 - A. overvalued.
 - B. undervalued.
 - C. correctly valued.
2. The strategy of rolling down the yield curve is *most likely* to produce superior returns for a fixed income portfolio manager investing in bonds with maturity higher than the manager's investment horizon when the spot rate curve:
 - A. is downward sloping.
 - B. in the future matches that projected by today's forward curves.
 - C. is upward sloping.

MODULE 23.3: THE SWAP RATE CURVE



Video covering this content is available online.

LOS 23.e: Explain the swap rate curve and why and how market participants use it in valuation.

The Swap Rate Curve

In a plain vanilla interest rate swap, one party makes payments based on a fixed rate while the counterparty makes payments based on a floating rate. The fixed rate in an interest rate swap is called the **swap fixed rate** or **swap rate**.

If we consider how swap rates vary for various maturities, we get the **swap rate curve**, which has become an important interest-rate benchmark for credit markets.

Market participants prefer the swap rate curve as a benchmark interest rate curve rather than a government bond yield curve for the following reasons:

- Swap rates reflect the credit risk of commercial banks rather than the credit risk of governments.
- The swap market is not regulated by any government, which makes swap rates in different countries more comparable. (Government bond yield curves additionally reflect sovereign risk unique to each country.)
- The swap curve typically has yield quotes at many maturities, while the U.S. government bond yield curve has on-the-run issues trading at only a small number of maturities.

Wholesale banks that manage interest rate risk with swap contracts are more likely to use swap curves to value their assets and liabilities. Retail banks, on the other hand, are more likely to use a government bond yield curve.

Given a notional principal of \$1 and a swap fixed rate SFR_T , the value of the fixed rate payments on a swap can be computed using the relevant (e.g., MRR) spot rate curve. For a given swap tenor T , we can solve for SFR in the following equation.

$$\sum_{t=1}^T \frac{SFR_T}{(1+S_t)^t} + \frac{1}{(1+S_T)^T} = 1$$

In the equation, SFR can be thought of as the coupon rate of a \$1 par value bond given the underlying spot rate curve.



PROFESSOR'S NOTE

Prior to the 2023 curriculum, CFA Institute referenced LIBOR (the “London Interbank Offered Rate”) as the underlying interest rate for derivative instruments. However, concerns about LIBOR being manipulated have led to LIBOR being replaced in recent years with other reference rates, such as the Federal Reserve Bank of New York’s “SOFR” (Secured Overnight Financing Rate), and the Bank of England’s “SONIA” (Sterling Overnight Index Average).

To reflect this change, the CFA curriculum now uses the generic term MRR (“market reference rate”) to refer to these various LIBOR replacements.

EXAMPLE: Swap rate curve

Given the following MRR spot rate curve, compute the swap fixed rate for a tenor of 1, 2, and 3 years (i.e., compute the swap rate curve).

Maturity	Spot Rate
1	3.00%
2	4.00%
3	5.00%

Answer:

1. SFR_1 can be computed using the equation:

$$\frac{SFR_1}{(1+S_1)} + \frac{1}{(1+S_1)} = 1$$

$$\frac{SFR_1}{(1.03)} + \frac{1}{(1.03)} = 1 \rightarrow SFR_1 = 3.00\%$$

2. SFR_2 can be similarly computed:

$$\frac{SFR_2}{(1+S_1)} + \frac{SFR_2}{(1+S_2)^2} + \frac{1}{(1+S_2)^2} = 1$$

$$\frac{SFR_2}{(1.03)} + \frac{SFR_2}{(1.04)^2} + \frac{1}{(1.04)^2} = 1 \rightarrow SFR_2 = 3.98\%$$

3. Finally, SFR_3 can be computed as:

$$\frac{SFR_3}{(1+S_1)} + \frac{SFR_3}{(1+S_2)^2} + \frac{SFR_3}{(1+S_3)^3} + \frac{1}{(1+S_3)^3} = 1$$

$$\frac{SFR_3}{(1.03)} + \frac{SFR_3}{(1.04)^2} + \frac{SFR_3}{(1.05)^3} + \frac{1}{(1.05)^3} = 1 \rightarrow SFR_3 = 4.93\%$$



PROFESSOR'S NOTE

A different (and better) method of computing swap fixed rates is discussed in detail in the Derivatives portion of the curriculum.



MODULE QUIZ 23.3

1. Which of the following statements about the swap rate curve is *most accurate*?
 - A. The swap rate reflects the interest rate for the floating-rate leg of an interest rate swap.
 - B. Retail banks are more likely to use the swap rate curve as a benchmark than the government spot curve.
 - C. Swap rates are comparable across different countries because the swap market is not controlled by governments.

MODULE 23.4: SPREAD MEASURES



Video covering this content is available online.

LOS 23.f: Calculate and interpret the swap spread for a given maturity.

Swap spread refers to the amount by which the swap rate exceeds the yield of a government bond with the same maturity.

$$\text{swap spread}_t = \text{swap rate}_t - \text{Treasury yield}_t$$

For example, if the fixed rate of a one-year fixed-for-floating MRR swap is 0.57% and the one-year Treasury is yielding 0.11%, the 1-year swap spread is $0.57\% - 0.11\% = 0.46\%$, or 46 bps.

Swap spreads are almost always positive, reflecting the lower credit risk of governments compared to the credit risk of surveyed banks that determines the swap rate.

The swap rate curve is arguably the most commonly used interest rate curve. This rate curve roughly reflects the default risk of a commercial bank with a credit rating of A1/A+.

EXAMPLE: Swap spread

The two-year fixed-for-floating swap rate is 2.02% and the two-year U.S. Treasury bond is yielding 1.61%. What is the swap spread?

Answer:

$$\begin{aligned}\text{swap spread} &= (\text{swap rate}) - (\text{T-bond yield}) \\ &= 2.02\% - 1.61\% = 0.41\% \text{ or } 41 \text{ bps}\end{aligned}$$

LOS 23.g: Describe short-term interest rate spreads used to gauge economy-wide credit risk and liquidity risk.

I-Spread

The **I-spread**, or interpolated spread, for a credit-risky bond is the amount by which the yield on the risky bond exceeds the swap rate for the same maturity. In a case where the swap rate for a specific maturity is not available, the missing swap rate can be estimated from the swap rate curve using linear interpolation.

EXAMPLE: I-spread

6% Zinni, Inc., bonds are currently yielding 2.35% and mature in 1.6 years. From the provided swap curve, compute the I-spread.

Swap curve:

Tenor	Swap Rate
0.5	1.00%
1	1.25%
1.5	1.35%
2	1.50%

Answer:

Linear interpolation:

First, recognize that 1.6 years falls in the 1.5-to-2-year interval.

Interpolated rate = rate for lower bound + (# of years for interpolated rate – # of years for lower bound)(higher bound rate – lower bound rate) / (# of years for upper bound – # of years for lower bound)

1.6-year swap rate =

$$\begin{aligned}&1.5\text{-year swap rate} + \frac{0.10(2\text{-year swap rate} - 1.5\text{-year swap rate})}{0.50} \\ &= 1.35 + \frac{0.10(1.50 - 1.35)}{0.50} = 1.38\%\end{aligned}$$

I-spread = yield on the bond – swap rate = 2.35% – 1.38% = 0.97% or 97 bps

While a bond's yield reflects time value as well as compensation for credit and liquidity risk, I-spread only reflects compensation for credit and liquidity risks. The higher the I-spread, the higher the compensation for liquidity and credit risk.

The Z-Spread

The **Z-spread** is the spread that, when added to each spot rate on the default-free spot curve, makes the present value of a bond's cash flows equal to the bond's market price. Therefore, the Z-spread is a spread over the entire spot rate curve.

For example, suppose the one-year spot rate is 4% and the two-year spot rate is 5%. The market price of a two-year bond with annual coupon payments of 8% is \$104.12. The Z-spread is the spread that balances the following equality:

$$\$104.12 = \frac{\$8}{(1 + 0.04 + Z)} + \frac{\$108}{(1 + 0.05 + Z)^2}$$

In this case, the Z-spread is 0.008, or 80 basis points. (Plug $Z = 0.008$ into the right-hand-side of the equation just listed to reassure yourself that the present value of the bond's cash flows equals \$104.12).

The term *zero volatility* in the Z-spread refers to the assumption of zero interest rate volatility. Z-spread is not appropriate to use to value bonds with embedded options; without any interest rate volatility options are meaningless. If we ignore the embedded options for a bond and estimate the Z-spread, the estimated Z-spread will include the cost of the embedded option (i.e., it will reflect compensation for option risk as well as compensation for credit and liquidity risk).

EXAMPLE: Computing the price of an option-free risky bond using Z-spread

A three-year, 5% annual-pay ABC, Inc., bond trades at a Z-spread of 100 bps over the benchmark spot rate curve.

The benchmark one-year spot rate, one-year forward rate in one year and one-year forward rate in year 2 are 3%, 5.051%, and 7.198%, respectively.

Compute the bond's price.

Answer:

First derive the spot rates:

$$S_1 = 3\% \text{ (given)}$$

$$(1 + S_2)^2 = (1 + S_1)[1 + f(1,1)] = (1.03)(1.05051) \rightarrow S_2 = 4.02\%$$

$$(1 + S_3)^3 = (1 + S_1)[1 + f(1,1)][1 + f(2,1)] = (1.03)(1.05051)(1.07198) \rightarrow S_3 = 5.07\%$$

value (with Z-spread) =

$$\frac{5}{(1.03 + 0.01)} + \frac{5}{(1.0402 + 0.01)^2} + \frac{105}{(1.0507 + 0.01)^3} = 97.33$$

TED Spread

The “TED” in “**TED spread**” is an acronym that combines the “T” in “T-bill” with “ED” (the ticker symbol for the Eurodollar futures contract).

Conceptually, the TED spread is the amount by which the MRR exceeds the interest rate on short-term U.S. government debt of the same maturity.

For example, if three-month MRR is 0.33% and the three-month T-bill rate is 0.03%, then:

$$\text{TED spread} = (\text{3-month MRR}) - (\text{3-month T-bill rate}) = 0.33\% - 0.03\% \\ = 0.30\% \text{ or } 30 \text{ bps.}$$

Because T-bills are considered to be risk free while MRR reflects the risk of lending to commercial banks, the TED spread is seen as an indication of the credit and liquidity risk in the banking sector. A rising TED spread indicates that market participants believe banks are increasingly likely to default on loans and that risk-free T-bills are becoming more valuable in comparison. The TED spread captures the risk in the banking system more accurately than does the 10-year swap spread.

MRR-OIS Spread

OIS stands for overnight indexed swap and represents interest rate on unsecured overnight lending between banks. The OIS rate roughly reflects the federal funds rate and includes minimal counterparty credit risk.

The **MRR-OIS spread** (formerly the LIBOR-OIS spread) is the amount by which the MRR (which includes some credit risk) exceeds the OIS rate (which includes only minimal credit risk) and also indicates the level of credit and liquidity risk in the banking system.



MODULE QUIZ 23.4

1. The swap spread for a default-free bond is *least likely* to reflect the bond's:
 - A. mispricing in the market.
 - B. illiquidity.
 - C. time value.
2. Which of the following statements about the Z-spread is *most accurate*? The Z-spread is the:
 - A. difference between the yield to maturity of a bond and the linearly interpolated swap rate.
 - B. spread over the Treasury spot curve that a bond would trade at if it had zero embedded options.
 - C. spread over the Treasury spot curve required to match the value of a bond to its current market price.
3. The TED spread is calculated as the difference between:
 - A. the three-month MRR and the three-month T-bill rate.
 - B. MRR and the overnight indexed swap rate.

C. the three-month T-bill rate and the overnight indexed swap rate.

MODULE 23.5: TERM STRUCTURE THEORY



Video covering
this content is
available online.

LOS 23.h: Explain traditional theories of the term structure of interest rates and describe the implications of each theory for forward rates and the shape of the yield curve.

We'll explain each of the theories of the **term structure of interest rates**, paying particular attention to the implications of each theory for the shape of the yield curve and the interpretation of forward rates.

Unbiased Expectations Theory

Under the **unbiased expectations theory** or the **pure expectations theory**, we hypothesize that it is investors' expectations that determine the shape of the interest rate term structure.

Specifically, this theory suggests that forward rates are solely a function of expected future spot rates, and that every maturity strategy has the same expected return over a given investment horizon. In other words, long-term interest rates equal the mean of future *expected* short-term rates. This implies that an investor should earn the same return by investing in a five-year bond or by investing in a three-year bond and then a two-year bond after the three-year bond matures. Similarly, an investor with a three-year investment horizon would be indifferent between investing in a three-year bond or in a five-year bond that will be sold two years prior to maturity. The underlying principle behind the pure expectations theory is risk neutrality: Investors don't demand a risk premium for maturity strategies that differ from their investment horizon.

For example, suppose the one-year spot rate is 5% and the two-year spot rate is 7%. Under the unbiased expectations theory, the one-year forward rate in one year must be 9% because investing for two years at 7% yields approximately the same annual return as investing for the first year at 5% and the second year at 9%. In other words, the two-year rate of 7% is the average of the expected future one-year rates of 5% and 9%. This is shown in Figure 23.3.

Figure 23.3: Spot and Future Rates



Notice that in this example, because short-term rates are expected to rise (from 5% to 9%), the yield curve will be upward sloping.

Therefore, the implications for the shape of the yield curve under the pure expectations theory are:

- If the yield curve is upward sloping, short-term rates are expected to rise.
- If the curve is downward sloping, short-term rates are expected to fall.
- A flat yield curve implies that the market expects short-term rates to remain constant.

Local Expectations Theory

The **local expectations theory** is similar to the unbiased expectations theory with one major difference: the local expectations theory preserves the risk-neutrality assumption only for short holding periods. In other words, over longer periods, risk premiums should exist. This implies that over short time periods, every bond (even long-maturity risky bonds) should earn the risk-free rate.

The local expectations theory can be shown not to hold because the short-holding-period returns of long-maturity bonds can be shown to be higher than short-holding-period returns on short-maturity bonds due to liquidity premiums and hedging concerns.

Liquidity Preference Theory

The **liquidity preference theory** of the term structure addresses the shortcomings of the pure expectations theory by proposing that forward rates reflect investors' expectations of future spot rates, plus a liquidity premium to compensate investors for exposure to interest rate risk. Furthermore, the theory suggests that this liquidity premium is positively related to maturity: a 25-year bond should have a larger liquidity premium than a five-year bond.

Thus, the liquidity preference theory states that forward rates are *biased* estimates of the market's expectation of future rates because they include a liquidity premium. Therefore, a positive-sloping yield curve may indicate that either: (1) the market expects future interest rates to rise or (2) rates are expected to remain constant (or even fall), but the addition of the liquidity premium results in a positive slope. A downward-sloping yield curve indicates steeply falling short-term rates according to the liquidity preference theory.

The size of the liquidity premiums need not be constant over time. They may be larger during periods of greater economic uncertainty when risk aversion among investors is higher.

Segmented Markets Theory

Under the **segmented markets theory**, yields are not determined by liquidity premiums and expected spot rates. Rather, the shape of the yield curve is determined by the preferences of borrowers and lenders, which drives the balance between supply of and demand for loans of different maturities. This is called the segmented markets theory because the theory suggests that the yield at each maturity is determined independently of the yields at other maturities; we can think of each maturity to be essentially unrelated to other maturities.

The segmented markets theory supposes that various market participants only deal in securities of a particular maturity because they are prevented from operating at different

maturities. For example, pension plans and insurance companies primarily purchase long-maturity bonds for asset-liability matching reasons and are unlikely to participate in the market for short-term funds.

Preferred Habitat Theory

The **preferred habitat theory** also proposes that forward rates represent expected future spot rates plus a premium, but it does not support the view that this premium is directly related to maturity.

Instead, the preferred habitat theory suggests that the existence of an imbalance between the supply and demand for funds in a given maturity range will induce lenders and borrowers to shift from their preferred habitats (maturity range) to one that has the opposite imbalance. However, to entice investors to do so, the investors must be offered an incentive to compensate for the exposure to price and/or reinvestment rate risk in the less-than-preferred habitat. Borrowers require cost savings (i.e., lower yields) and lenders require a yield premium (i.e., higher yields) to move out of their preferred habitats.

Under this theory, premiums are related to supply and demand for funds at various maturities. Unlike the liquidity preference theory, under the preferred habitat theory a 10-year bond might have a higher or lower risk premium than the 25-year bond. It also means that the preferred habitat theory can be used to explain almost any yield curve shape.



MODULE QUIZ 23.5

1. Which of the following statements regarding the traditional theories of the term structure of interest rates is *most accurate*?
 - A. The segmented markets theory proposes that market participants have strong preferences for specific maturities.
 - B. The liquidity preference theory hypothesizes that the yield curve must always be upward sloping.
 - C. The preferred habitat theory states that yields at different maturities are determined independently of each other.

MODULE 23.6: YIELD CURVE RISKS AND ECONOMIC FACTORS



Video covering this content is available online.

LOS 23.i: Explain how a bond's exposure to each of the factors driving the yield curve can be measured and how these exposures can be used to manage yield curve risks.

Managing Yield Curve Risks

Yield curve risk refers to risk to the value of a bond portfolio due to unexpected changes in the yield curve.

To counter yield curve risk, we first identify our portfolio's sensitivity to yield curve changes using one or more measures. Yield curve sensitivity can be generally measured by **effective**

duration, or more precisely using **key rate duration**, or a three-factor model that decomposes changes in the yield curve into changes in **level**, **steepness**, and **curvature**.

Effective Duration

Effective duration measures price sensitivity to small *parallel* shifts in the yield curve. It is important to note that effective duration is not an accurate measure of interest rate sensitivity to *non-parallel* shifts in the yield curve like those described by **shaping risk**. Shaping risk refers to changes in portfolio value due to changes in the *shape* of the benchmark yield curve. (Note, however, that parallel shifts explain more than 75% of the variation in bond portfolio returns.)

Key Rate Duration

A more precise method used to quantify bond price sensitivity to interest rates is key rate duration. Compared to effective duration, key rate duration is superior for measuring the impact of nonparallel yield curve shifts.

Key rate duration is the sensitivity of the value of a security (or a bond portfolio) to changes in a single par rate, holding all other par rates constant. In other words, key rate duration isolates price sensitivity to a change in the yield at a particular maturity only.

Numerically, key rate duration is defined as the approximate percentage change in the value of a bond portfolio in response to a 100 basis point change in the corresponding key par rate, holding all other par rates constant. Conceptually, we could determine the key rate duration for the five-year segment of the yield curve by changing only the five-year par rate and observing the change in value of the portfolio. Keep in mind that every security or portfolio has a set of key rate durations—one for each key rate.

For example, a bond portfolio has interest rate risk exposure to only three maturity points on the par rate curve: the 1-year, 5-year, and 25-year maturities, with key rate durations represented by $D_1 = 0.7$, $D_5 = 3.5$, and $D_{25} = 9.5$, respectively. The sum total of these key rate durations is the effective duration of the portfolio.

The model for yield curve risk using these key rate durations would be:

$$\frac{\Delta P}{P} \approx -D_1 \Delta r_1 - D_5 \Delta r_5 - D_{25} \Delta r_{25}$$

$$\frac{\Delta P}{P} \approx -(0.7) \Delta r_1 - (3.5) \Delta r_5 - (9.5) \Delta r_{25}$$

Sensitivity to Parallel, Steepness, and Curvature Movements

An alternative to decomposing yield curve risk into sensitivity to changes at various maturities (key rate duration) is to decompose the risk into sensitivity to the following three categories of yield curve movements:

- **Level** (Δx_L). A parallel increase or decrease of interest rates.
- **Steepness** (Δx_S). Long-term interest rates increase while short-term rates decrease.
- **Curvature** (Δx_C). Increasing curvature means short- and long-term interest rates increase while intermediate rates do not change.

It has been found that all yield curve movements can be described using a combination of one or more of these movements.

We can then model the change in the value of our portfolio as follows:

$$\frac{\Delta P}{P} \approx -D_L \Delta x_L - D_S \Delta x_S - D_C \Delta x_C$$

where D_L , D_S , and D_C are respectively the portfolio's sensitivities to changes in the yield curve's level, steepness, and curvature.

For example, for a particular portfolio, yield curve risk can be described as:

$$\frac{\Delta P}{P} \approx -4\Delta x_L - 5\Delta x_S - 3\Delta x_C$$

If the following changes in the yield curve occurred: $\Delta x_L = -0.004$, $\Delta x_S = 0.001$, and $\Delta x_C = 0.002$, then the percentage change in portfolio value could be calculated as:

$$\frac{\Delta P}{P} \approx -4(-0.004) - 5(0.001) - 3(0.002) = 0.005$$

This predicts a +0.5% increase in the portfolio value resulting from the yield curve movements.

LOS 23.j: Explain the maturity structure of yield volatilities and their effect on price volatility.

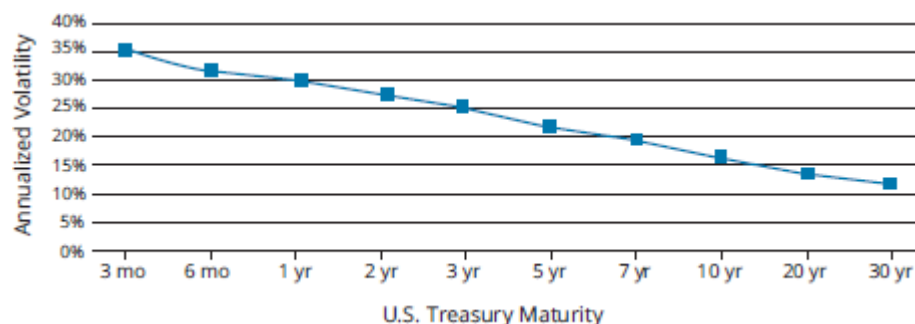
Maturity Structure of Yield Curve Volatilities

Interest rate volatility is a key concern for bond managers because interest rate volatility drives price volatility in a fixed income portfolio. Interest rate volatility becomes particularly important when securities have embedded options, which are especially sensitive to volatility.

The **term structure of interest rate volatility** is the graph of yield volatility versus maturity.

Figure 23.4 shows a typical term structure of interest rate volatility. Note that, as shown here, short-term interest rates are generally more volatile than are long-term rates.

Figure 23.4: Historical Volatility Term Structure: U.S. Treasuries, August 2005–December 2007



Volatility at the long-maturity end is thought to be associated with uncertainty regarding the real economy and inflation, while volatility at the short-maturity end reflects risks

regarding monetary policy.

Interest rate volatility at time t for a security with maturity of T is denoted as $\sigma(t,T)$. This variable measures the annualized standard deviation of the change in bond yield.

LOS 23.k: Explain how key economic factors are used to establish a view on benchmark rates, spreads, and yield curve changes.

Macroeconomic factors that affect bond yields include inflation forecasts, GDP growth, and monetary policy. Two-thirds of the *variation* in short- and intermediate-term yields is explained by monetary policy, and the remaining is explained by the other factors. In contrast, inflation explains two-thirds of the variation in long-term yields, with the remaining mostly explained by monetary policy.

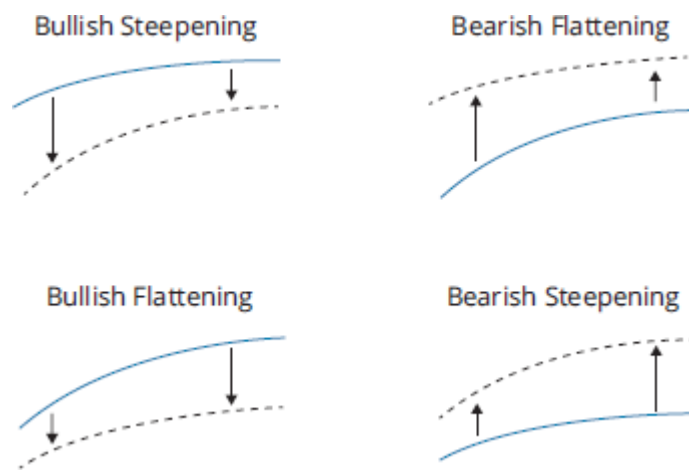
Bond risk premium is the excess return (over the one-year risk-free rate) earned by investors for investing in long-term government bonds. Bond risk premium (also known as the *term premium* or *duration premium*) is a forward-looking expectation. For example, suppose that the expected one-year holding period return on a 5-year government bond is 4%, while the one-year risk-free rate is 1%. The 5-year bond risk premium in this example would be $4\% - 1\% = 3\%$.

Monetary policy influences the shape of the yield curve. During economic expansions, to combat rising inflation, central banks may raise short-term rates, leading to a **bearish flattening** of the yield curve. During recessionary times, central banks may reduce short-term rates, leading to a **bullish steepening**. The U.S. Federal Reserve has dramatically expanded its balance sheet in recent years, purchasing U.S. Treasuries and mortgage-backed and municipal securities, reducing long-term bond yields.

Other factors affecting bond prices include the following:

- **Fiscal policy.** Expansionary (restrictive) fiscal policies increase (decrease) yields.
- **Maturity structure.** The government's choice of maturity when issuing new securities affects the supply (and yield) of bonds in those maturity segments. An increase in offerings in a specific segment of the market increases the supply and increases the yield in that segment (the market segmentation theory).
- **Investor demand.** Domestic and foreign investor demand and preferences for specific maturity segments affect the yield in that segment. For example, pension plans and insurance companies may prefer longer maturity bonds, driving down longer-term yields. During periods of market turmoil, a **flight to safety** may reduce long-term government bond yields, resulting in a **bullish flattening** of the yield curve.

Figure 23.5: Categories of Shifts in the Yield Curve



Investor Actions

In expectation of a rise (fall) in rates, investors will lower (extend) the duration of their bond portfolios. Expectations of a steepening of the yield curve may lead to investors going long short-term bonds and short longer-term bonds. Trades may be designed to be duration-neutral so that a change in the *level* of interest rates does not affect the value of the portfolio. Investors with long-only mandates may rotate between a **bullet portfolio** (a portfolio concentrated in a single maturity) and a **barbell portfolio** (a portfolio with short and long maturities). An investor that expects a bullish flattening of the yield curve may rotate out of a bullet portfolio and into a barbell portfolio.



MODULE QUIZ 23.6

1. The *least appropriate* measure to use to identify and manage “shaping risk” is a portfolio’s:
 - A. effective duration.
 - B. key rate durations.
 - C. sensitivities to level, steepness, and curvature factors.
2. Regarding the volatility term structure, research indicates that volatility in short-term rates is *most strongly* linked to uncertainty regarding:
 - A. the real economy.
 - B. monetary policy.
 - C. inflation.
3. Restrictive monetary policy is *most likely* to be associated with:
 - A. bearish flattening.
 - B. bullish steepening.
 - C. bullish flattening.

KEY CONCEPTS

LOS 23.a

The spot rate for a particular maturity is equal to a geometric average of the one-period spot rate and a series of one-period forward rates.

When the spot curve is flat, forward rates will equal spot rates and yields. When the spot curve is upward sloping (downward sloping), forward rate curves will be above (below) the spot curve and the yield for a maturity of T will be less than (greater than) the spot rate S_T .

The forward *pricing* model values forward contracts by using an arbitrage-free framework that equates buying a zero-coupon bond to entering into a forward contract to buy a zero-coupon bond in the future that matures at the same time:

$$P_{(j+k)} = P_j F_{(j,k)}$$

The forward *rate* model tells us that the investors will be indifferent between buying a long-maturity zero-coupon bond versus buying a shorter-maturity zero-coupon bond and reinvesting the principal at the locked in forward rate $f(j,k)$.

$$[1 + S_{(j+k)}]^{(j+k)} = (1 + S_j)^j [1 + f(j,k)]^k$$

LOS 23.b

By using a process called bootstrapping, spot rates (i.e., zero-coupon rates) can be derived from the par curve iteratively—one spot rate at a time.

LOS 23.c

If spot rates evolve as predicted by forward rates, bonds of all maturities will realize a one-period return equal to the one-period spot rate and the forward price will remain unchanged.

Active bond portfolio management is built on the presumption that the current forward curve may not accurately predict future spot rates. Managers attempt to outperform the market by making predictions about how spot rates will change relative to the rates suggested by forward rate curves.

If an investor believes that future spot rates will be lower than corresponding forward rates, then the investor will purchase bonds (at a presumably attractive price) because the market appears to be discounting future cash flows at “too high” of a discount rate.

LOS 23.d

When the yield curve is upward sloping, bond managers may use the strategy of “rolling down the yield curve” to chase above-market returns. By holding long-maturity rather than short-maturity bonds, the manager earns an excess return as the bond “rolls down the yield curve” (i.e., approaches maturity and increases in price). As long as the yield curve remains upward sloping and the spot rates continue to be lower than previously implied by their corresponding forward rates, this strategy will add to the return of a bond portfolio.

LOS 23.e

The swap rate curve provides a benchmark measure of interest rates. It is similar to the yield curve except that the rates used represent the interest rates of the fixed-rate leg in an interest rate swap.

Market participants prefer the swap rate curve as a benchmark interest rate curve rather than a government bond yield curve for the following reasons:

1. Swap rates reflect the credit risk of commercial banks rather than that of governments.

2. The swap market is not regulated by any government.
3. The swap curve typically has yield quotes at many maturities.

LOS 23.f

We define swap spread as the additional interest rate paid by the fixed-rate payer of an interest rate swap over the rate of the “on-the-run” government bond of the same maturity.

$$\text{swap spread} = (\text{swap rate}) - (\text{Treasury bond yield})$$

Investors use the swap spread to separate the time value portion of a bond’s yield from the risk premiums for credit and liquidity risk. The higher the swap spread, the higher the compensation for liquidity and credit risk.

For a default-free bond, the swap spread provides an indication of (1) the bond’s liquidity and/or (2) possible mispricing.

LOS 23.g

The Z-spread is the spread that when added to each spot rate on the yield curve makes the present value of a bond’s cash flows equal to the bond’s market price. The Z refers to zero volatility—a reference to the fact that the Z-spread assumes interest rate volatility is zero. Z-spread is not appropriate to use to value bonds with embedded options.

TED spreads

TED is an acronym that combines T-bill and ED (*“ED” is the ticker symbol for the Eurodollar futures contract*).

$$\text{TED spread} = (\text{MRR}) - (\text{T-bill rate})$$

The TED spread is used as an indication of the perceived credit and liquidity risk in the economy. Because it captures the spread between rates paid by large banks and those paid by the U.S. Treasury, the TED spread reflects the default risk in the banking sector.

MRR-OIS spread

The MRR-OIS spread is the amount by which the MRR (which includes some credit risk) exceeds the overnight indexed swap (OIS) rate (which includes only minimal credit risk).

LOS 23.h

There are several traditional theories that attempt to explain the term structure of interest rates:

Unbiased expectations theory—Forward rates are an unbiased predictor of future spot rates. Also known as the *pure expectations theory*.

Local expectations theory—Bond maturity does not influence returns for short holding periods.

Liquidity preference theory—Investors demand a liquidity premium that is positively related to a bond’s maturity.

Segmented markets theory—The shape of the yield curve is the result of the interactions of supply and demand for funds in different market (i.e., maturity) segments.

Preferred habitat theory—Similar to the segmented markets theory, but recognizes that market participants will deviate from their preferred maturity habitat if compensated adequately.

LOS 23.i

We can measure a bond's exposures to the factors driving the yield curve in a number of ways:

1. **Effective duration**—Measures the sensitivity of a bond's price to *parallel* shifts in the benchmark yield curve.
2. **Key rate duration**—Measures bond price sensitivity to a change in a specific spot rate keeping everything else constant.
3. **Sensitivity to parallel, steepness, and curvature movements**—Measures sensitivity to three distinct categories of changes in the shape of the benchmark yield curve.

LOS 23.j

The maturity structure of yield volatilities indicates the level of yield volatilities at different maturities. This term structure thus provides an indication of yield curve risk. The volatility term structure usually indicates that short-term rates (which are linked to uncertainty over monetary policy) are more volatile than long-term rates (which are driven by uncertainty related to the real economy and inflation). Fixed income instruments with embedded options can be especially sensitive to interest rate volatility.

LOS 23.k

Inflation forecasts, GDP growth, and monetary policy affect bond yields. Bond risk premium is the excess return (over the one-year risk-free rate) from investing in longer-term government bonds. Additionally, fiscal policy, maturity structure, and investor demand all affect yields.

In expectation of a rise (fall) in rates, investors will lower (extend) the duration of their bond portfolios. An investor will rotate out of a bullet portfolio and into a barbell portfolio in expectation of a bullish flattening of the yield curve.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 23.1

1. **B** When the yield curve is *upward* sloping, the forward curves will lie above the spot curve. The opposite is true when the yield curve is downward sloping. (LOS 23.a)
2. **B** The forward *pricing* model values forward contracts by using an arbitrage argument that equates buying a zero-coupon bond to entering into a forward contract to buy a zero-coupon bond that matures at the same time:

$$P_{(j+k)} = P_j F_{(j,k)}$$

The forward *rate* model tells us that the forward rate $f(j,k)$ should make investors indifferent between buying a long-maturity zero-coupon bond versus buying a shorter-maturity zero-coupon bond and reinvesting the principal. (LOS 23.a)

Module Quiz 23.2

1. **B** If an investor believes that future spot rates will be lower than the current forward rates, then the investor will perceive an opportunity to purchase bonds at an attractive price, as the market is discounting future cash flows at “too high” a discount rate. The bonds are thus undervalued in the market. (LOS 23.c)
2. **C** Fixed income managers will earn an extra return through rolling down the yield curve if the spot rate curve is upward sloping and remains unchanged over time. (LOS 23.d)

Module Quiz 23.3

1. **C** The swap market is not controlled by governments, which makes swap rates more comparable across different countries. The swap rate is the interest rate for the fixed-rate leg of an interest rate swap. *Wholesale* banks frequently use the swap curve to value their assets and liabilities, while *retail* banks with little exposure to the swap market are more likely to use the government spot curve as their benchmark. (LOS 23.e)

Module Quiz 23.4

1. **C** The swap spread of a default free bond should provide an indication of the bond’s illiquidity—or, alternatively, that the bond is mispriced. Time value is reflected in the government bond yield curve; the swap spread is an additional amount of interest above this benchmark. (LOS 23.f)
2. **C** The Z-spread is the constant spread that must be added to the default-free spot curve to match the valuation of a risky bond to its market price. A higher Z-spread implies a riskier bond. (LOS 23.g)
3. **A** The TED spread (from *T*-bill and *Eurodollar*) is computed as the difference between the three-month MRR and the three-month T-bill rate. The MRR-OIS spread is the difference between MRR and the overnight indexed swap rate (OIS) rates. (LOS 23.g)

Module Quiz 23.5

1. **A** The segmented markets theory (and the preferred habitat theory) propose that borrowers and lenders have strong preferences for particular maturities. The liquidity preference theory argues that there are liquidity premiums that increase with maturity; however, the liquidity preference theory does not preclude the existence of other factors that could lead to an overall downward-sloping yield curve. The segmented markets theory—not the preferred habitat theory—proposes that yields at different maturities are determined independently of each other. (LOS 23.h)

Module Quiz 23.6

1. **A** Effective duration is an inappropriate measure for identifying and managing shaping risk. Shaping risk refers to risk to portfolio value from changes in the shape of the benchmark yield curve. Effective duration can be used to accurately measure the risk associated with parallel yield curve changes but is not appropriate for measuring the risk from other changes in the yield curve. (LOS 23.i)
2. **B** It is believed that short-term volatility reflects uncertainty regarding monetary policy while long-term volatility is most closely associated with uncertainty regarding the real economy and inflation. Short-term rates in the volatility term structure tend to be more volatile than long-term rates. (LOS 23.j)
3. **A** During economic expansions, restrictive monetary policy to control inflation leads to an increase in short-term rates and a bearish flattening of the yield curve. (LOS 23.k)

READING 24

THE ARBITRAGE-FREE VALUATION FRAMEWORK

EXAM FOCUS

This topic review discusses valuation of fixed-income securities using spot rates as well as using the backward induction methodology in a binomial interest rate tree framework. Understand how embedded options impact the suitability of the binomial model or the Monte Carlo simulation method.

MODULE 24.1: BINOMIAL TREES, PART 1

LOS 24.a: Explain what is meant by arbitrage-free valuation of a fixed-income instrument.



Video covering this content is available online.

The arbitrage-free valuation framework is used extensively in the pricing of securities. The basic principle of the “law of one price” in freely functioning markets drives this analytical framework.

Arbitrage-free valuation methods value securities such that no market participant can earn an arbitrage profit in a trade involving that security. An arbitrage transaction involves no initial cash outlay but a positive riskless profit (cash flow) at some point in the future.

There are two types of arbitrage opportunities: **value additivity** (when the value of whole differs from the sum of the values of parts) and **dominance** (when one asset trades at a lower price than another asset with identical characteristics).

If the principle of value additivity does not hold, arbitrage profits can be earned by **stripping** or **reconstitution**. A five-year, 5% Treasury bond should be worth the same as a portfolio of its coupon and principal strips. If the portfolio of strips is trading for less than an intact bond, one can purchase the strips, combine them (reconstituting), and sell them as a bond. Similarly, if the bond is worth less than its component parts, one could purchase the bond, break it into a portfolio of strips (stripping), and sell those components.

EXAMPLE: Arbitrage opportunities

The following information has been collected:

Security	Current Price	Payoff in 1 Year
A	\$99	\$100
B	\$990	\$1,010
C	\$100	\$102
D	\$100	\$103

Securities A and B are identical in every respect other than as noted. Similarly, securities C and D are identical in every other respect.

Demonstrate the exploitation of any arbitrage opportunities.

Answer:

1. Arbitrage due to violation of the value additivity principle:

	Cash Flow	
	t = 0	t = 1
Short 10 units of Security A	+\$990	–\$1,000
Long 1 unit of Security B	–\$990	+\$1,010
Net cash flow	–0–	+\$10

2. Arbitrage due to the occurrence of dominance:

	Cash Flow	
	t = 0	t = 1
Short 1 unit of Security C	+\$100	-\$102
Long 1 unit of Security D	-\$100	+\$103
Net cash flow	-0-	+\$1

LOS 24.b: Calculate the arbitrage-free value of an option-free, fixed-rate coupon bond.

Arbitrage-free valuation of a fixed-rate, option-free bond entails discounting each of the bond's future cash flows (i.e., each coupon payment and the par value at maturity) using the corresponding spot rate.

EXAMPLE: Arbitrage-free valuation

Sam Givens, a fixed income analyst at GBO Bank, has been asked to value a three-year, 3% annual pay, €100 par bond with the same liquidity and risk as the benchmark. What is the value of the bond using the spot rates provided in the following?

€ Benchmark Spot Rate Curve:

Year	Spot Rate
1	3.00%
2	3.25%
3	3.50%

Answer:

$$\text{Bond value} = \frac{3}{(1.03)} + \frac{3}{(1.0325)^2} + \frac{103}{(1.0350)^3} = \text{€}98.63$$

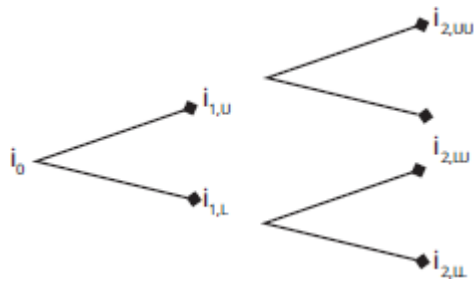
While we can value option-free bonds with a simple spot rate curve, for bonds with embedded options, changes in future rates will affect the probability of the option being exercised and the underlying future cash flows. Thus, a model that allows both rates and the underlying cash flows to vary should be used to value bonds with embedded options. One such model, the binomial interest rate tree framework, is discussed next.

LOS 24.c: Describe a binomial interest rate tree framework.

The **binomial interest rate tree** framework assumes that interest rates have an equal probability of taking one of two possible values in the next period (hence the term binomial).

Over multiple periods, the set of possible interest rate paths that are used to value bonds with a binomial model is called a binomial interest rate tree. The diagram in Figure 24.1 depicts a binomial interest rate tree.

Figure 24.1: Binomial Interest Rate Tree



The starting point i_0 in this tree is the current 1-period spot rate and is called the root of the tree. The future nodes are indicated with the bold dots (◆). A node is a point in time when interest rates can take one of two possible paths, an upper path, U , or a lower path, L . Now, consider the node on the right side of the diagram where the interest rate $i_{2,LU}$ appears. This is the rate that will occur if the initial rate, i_0 , follows the lower path from node 0 to node 1 to become $i_{1,L}$, then follows the upper of the two possible paths to node 2, where it takes on the value $i_{2,LU}$. At the risk of stating the obvious, the upper path from a given node leads to a higher rate than the lower path. In fact,

$$i_{2,LU} = i_{2,LL} e^{2\sigma}$$

where:

$e \approx 2.7183$ (i.e., the base of natural log)

σ = standard deviation of interest rates (i.e., the interest rate volatility used in the model)

Note that an upward move followed by a downward move, or a down-then-up move, produces the same result: $i_{2,LU} = i_{2,UL}$ (the math term for this is lattice).

The interest rates at each node in this interest rate tree are one-period forward rates corresponding to the nodal period. Each forward rate is related to (i.e., is a multiple of) the other forward rates in the same nodal period. Adjacent forward rates (at the same period) are two standard deviations apart. For the first period, there are two forward rates and hence:

$$i_{1,U} = i_{1,L} e^{2\sigma}$$



PROFESSOR'S NOTE

Consistent with the curriculum, I use the term “forward rates” to denote the “future one-period expected spot rates,” even though they are not the same.

Beyond the first nodal period, non-adjacent forward rates are a multiple of the two standard deviations (depending on how separated the forward rates are).

For example,

$$i_{2,UU} = i_{2,LL} e^{4\sigma}$$

The relationship among the set of rates associated with each individual nodal period is a function of the interest rate volatility assumed to generate the tree. Volatility estimates can be based on historical data or can be implied volatility derived from interest rate derivatives.

The binomial interest rate tree framework is a lognormal random walk model with two properties: (1) higher volatility at higher rates and (2) non-negative interest rates.

LOS 24.e: Describe the backward induction valuation methodology and calculate the value of a fixed-income instrument given its cash flow at each node.

Valuing an Option-Free Bond with the Binomial Model

Backward induction refers to the process of valuing a bond using a binomial interest rate tree. The term “backward” is used because in order to determine the value of a bond today at Node 0, you need to know the values that the bond can take at the Year 1 node. But to determine the values of the bond at the Year 1 nodes, you need to know the possible values of the bond at the Year 2 nodes. Thus, for a bond that has N compounding periods, the current value of the bond is determined by computing the bond’s possible values at Period N and working backwards to Node 0.

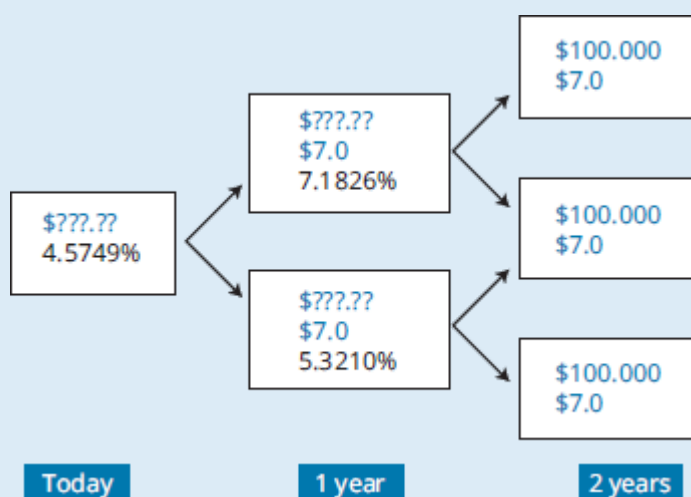
Because the probabilities of an up move and a down move are both 50%, the value of a bond at a given node in a binomial tree is the average of the present values of the two possible values from the next period. The appropriate discount rate is the forward rate associated with the node.

The following example should make this all clear.

EXAMPLE: Valuing an option-free bond with the binomial model

A 7% annual coupon bond has two years to maturity. The interest rate tree is shown in the following figure. Fill in the tree and calculate the value of the bond today.

Valuing a 2-Year, 7.0% Coupon, Option-Free Bond



Answer:

Consider the value of the bond at the *upper* node for Period 1 ($V_{1,U}$):

$$V_{1,U} = \frac{1}{2} \times \left[\frac{\$100 + \$7}{1.071826} + \frac{\$100 + \$7}{1.071826} \right] = \$99.830$$

Similarly, the value of the bond at the *lower* node for Period 1 ($V_{1,L}$) is:

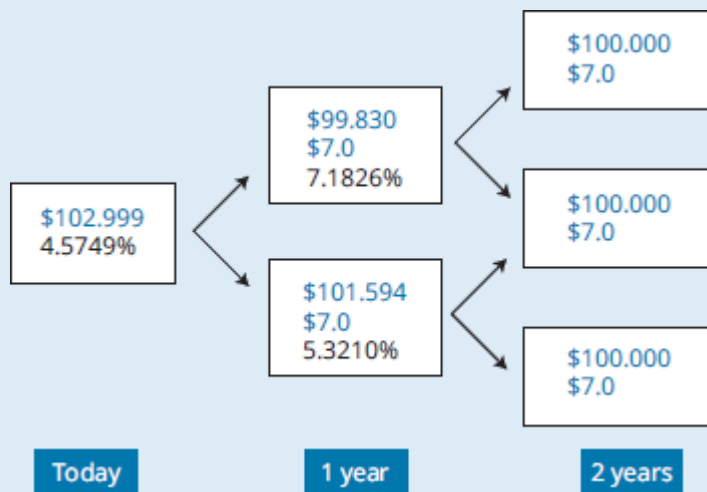
$$V_{1,L} = \frac{1}{2} \times \left[\frac{\$100 + \$7}{1.053210} + \frac{\$100 + \$7}{1.053210} \right] = \$101.594$$

Now calculate V_0 , the current value of the bond at Node 0.

$$V_0 = \frac{1}{2} \times \left[\frac{\$99.830 + \$7}{1.045749} + \frac{\$101.594 + \$7}{1.045749} \right] = \$102.999$$

The completed binomial tree is shown as follows:

Valuing a 2-Year, 7.0% Coupon, Option-Free Bond



MODULE QUIZ 24.1

Use the following information to answer Questions 1 through 5.

Dan Green, CFA, is currently working for FIData, a company specializing in the provision of price data for fixed-income instruments.

Green heads a team that inputs raw data into an instructional area on FIData's website. FIData uses these pages to provide hypothetical bond information along with a description of how to read and interpret the information.

One of FIData's customers has questioned whether some of the data used to demonstrate pricing concepts is correct. The customer's email stated, "The prices of the three bonds used are not consistent with each other and hence may not be accurate. If they were, I would be able to make a significant arbitrage profit (i.e., I could secure the current risk-free rate of return with zero net investment)." Green thinks the customer is misinformed regarding arbitrage gains, but

wants to check the data anyway.

The relevant data for three hypothetical risk-free bonds is shown in Figure 1 given that the benchmark yield curve is flat at 1.50%. (FFPQ is an annual-pay bond.)

Figure 1: Bond Pricing Data

Bond Issue	Maturity	Coupon	Par (\$)	Price (\$)
FFPQ	2 years	10%	\$1,000	1,170.12
DALO	1 year	0%	\$1,000	985.22
NKDS	2 years	0%	\$11,000	10,677.28

In another area of the company's instructional website, FIData has an explanation of the binomial interest rate tree framework that its analysts use in their valuation process. All of FIData's models populate a binomial interest rate tree assuming interest rates follow a lognormal random walk.

The web page makes two statements regarding the assumptions that underpin the construction and population of such trees:

Assumption 1: The lognormal model ensures that interest rates are non-negative.

Assumption 2: The lognormal model ensures a constant volatility of interest rates at all levels of interest rates.

An example of a standard tree used by FIData is given in Figure 2.

Figure 2: Binomial Interest Rate Tree

Year 0	Year 1
4.5749%	7.1826%
	5.3210%

FIData's website uses rates in Figure 2 to value a two-year, 5% annual-pay coupon bond with a par value of \$1,000 using the backward induction method.

1. Green is correct in stating that the customer who sent the email regarding arbitrage gains is misinformed because:
 - A. an arbitrage gain always requires a net investment, although this may be small compared to the potential gains.
 - B. an arbitrage gain is not constrained by the risk-free rate.
 - C. arbitrage gains require a net investment in proportion to returns.
2. Given the three bonds in Figure 1, it is possible to make an arbitrage gain by:
 - A. selling 1 FFPQ bond and simultaneously purchasing 10 DALO and 10 NKDS bonds.
 - B. selling 10 FFPQ bonds and simultaneously purchasing 1 DALO and 1 NKDS bond.
 - C. selling 1 DALO bond and 1 NKDS bond and simultaneously purchasing 10 FFPQ bonds.
3. Which of the following statements regarding the FFPQ bond in Figure 1 is *most likely*?
 - A. It is priced above its no arbitrage price.
 - B. It is priced at its no arbitrage price.
 - C. It is priced below its no arbitrage price.
4. Which of the assumptions regarding the construction of FIData's binomial interest rate trees is *most accurate*?
 - A. Assumption 1 only.
 - B. Assumption 2 only.
 - C. Neither assumption is correct.
5. Using the backward induction method, the value of the 5% annual-pay bond using the interest rate tree given in Figure 2 is *closest* to:

- A. \$900.
- B. \$945.
- C. \$993.



Video covering
this content is
available online.

MODULE 24.2: BINOMIAL TREES, PART 2

LOS 24.d: Describe the process of calibrating a binomial interest rate tree to match a specific term structure.

The construction of a binomial interest rate tree is a tedious process. In practice, the interest rate tree is usually generated using specialized computer software. The underlying process conforms to three rules:

1. The interest rate tree should generate **arbitrage-free values** for the benchmark security. This means that the value of bonds produced by the interest rate tree must be equal to their market price, which excludes arbitrage opportunities. This requirement is very important because without it, the model will not properly price more complex callable and putable securities, which is the intended purpose of the model.
2. As stated earlier, adjacent forward rates (for the same period) are two standard deviations apart (calculated as $e^{2\sigma}$). Hence, knowing one of the forward rates (out of many) for a particular nodal period allows us to compute the other forward rates for that period in the tree.
3. The middle forward rate (or mid-point in case of even number of rates) in a period is approximately equal to the implied (from the benchmark spot rate curve) one-period forward rate for that period.

EXAMPLE: Binomial interest rate tree

Xi Nguyen, CFA, has collected the following information on the par rate curve, spot rates, and forward rates. Nguyen had asked a colleague, Alok Nath, to generate a binomial interest rate tree consistent with this data and an assumed volatility of 20%. Nath completed a partial interest rate tree shown as follows:

Maturity	Par Rate	Spot Rate
1	3%	3.000%
2	4%	4.020%
3	5%	5.069%

Binomial Tree With $\sigma = 20\%$ (One-Year Forward Rates)

Time 0	Time 1	Time 2
3%	5.7883%	B
	A	C
		D

1. Calculate the forward rate indicated by A.
2. Estimate the forward rate indicated by C.
3. Estimate forward rates B and D.

Answer:

1. Forward rate $i_{1,L}$ is indicated by A and is related to forward rate $i_{1,U}$ given as 5.7883%.

$$i_{1,L} = i_{1,U} e^{-2\sigma} = (0.057883)e^{-(2 \times 0.20)} = 0.0388 \text{ or } 3.88\%$$

2. Forward rate C is the middle rate for Period 3 and hence the best estimate for that rate is the one-year forward rate in two years $f(2,1)$.

Using the spot rates, we can bootstrap the forward rate:

$$(1 + S_3)^3 = (1 + S_2)^2(1 + f(2,1)) \rightarrow (1.05069)^3 = (1.0402)^2(1 + f(2,1)) \rightarrow f(2,1) = 7.199\%$$

1. The forward rates B and D are related to C as follows (note that $C = i_{2,LU}$):

$$D = i_{2,LL} = i_{2,LU} e^{-2\sigma} = (0.07199) e^{-0.40} = 0.0483 \text{ or } 4.83\%$$

$$B = i_{2,UU} = i_{2,LU} e^{+2\sigma} = (0.07199) e^{+0.40} = 0.1074 \text{ or } 10.74\%$$



PROFESSOR'S NOTE

To calculate the value of $e^{-0.40}$ on your TI BA II PLUS calculator, use the following keystrokes: 0.4 [+|-] [2ND] [LN]

LOS 24.f: Compare pricing using the zero-coupon yield curve with pricing using an arbitrage-free binomial lattice.

We have already discussed arbitrage-free valuation of benchmark option-free bonds using the **zero-coupon yield curve** (also known as the spot rate curve). Each known future cash flow is discounted at the underlying spot rate (also known as the zero-coupon yield).

EXAMPLE: Valuation of option-free bond

Samuel Favre is interested in valuing a three-year, 3% annual-pay Treasury bond. He has compiled the following information on Treasury spot rates:

Treasury Spot Rate Curve

Maturity	Spot Rate
1	3.000%
2	4.020%
3	5.069%

Compute the value of the \$100 face value option-free bond.

Answer:

$$\text{value of the bond} = \frac{3}{1.03} + \frac{3}{(1.0402)^2} + \frac{103}{(1.05069)^3} = \$94.485$$

For bonds with embedded options, the future cash flows are uncertain as they depend on whether the embedded option will be in the money (and hence exercised). Because the value of the option depends on uncertain future interest rates, the underlying cash flows are also dependent on the same future interest rates. Hence, to value bonds with embedded options, we have to allow for rates to fluctuate. One way to accomplish this is to use the binomial interest rate tree.

**PROFESSOR'S NOTE**

The binomial tree approach can value either option-free bonds or bonds with embedded options. Valuation of bonds with embedded options is covered in the next topic review.

EXAMPLE: Valuation of an option-free bond using binomial tree

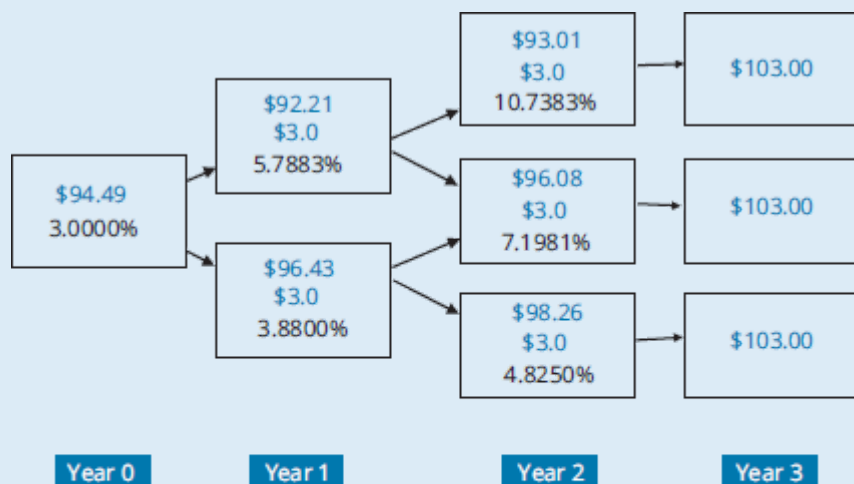
Samuel Favre is interested in valuing the same three-year, 3% annual-pay Treasury bond. The spot rate curve is as before, but this time Favre wants to use a binomial interest rate tree with the following rates:

One-Period Forward Rate in Year

0	1	2
3%	5.7883%	10.7383%
	3.8800%	7.1981%
		4.8250%

Compute the value of the \$100 par option-free bond.

Answer:



$$V_{2,uu} = \frac{103}{(1.107383)} = \$93.01$$

$$V_{2,ul} = \frac{103}{(1.071981)} = \$96.08$$

$$V_{2,ll} = \frac{103}{(1.048250)} = \$98.26$$

$$V_{1,u} = \frac{1}{2} \times \left[\frac{93.01 + 3}{1.057883} + \frac{96.08 + 3}{1.057883} \right] = \$92.21$$

$$V_{1,l} = \frac{1}{2} \times \left[\frac{96.08 + 3}{1.038800} + \frac{98.26 + 3}{1.038800} \right] = \$96.43$$

$$V_0 = \frac{1}{2} \times \left[\frac{92.21 + 3}{1.03} + \frac{96.43 + 3}{1.03} \right] = \$94.485$$

Note that the underlying interest rate tree in this example was calibrated to generate arbitrage-free values consistent with the benchmark spot rate curve and hence produced the same value for the option-free bond as in the earlier example.

LOS 24.g: Describe pathwise valuation in a binomial interest rate framework and calculate the value of a fixed-income instrument given its cash flows along each path.

Another mathematically identical approach to valuation using the backward induction method in a binomial tree is the **pathwise valuation** approach. For a binomial interest rate tree with n periods, there will be $2^{(n-1)}$ unique paths. For example, for a three-period tree, there will be 2^2 or four paths comprising one known spot rate and varying combinations of two unknown forward rates. If we use the label of S for the one-period spot rate and U and L for upper and lower outcomes, respectively, for forward rates in our binomial framework, the four paths would be SUU, SUL, SLU, and SLL.

EXAMPLE: Valuation of option-free bond using pathwise valuation

Samuel Favre wants to value the same three-year, 3% annual-pay Treasury bond. The interest rate tree (shown next) is the same as before but this time, Favre wants to use a pathwise valuation approach.

One-Period Forward Rate in Year

0	1	2
3%	5.7883%	10.7383%
	3.8800%	7.1981%
		4.8250%

Compute the value of the \$100 par option-free bond.

Answer:

For a three-year bond, there are four potential interest rate paths. The value of the bond for each path is computed as the sum of the present values of each cash flow discounted at its respective path-specified rate. Pathwise valuation discounts cash flows one year at a time using one-year forward rates (similar to backward induction) rather than spot rates.

Path	Year 1	Year 2	Year 3	Value
1	3%	5.7883%	10.7383%	\$91.03
2	3%	5.7883%	7.1981%	\$93.85
3	3%	3.8800%	7.1981%	\$95.52
4	3%	3.8800%	4.8250%	\$97.55
			Average	\$94.49

For example, the value of the bond in Path 1 is computed as:

$$\begin{aligned} \text{value}_1 &= \frac{3}{(1.03)} + \frac{3}{(1.03)(1.057883)} + \frac{103}{(1.03)(1.057883)(1.107383)} \\ &= \$91.03 \end{aligned}$$

LOS 24.h: Describe a Monte Carlo forward-rate simulation and its application.

Path Dependency

Prepayments on underlying residential mortgages affect the cash flows of a mortgage-backed security. Prepayment risk is similar to call risk in a callable bond. However, unlike call risk, prepayment risk is affected not only by the level of interest rate at a particular point in time, but also by the path rates took to get there.

Consider a mortgage pool that was formed when rates were 6%, then interest rates dropped to 4%, rose to 6%, and then dropped again to 4%. Many homeowners will have refinanced when interest rates dipped the first time. On the second occurrence of 4% interest rates, most homeowners in the pool who were able to refinance would have already taken advantage of the opportunity, leading to lower prepayments than would be observed had 4% interest rates not occurred previously.

An important assumption of the binomial valuation process is that the value of the cash flows at a given point in time is independent of the path that interest rates followed up to that point. In other words, cash flows are not **path dependent**; cash flows at any node do not depend on the path rates took to get to that node. Because of path dependency of cash flows of mortgage-backed securities, the binomial tree backward induction process cannot be used to value such securities. We instead use the Monte Carlo simulation method to value mortgage-backed securities.

A **Monte Carlo forward-rate simulation** involves randomly generating a large number of interest rate paths, using a model that incorporates a volatility assumption and an assumed probability distribution. A key feature of the Monte Carlo method is that the underlying cash flows can be path dependent.

As with pathwise valuation discussed earlier, the value of the bond is the average of values from the various paths. The simulated paths should be calibrated so benchmark interest rate paths value benchmark securities at their market price (i.e., arbitrage-free valuation). The calibration process entails adding (subtracting) a constant to all rates when the value obtained from the simulated paths is too high (too low) relative to market prices. This calibration process results in a **drift adjusted** model.

A Monte Carlo simulation may impose upper and lower bounds on interest rates as part of the model generating the simulated paths. These bounds are based on the notion of mean reversion; rates tend to rise when they are too low and fall when they are too high.

EXAMPLE: Valuation of option-free bond using Monte Carlo simulation

Samuel Favre is interested in valuing the same three-year, 3% annual-pay Treasury bond as discussed before. Favre wants to use Monte Carlo simulation and has generated the

following rate paths:

Monte Carlo Simulation (Drift-Adjusted)

Path	Year 1	Year 2	Year 3
1	3%	5.32%	10.59%
2	3%	5.11%	10.33%
3	3%	4.79%	9.89%
4	3%	4.56%	9.10%
5	3%	4.11%	8.22%
6	3%	3.79%	6.54%
7	3%	3.58%	5.11%
8	3%	3.62%	4.11%

Answer:

Valuation using the simulated interest rate paths is shown as follows:

Path	Year 1	Year 2	Year 3	Value
1	3%	5.32%	10.59%	\$91.53
2	3%	5.11%	10.33%	\$91.91
3	3%	4.79%	9.89%	\$92.53
4	3%	4.56%	9.10%	\$93.36
5	3%	4.11%	8.22%	\$94.47
6	3%	3.79%	6.54%	\$96.15
7	3%	3.58%	5.11%	\$97.57
8	3%	3.62%	4.11%	\$98.42
Average				\$94.49

Where, for example:

$$\text{value}_{\text{path2}} = \frac{3}{(1.03)} + \frac{3}{(1.03)(1.0511)} + \frac{103}{(1.03)(1.0511)(1.1033)} = 91.91$$

Note that the interest rates in the example were calibrated to ensure that the valuation obtained was consistent with market prices (i.e., arbitrage-free) and hence the value obtained is the same as before.

MODULE 24.3: INTEREST RATE MODELS

LOS 24.i: Describe term structure models and how they are used.

Term structure models attempt to capture the statistical properties of interest rate movements and provide us with quantitatively precise descriptions of how interest rates will change.



Video covering this content is available online.

Equilibrium Term Structure Models

Equilibrium term structure models attempt to describe changes in the term structure through the use of fundamental economic variables that drive interest rates. While equilibrium term structure models can rely on multiple factors, the two famous models discussed in the curriculum, the Cox-Ingersoll-Ross (CIR) model and the Vasicek model, are both single-factor models. The single factor in the CIR and Vasicek models is the short-term interest rate.

The Cox-Ingersoll-Ross Model

The **Cox-Ingersoll-Ross (CIR) model** is based on the idea that interest rate movements are driven by individuals choosing between consumption today versus investing and consuming at a later time.

Mathematically, the CIR model consists of two components: the first part of this expression is a drift term, while the second part is the random component:

$$dr_t = a(b - r) dt + \sigma \sqrt{r} dz$$

where:

dr_t = change in the short-term interest rate

a = speed of mean reversion parameter (a high a means fast mean reversion)

b = long-run value of the short-term interest rate

r = the short-term interest rate

t = time

dt = a small increase in time

σ = volatility

dz = a small random walk movement

The $a(b - r)dt$ term forces the interest rate to mean-revert toward its long-run value (b) at a speed determined by the mean reversion parameter (a).

Under the CIR model, volatility increases with the interest rate, as can be seen in the $\sigma \sqrt{r} dz$ term. In other words, at high interest rates the amount of period-over-period fluctuation in rates is also high.

The Vasicek Model

Like the CIR model, the **Vasicek model** suggests that interest rates are mean reverting to some long-run value.

The Vasicek model is expressed as:

$$dr_t = a(b - r)dt + \sigma dz$$

The difference from the CIR model that you will notice is that no interest rate (r) term appears in the second term σdz , meaning that volatility in this model does not increase as the level of interest rates increases (i.e., volatility is constant).

The main disadvantage of the Vasicek model is that the model does not force interest rates to be nonnegative.

Arbitrage-Free Models

Arbitrage-free models of the term structure of interest rates begin with the assumption that bonds trading in the market are correctly priced, and the model is calibrated to value such bonds consistent with their market price (hence the “arbitrage-free” label). These models do not try to justify the current yield curve; rather, they take this curve as given.

The ability to calibrate arbitrage-free models to match current market prices is one advantage of arbitrage-free models over the equilibrium models.

The Ho-Lee Model

The **Ho-Lee model** takes the following form:

$$dr_t = \theta_t dt + \sigma dz_t$$

where:

θ_t = a time-dependent drift term

Derived using the relative pricing concepts of the Black-Scholes model, this model assumes that changes in the yield curve are consistent with a no-arbitrage condition.

The Ho-Lee model is calibrated by using market prices to find the time-dependent drift term θ_t that generates the current term structure. The Ho-Lee model can then be used to price zero-coupon bonds and to determine the spot curve. The model assumes constant volatility and produces a symmetrical (normal) distribution of future rates.

The Kalotay-Williams-Fabozzi (KWF) Model

The **Kalotay-Williams-Fabozzi (KWF) model** does not assume mean reversion and, like the Ho-Lee model, assumes a constant volatility. However, the KWF model assumes that the short-term rate is lognormally distributed. The model takes the following form:

$$d\ln(r_t) = \theta_t dt + \sigma dz_t$$

Note that the right-hand side of the equation is the same as that of the Ho-Lee model. The only difference is that the KWF model assumes lognormal distribution of rates while the Ho-Lee model assumed normal distribution.

Other Models

Gauss+ is a multifactor model that incorporates short-, medium-, and long-term rates, where the long-term rate is designed to be mean reverting and depends on macroeconomic variables. Medium-term rates revert to the long-term rate, while the short-term rate is devoid of a random component—consistent with the role of the central bank controlling the short-term rate. This results in a hump-shaped volatility curve across tenors, with the volatility of the medium term rates being the highest.



MODULE QUIZ 24.2, 24.3

Use the following information to answer Questions 1 through 6.

Farah Dane, CFA, works for Geodesic Investing, a small hedge fund that offers investment services for a handful of clients known personally by the owner, Mike DeGrekker. The fund makes few trades, preferring to wait for what it perceives to be arbitrage opportunities before investing. Last year, the fund managed a return of more than 45%, thanks largely to a single

transaction on which the company made a profit of \$9.4 million.

The transaction, which DeGrekker described as a “valuation farming exercise” involved simultaneously purchasing a government Treasury and selling the corresponding strips for a higher price than the cost of the Treasury.

Dane is currently using a binomial lattice and a pathwise method to value fixed-income bonds in order to identify potential trading opportunities. She has used the binomial lattice shown in Figure 1 to value a three-year, annual pay, 4% coupon risk-free government bond with a par value of \$1,000. Her pathwise valuation is also shown.

Figure 1: Binomial Lattice

	Year 1	Year 2	Year 3	Value
Path 1	3%	5.7883%	10.7383%	\$937.45
Path 2	3%	5.7883%	7.1981%	\$965.92
Path 3	3%	5.7883%	4.8250%	\$986.07
Path 4	3%	3.8800%	7.1981%	\$982.95
Path 5	3%	3.8800%	4.8250%	\$1,003.48
			Average	\$975.17

Dane is not satisfied with this method of valuation and has put together a report for DeGrekker on the use of the Monte Carlo method, which she feels will lead to more accurate valuations. She quotes the following advantages of using Monte Carlo method:

Advantage 1: The Monte Carlo method will estimate the value of the bond using multiple interest rate paths and hence there is no need to make an assumption about volatility of rates.

Advantage 2: The method could be applied to get more accurate valuations for securities that are interest rate path dependent.

DeGrekker is resistant to the idea as he is concerned about the amount of computing time the model may require. He accepts, however, that the idea of using many paths is attractive. He concedes that, “increasing the number of paths used in the model increases the statistical accuracy of the estimated value and produces a value closer to the true fundamental value of the security.”

1. Which of the following statements regarding the valuation of an option-free bond using an arbitrage-free binomial lattice is *most accurate*?
 - A. If the binomial lattice is correctly calibrated, it should give the same value for an option-free bond as using the par curve used to calibrate the tree.
 - B. The binomial lattice will only produce the same value for an option-free bond as the par curve that was used to calibrate it if the bond is priced at par.
 - C. The binomial lattice will only produce the same value for an option-free bond as the par curve that was used to calibrate it if the yield curve is flat.
2. Which of the following statements *most accurately* describes the “valuation farming exercise” undertaken by DeGrekker?
 - A. DeGrekker used the process of stripping and the law of one price to make an arbitrage gain.
 - B. DeGrekker used the process of reconstitution and the principle of no arbitrage to make a risk-free gain.
 - C. DeGrekker’s profit is not an arbitrage profit as the securities involved are risk free.
3. Which of the following is *most accurate* regarding the value Dane would obtain using the backward induction method as opposed to the pathwise valuation method for the bond in Figure 1?

- A. Both methods would produce the same value.
 - B. The pathwise valuation method will give lower values when interest rates are rising because the backward induction method places a higher weighting on earlier cash flows.
 - C. The backward induction method will give a different value compared to the pathwise method when the volatility of interest rates is high as the pathwise method uses equal weights.
4. Dane's pathwise valuation is:
- A. correct.
 - B. incorrect, as the correct value is lower than \$975.17.
 - C. incorrect, as the correct value is higher than \$975.17.
5. Which of the advantages of the Monte Carlo method stated by Dane is *most accurate*?
- A. Advantage 1 only.
 - B. Advantage 2 only.
 - C. Neither advantage is correct.
6. DeGrekker's comment on increasing the number of paths is *most likely*:
- A. correct.
 - B. incorrect in asserting that a larger number of paths will produce an estimate that is statistically more accurate.
 - C. incorrect in asserting that a larger number of paths will produce a value closer to the true fundamental value.
7. When calibrating a binomial interest rate tree to match a specific term structure, which of the following statements is *least accurate*?
- A. Interest rates in the tree should produce an arbitrage-free valuation for benchmark securities.
 - B. Adjacent spot rates at each node of the tree are related by the multiplier $e^{2\sigma}$.
 - C. The middle forward rate in a period is approximately equal to the implied (from the benchmark spot rate curve) one-period forward rate for that period.
8. The modern term structure model that is *most likely* to precisely generate the current term structure is the:
- A. Cox-Ingersoll-Ross model.
 - B. Vasicek model.
 - C. Ho-Lee model.

KEY CONCEPTS

LOS 24.a

Arbitrage-free valuation leads to a security value such that no market participant can earn an arbitrage profit in a trade involving that security. In other words, the valuation is consistent with the value additivity principle and without dominance of any security relative to others in the market.

LOS 24.b

Arbitrage-free valuation of fixed-rate, option-free bonds entails discounting each of the bond's future cash flows at its corresponding spot rate.

LOS 24.c

The binomial interest rate tree framework is a lognormal model with two equally likely outcomes for one-period forward rates at each node. A volatility assumption drives the spread of the nodes in the tree.

LOS 24.d

A binomial interest rate tree is calibrated such that (1) the values of benchmark bonds using the tree are equal to the bonds' market prices, (2) adjacent forward rates at any nodal period are two standard deviations apart and (3) the midpoint for each nodal period is approximately equal to the implied one-period forward rate for that period.

LOS 24.e

Backward induction is the process of valuing a bond using a binomial interest rate tree. The term backward is used because in order to determine the value of a bond at Node 0, we need to know the values that the bond can take on at nodal period 1, and so on.

LOS 24.f

Valuation of bonds using a zero-coupon yield curve (also known as the spot rate curve) is suitable for option-free bonds. However, for bonds with embedded options where the value of the option varies with outcome of unknown forward rates, a model that allows for variability of forward rates is necessary. One such model is the binomial interest rate tree framework.

LOS 24.g

In the pathwise valuation approach, the value of the bond is simply the average of the values of the bond at each path. For a n -period binomial tree, there are $2^{(n-1)}$ possible paths.

LOS 24.h

The Monte Carlo simulation method uses pathwise valuation and a large number of randomly generated simulated paths. Mortgage-backed securities have path-dependent cash flows on account of the embedded prepayment option. The Monte Carlo simulation method should be used for valuing MBS as the binomial tree backward induction process is inappropriate for securities with path-dependent cash flows.

LOS 24.i

Two major classes of term structure models are as follows:

1. Equilibrium term structure models—Attempt to model the term structure using fundamental economic variables that are thought to determine interest rates.

a. Cox-Ingersoll-Ross model: $dr_t = a(b - r)dt + \sigma\sqrt{r}dz$

Assumes the economy has a natural long-run interest rate (b) that the short-term rate (r) converges to at a speed of (a). Interest rate volatility varies with r and is not constant. Produces non negative rates only.

b. Vasicek model: $dr_t = a(b - r)dt + \sigma dz$

Similar to the CIR model, but assumes that the interest rate volatility level is constant and independent of the level of short-term interest rates. Can produce negative rates.

2. Arbitrage-free models—Begin with observed market prices and the assumption that securities are correctly priced.

- a. Ho-Lee model: $dr_t = \theta_t dt + \sigma dz_t$

Calibrated by using market prices to find the time-dependent drift term θ_t that generates the current term structure. Assumes that short-term rates are normally distributed with a constant volatility. Produces a normal distribution of rates and rates can be negative.

- b. Kalotay-Williams-Fabozzi (KWF) model: $d\ln(r_t) = \theta_t dt + \sigma dz_t$

Similar to the Ho-Lee model, but assumes that short rate has a lognormal distribution.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 24.1

1. **B** An arbitrage gain is a risk-free profit and hence requires no net investment. The returns therefore are not simply the risk-free rate. As there is no initial investment, the gains cannot be measures as percentage of initial cost. (LOS 24.a)
2. **B** An up-front arbitrage profit of \$38.70 can be earned by selling 10 FFPQ bonds short and purchasing 1 DALO and 1 NKDS bonds as shown here.

Position	Initial Cash Flow	Year 1 Cash Flow	Year 2 Cash Flow
Short 10 FFPQ	\$ 11,701.20	\$(1,000.00)	\$(11,000.00)
Long 1 DALO	\$ (985.22)	\$ 1,000.00	\$ -
Long 1 NKDS	\$(10,667.28)	\$ -	\$ 11,000.00
Net	\$38.70	0	0

(LOS 24.a)

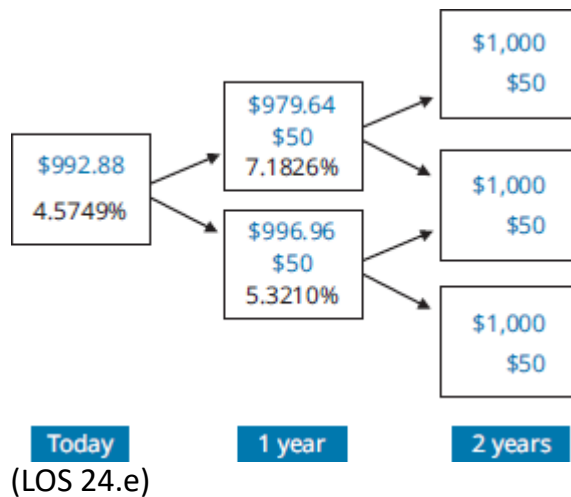
3. **A** FFPQ is overpriced. Based on the 1.5% benchmark yield, the other two bonds are correctly priced.

Arbitrage-free price = (PV Year 1 Cash Flow) + (PV Year 2 Cash Flow)

	PV Year 1 Cash Flow	PV Year 2 Cash Flow	Arbitrage-Free Price
FFPQ	98.52	1,067.73	1,166.25
DALO	985.22	0	985.22
NKDS	0	10,677.28	10,677.28

(LOS 24.a)

4. **A** A lognormal random walk will ensure non-negativity but will lead to higher volatility at higher interest rates. (LOS 24.c)
5. **C** The value of the 5%, two-year annual pay \$1000 par bond is \$992.88.



Module Quiz 24.2, 24.3

1. **A** A correctly calibrated tree will value the bond at the same price as the par and spot curves used to derive it. (Module 24.2, LOS 24.d)
2. **A** DeGrekker purchased the bonds and stripped them into constituent parts before selling them. The strategy involved no initial net investment and yet results in an arbitrage profit. (Module 24.1, LOS 24.a)
3. **A** The two methods are identical and will always give the same result. (Module 24.2, LOS 24.g)
4. **B** This is a tricky question. There are only four possible paths that Dane should have used. The possible paths are UU (Path 1), UD (Path 2), DU (Path 4), and DD (Path 5). Path 3 isn't a valid path.

Path	Year 1	Year 2	Year 3
Path 1	3%	5.7883%	10.7383%
Path 2	3%	5.7883%	7.1981%
Path 4	3%	3.8800%	7.1981%
Path 5	3%	3.8800%	4.8250%

So the value should be the average of the values for Paths 1, 2, 4, and 5. The correct average value is \$972.45. (Module 24.2, LOS 24.g)

5. **B** Monte Carlo simulation also requires an assumed level of volatility as an input. (Module 24.2, LOS 24.h)
6. **C** The larger the number of paths, the more accurate the value in a statistical sense. However, whether the value is closer to the true fundamental value depends on the accuracy of the model inputs. (Module 24.2, LOS 24.h)
7. **B** The stated multiplier is correct but it is important to note that the rates given at each node of the tree are forward rates not spot rates. (Module 24.2, LOS 24.d)
8. **C** The Ho-Lee model is calibrated by using market prices to find the time-dependent drift term, θ_t that generates the current term structure. (One of the drawbacks of the Vasicek and Cox-Ingersoll-Ross models is that the model prices generated by these

models generally do not coincide with observed market prices.) (Module 24.3, LOS 24.i)

READING 25

VALUATION AND ANALYSIS OF BONDS WITH EMBEDDED OPTIONS

EXAM FOCUS

This topic review extends the arbitrage-free valuation framework to valuation of bonds with embedded options. Understand the risk/return dynamics of embedded options, including their impact on a bond's duration and convexity. You should also know the adjustment required for the valuation of credit-risky bonds, including the process to estimate an OAS. Finally, understand the terminology and risk/return characteristics of convertibles.

MODULE 25.1: TYPES OF EMBEDDED OPTIONS



Video covering this content is available online.

LOS 25.a: Describe fixed-income securities with embedded options.

Embedded options in a bond allow an issuer to (1) manage interest rate risk and/or (2) issue the bonds at an attractive coupon rate. The embedded options can be a simple call or put option, or more complex options such as provisions for a sinking fund or an estate put.

Simple Options

Callable bonds give the *issuer* the option to call back the bond; the *investor* is *short* the call option. Most callable bonds have a **call protection period** during which the bond cannot be called. The call option can be a **European-style** option (whereby the option can only be exercised on a single day immediately after the protection period), an **American-style** option (whereby the option can be exercised at any time *after* the protection period), or even a **Bermudan-style** option (whereby the option can be exercised at fixed dates after the protection period).

Puttable bonds allow the *investor* to put (sell) the bond back to the issuer prior to maturity. The *investor* is *long* the underlying put option. A related bond is an **extendible bond**, which allows the investor to extend the maturity of the bond. An extendible bond can be evaluated as a puttable bond with longer maturity (i.e., the maturity if the bond is extended). A two-year, 3% bond extendible for an additional year at the same coupon rate would be valued the same as an otherwise identical three-year puttable (European style) bond with a protection period of two years.

Complex Options

More complex options include:

- An **estate put** which includes a provision that allows the heirs of an investor to put the bond back to the issuer upon the death of the investor. The value of this contingent put option is inversely related to the investor's life expectancy; the shorter the life expectancy, the higher the value.
- **Sinking fund bonds** (sinkers) which require the issuer to set aside funds periodically to retire the bond (a sinking fund). This provision reduces the credit risk of the bond. Sinkers typically have several related *issuer* options (e.g., call provisions, acceleration provisions, and delivery options).

LOS 25.b: Explain the relationships between the values of a callable or putable bond, the underlying option-free (straight) bond, and the embedded option.

In essence, the holder of a callable bond owns an option-free (straight) bond and is also short a call option written on the bond. The value of the embedded call option (V_{call}) is, therefore, simply the difference between the value of a straight (V_{straight}) bond and the value of the comparable callable bond (V_{callable}):

$$V_{\text{call}} = V_{\text{straight}} - V_{\text{callable}}$$

Conversely, investors are willing to pay a premium for a putable bond since its holder effectively owns an option-free bond plus a put option. The value of a putable bond can be expressed as:

$$V_{\text{putable}} = V_{\text{straight}} + V_{\text{put}}$$

Rearranging, the value of the embedded put option can be stated as:

$$V_{\text{put}} = V_{\text{putable}} - V_{\text{straight}}$$



Video covering
this content is
available online.

MODULE 25.2: VALUING BONDS WITH EMBEDDED OPTIONS, PART 1

LOS 25.c: Describe how the arbitrage-free framework can be used to value a bond with embedded options.

LOS 25.f: Calculate the value of a callable or puttable bond from an interest rate tree.

The basic process for valuing a callable (or puttable) bond is similar to the process for valuing a straight bond. However, instead of using spot rates, one-period forward rates are used in a binomial tree framework. This change in methodology computes the value of the bond at different points in time; these checks are necessary to determine whether the embedded option is in-the-money (and exercised).

When valuing a callable bond, the value at any node where the bond is callable must be either the price at which the issuer will call the bond (i.e., the call price) or the computed value if the bond is not called, *whichever is lower*. This is known as the call rule. Similarly, for a puttable bond, the value used at any node corresponding to a put date must be either the price at which the investor will put the bond (i.e., the put price) or the computed value if the bond is not put, *whichever is higher*. This is known as the put rule.



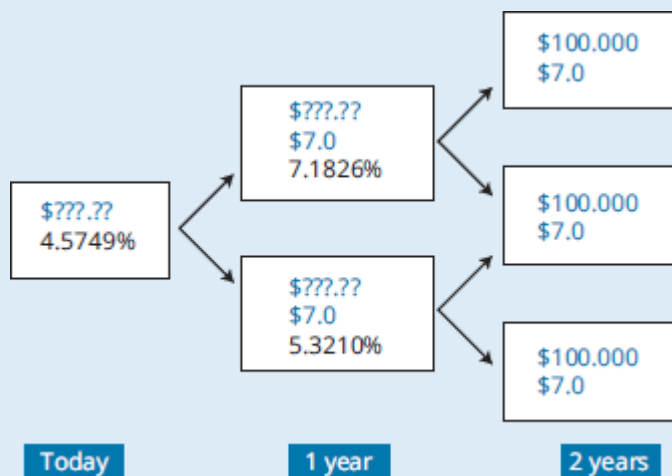
PROFESSOR'S NOTE

Call date and put date in this context vary depending on whether the option is European-, American-, or Bermudan-style.

EXAMPLE: Valuation of call and put options

Consider a two-year, 7% annual-pay, \$100 par bond callable in one year at \$100. Also consider a two-year, 7% annual-pay, \$100 par bond puttable in one year at \$100.

The interest rate tree at 15% assumed volatility is as follows:



Value the embedded call and put options.

Answer:

Value of the straight (option-free) bond:

Consider the value of the bond at the *upper* node for Period 1, $V_{1,U}$:

$$V_{1,U} = \frac{1}{2} \times \left[\frac{\$100 + \$7}{1.071826} + \frac{\$100 + \$7}{1.071826} \right] = \$99.830$$

Similarly, the value of the bond at the *lower* node for Period 1, $V_{1,L}$, is:

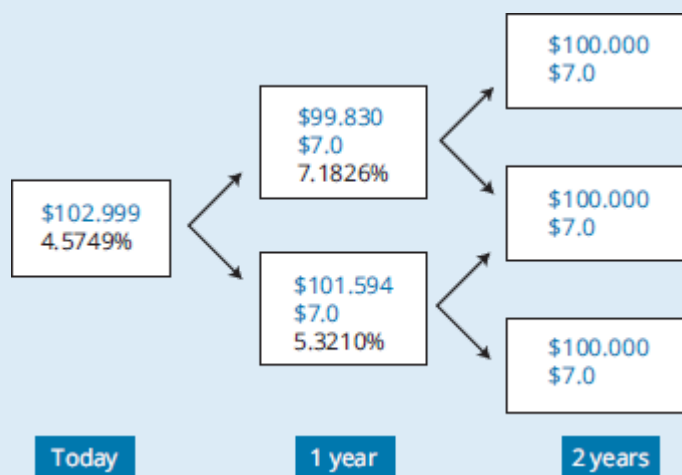
$$V_{1,L} = \frac{1}{2} \times \left[\frac{\$100 + \$7}{1.053210} + \frac{\$100 + \$7}{1.053210} \right] = \$101.594$$

Now calculate V_0 , the current value of the bond at Node 0.

$$V_0 = \frac{1}{2} \times \left[\frac{\$99.830 + \$7}{1.045749} + \frac{\$101.594 + \$7}{1.045749} \right] = \$102.999$$

The completed binomial tree is shown as follows:

Valuing a Two-Year, 7.0% Coupon, Option-Free Bond



Value of the callable bond:

The call rule (call the bond if the price exceeds \$100) is reflected in the boxes in the completed binomial tree, where the second line of the boxes at the one-year node is the lower of the call price or the computed value. For example, the value of the bond in one year at the lower node is \$101.594. However, in this case, the bond will be called, and the investor will only receive \$100. Therefore, for valuation purposes, the value of the bond in one year at this node is \$100.

$$V_{1,L} = \$100$$

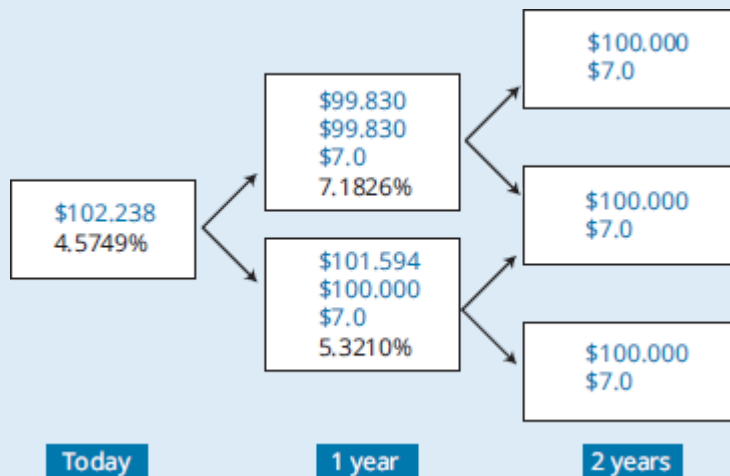
$$V_{1,U} = (107 / 1.071826) = \$99.830$$

The calculation for the current value of the bond at Node 0 (today), assuming the simplified call rules of this example, is:

$$V_0 = \frac{1}{2} \times \left[\frac{\$99.830 + \$7}{1.045749} + \frac{\$100.00 + \$7}{1.045749} \right] = \$102.238$$

The completed binomial tree is shown as follows:

Valuing a Two-Year, 7.0% Coupon, Callable Bond, Callable in One Year at 100



Value of the puttable bond:

Similarly, for a puttable bond, the put rule is to put the bond if the value falls below \$100. The put option would therefore be exercised at the upper-node in Year 1 and hence the \$99.830 computed value is replaced by the exercise price of \$100.

$$V_{1,U} = 100$$

$$V_{1,L} = (107 / 1.053210) = \$101.594$$

$$V_0 = \frac{1}{2} \times \left[\frac{100 + 7}{1.045749} + \frac{101.594 + 7}{1.045749} \right] = \$103.081$$

Value of the embedded options:

$$V_{\text{call}} = V_{\text{straight}} - V_{\text{callable}} = \$102.999 - \$102.238 = \$0.76$$

$$V_{\text{put}} = V_{\text{puttable}} - V_{\text{straight}} = \$103.081 - \$102.999 = 0.082$$

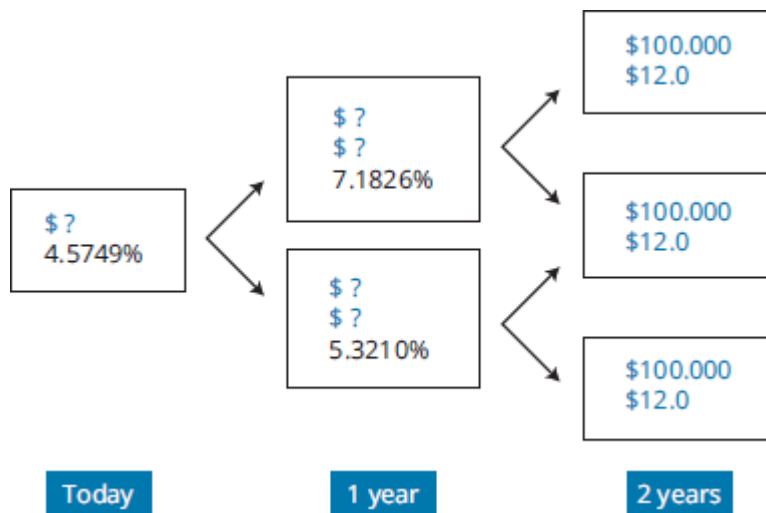


MODULE QUIZ 25.1, 25.2

- Which of the following statements concerning the calculation of value at a node in a binomial interest rate tree is *most accurate*? The value at each node is the:
 - present value of the two possible values from the next period.
 - average of the present values of the two possible values from the next period.

C. sum of the present values of the two possible values from the next period.

Use the following binomial interest rate tree to answer Questions 2 through 4.



2. The value today of an option-free, 12% annual coupon bond with two years remaining until maturity is *closest* to:

- A. 110.525.
- B. 111.485.
- C. 112.282.

3. The value of the bond and the value of the embedded call option, assuming the bond in Question 2 is callable at \$105 at the end of Year 1, are *closest* to:

	<u>Callable bond value</u>	<u>Embedded call option value</u>
A.	110.573	1.709
B.	110.573	0.642
C.	111.640	0.642

4. The value of the bond and the value of the embedded put option, assuming the bond in Question 2 is puttable at \$105 at the end of Year 1, are *closest* to:

	<u>Puttable bond value</u>	<u>Embedded put option value</u>
A.	112.523	0.241
B.	112.523	1.646
C.	113.928	1.646



Video covering
this content is
available online.

MODULE 25.3: VALUING BONDS WITH EMBEDDED OPTIONS, PART 2

LOS 25.d: Explain how interest rate volatility affects the value of a callable or putable bond.

Option values are positively related to the volatility of their underlying. Accordingly, when interest rate volatility increases, the values of both call and put options increase. The value of a straight bond is affected by changes in the level of interest rates but is *unaffected* by changes in the *volatility* of interest rates.

When interest rate volatility increases, the value of a callable bond (where the investor is short the call option) decreases and the value of a putable bond (where the investor is long the put option) increases.

LOS 25.e: Explain how changes in the level and shape of the yield curve affect the value of a callable or putable bond.

Level of Interest Rates

As interest rates decline, the short call in a callable bond limits the bond's upside, so the value of a callable bond rises less rapidly than the value of an otherwise-equivalent straight bond.

As interest rates increase, the long put in a putable bond hedges against the loss in value; the value of a putable bond falls less rapidly than the value of an otherwise-equivalent straight bond.

Call option value is inversely related to the level of interest rates, while put option value varies directly with the level of interest rates.

Shape of the Yield Curve

The value of an embedded call option increases as interest rates decline. When the yield curve is upward sloping (i.e., normal), the more distant one-period forward rates are higher than the one-period forward rates in the near future. Because a higher interest rate scenario limits the probability of the call option being in the money, the value of a call option will be lower for an upward sloping yield curve. As an upward-sloping yield curve becomes flatter, the call option value increases.

The value of a put option increases with interest rates. When the yield curve is upward sloping, the probability of the put option going in the money is higher. Put option value therefore declines as an upward-sloping yield curve flattens.



Video covering
this content is
available online.

So far our backward induction process has relied on the risk-free binomial interest rate tree; our valuation assumed that the underlying bond was risk-free. If risk-free rates are used to discount cash flows of a credit risky corporate bond, the calculated value will be too high. To correct for this, a constant spread must be added to all one-period rates in the tree such that the calculated value equals the market price of the risky bond. This constant spread is called the **option adjusted spread (OAS)**.



The OAS is added to the tree after the adjustment for the embedded option (i.e., the node values are adjusted according to the call/put rule). Hence the OAS is calculated after the option risk has been removed.

A \$100-par, three-year, 6% annual-pay ABC, Inc., callable bond trades at \$99.95. The underlying call option is a Bermudan-style option that can be exercised in one or two years at par.

One-Period Forward Rates		
Year 0	Year 1	Year 2
3.000%	5.7883%	10.7383%
	3.8800%	7.1981%
		4.8250%

Answer:

Year 0

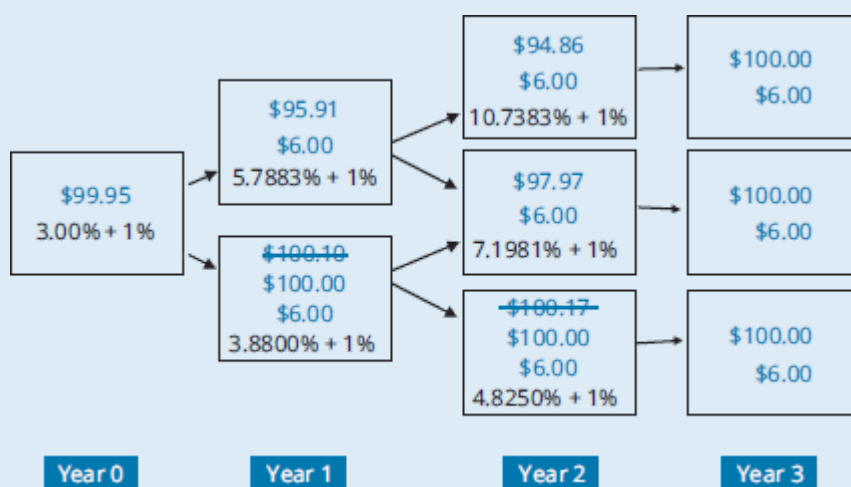
Year 1

Year 2

Year 3

Year	Price	Return
Year 0	\$101.77	3.0000%
Year 1 (Up)	\$97.65	5.7883%
Year 1 (Down)	\$101.56	3.8800%
Year 2 (Up-Up)	\$95.72	10.7383%
Year 2 (Up-Down)	\$98.88	7.1981%
Year 2 (Down-Up)	\$101.12	4.8250%
Year 2 (Down-Down)	\$101.12	4.8250%
Year 3 (All)	\$100.00	

To force the computed value to be equal to the current market price of \$99.95, a constant spread (OAS) of 100 bps is added to each interest rate in the tree, as shown here:



PROFESSOR'S NOTE

The OAS computed using the methodology just listed is the spread implied by the current market price and hence assumes that the bond is priced correctly. Note that the actual estimation of OAS is largely an iterative process and is beyond the scope of the exam.

OAS is used by analysts in relative valuation; bonds with similar credit risk should have the same OAS. If the OAS for a bond is higher than the OAS of its peers, it is considered to be undervalued and hence an attractive investment (i.e., it offers a higher compensation for a given level of risk). Conversely, bonds with low OAS (relative to peers) are considered to be overvalued.

LOS 25.h: Explain how interest rate volatility affects option-adjusted spreads.

Consider for a moment the difference between the calculated value of a bond that we obtain from a tree and the bond's actual market price. The greater this difference is, the greater the OAS we would need to add to the rates in the tree to force the calculated bond value down to the market price.

Suppose that a 7%, 10-year callable bond of XYZ Corp. is trading for \$958.

Analyst A assumes a 15% value for future volatility in generating her *benchmark* interest rate tree and calculates the value of the bond as \$1,050. She then computes the OAS (the increase in discount rate required to lower the calculated bond value to the market price of \$958) for this bond to be 80 bps.

Analyst B assumes a 20% value for future volatility, and because of higher assumed volatility, the computed value of the bond turns out to be \$992. (Note that this is lower than the \$1,050 calculated by A—the higher the volatility, the lower a callable bond's value.) The OAS calculated by B is accordingly lower at 54 bps.

Observe that for the same bond, the OAS calculated varied depending on the volatility assumption used.

When we use a higher estimate of volatility to value a callable bond, the calculated value of the call option increases, the calculated value of the straight bond is unaffected, and the *computed value* (not the market price) of the callable bond decreases (since the bondholder is short the option). Hence when the estimated (or assumed) volatility (of benchmark rates) used in a binomial tree is higher, the computed value of a callable bond will be lower—and therefore closer to its true market price. The constant spread that needs to be added to the benchmark rates to correctly price the bond (the OAS) is therefore lower.

Figure 25.1: Relationship Between Volatility and OAS

Assumed Level of Volatility	Value				OAS _{CALL}	OAS _{PUT}
	Calls	Puts	Callable	Putable		
High	High	High	Low	High	Low	High
Low	Low	Low	High	Low	High	Low

To summarize, as the *assumed* level of volatility used in an interest rate tree increases, the computed OAS (for a given market price) for a callable bond decreases. Similarly, the computed OAS of a putable bond increases as the assumed level of volatility in the binomial tree increases.



PROFESSOR'S NOTE

Notice that the columns for the value of callable and putable bonds in the preceding figure match the corresponding OAS columns.



MODULE QUIZ 25.3, 25.4

- The option adjusted spread (OAS) on a callable corporate bond is 73 basis points using on-the-run Treasuries as the benchmark rates in the construction of the binomial tree. The *best* interpretation of this OAS is the:
 - cost of the embedded option is 73 basis points.
 - cost of the option is 73 basis points over Treasury.
 - spread that reflects the credit risk is 73 basis points over Treasury.
- An increase in interest rate volatility increases the value of:
 - bonds with embedded call options.
 - bonds with embedded put options.
 - low-coupon bonds with embedded options, but decreases the value of high-coupon bonds with embedded options.

MODULE 25.5: DURATION



Video covering this content is available online.

LOS 25.i: Calculate and interpret effective duration of a callable or putable bond.

Recall from Level I that:

- Modified duration measures a bond's price sensitivity to interest rate changes, *assuming that the bond's cash flows do not change as interest rates change*.
- The standard measure of convexity can be used to improve price changes estimated from modified duration.

Modified duration and convexity are not useful for bonds with embedded options, however, because the cash flows from these bonds will change if the option is exercised. To overcome this problem, **effective duration** and **effective convexity** should be used, because these measures take into account how changes in interest rates may alter cash flows.

The following expressions can be used to compute effective duration and effective convexity for *any* bond:

$$\text{effective duration} = ED = \frac{BV_{-\Delta y} - BV_{+\Delta y}}{2 \times BV_0 \times \Delta y}$$

$$\text{effective convexity} = EC = \frac{BV_{-\Delta y} + BV_{+\Delta y} - (2 \times BV_0)}{BV_0 \times \Delta y^2}$$

where:

Δy = change in required yield, in decimal form

$BV_{-\Delta y}$ = estimated price if yield decreases by Δy

$BV_{+\Delta y}$ = estimated price if yield increases by Δy

BV_0 = initial observed bond price

Calculating effective duration and effective convexity for bonds with embedded options is a complicated undertaking because we must calculate values of $BV_{+\Delta y}$ and $BV_{-\Delta y}$. Here's how it is done:

Step 1: Given assumptions about benchmark interest rates, interest rate volatility, and any calls and/or puts, calculate the OAS for the issue using the current market price and the binomial model.

Step 2: Impose a small parallel shift in the benchmark yield curve by an amount equal to $+\Delta y$.

Step 3: Build a new binomial interest rate tree using the new yield curve.

Step 4: Add the OAS from step 1 to each of the one-year rates in the interest rate tree to get a "modified" tree.

Step 5: Compute $BV_{+\Delta y}$ using this modified interest rate tree.

Step 6: Repeat steps 2 through 5 using a parallel rate shift of $-\Delta y$ to obtain a value of $BV_{-\Delta y}$.

LOS 25.j: Compare effective durations of callable, putable, and straight bonds.

Both call and put options have the potential to reduce the life of a bond, so the duration of callable and putable bonds will be less than or equal to the duration of their straight counterparts.

- Effective duration (callable) $\leq$ effective duration (straight).
- Effective duration (putable) $\leq$ effective duration (straight).

- Effective duration (zero-coupon) $\approx$ maturity of the bond.
- Effective duration of fixed-rate coupon bond $<$ maturity of the bond.
- Effective duration of floater $\approx$ time (in years) to next reset.

While effective duration of straight bonds is relatively unaffected by changes in interest rates, an increase in rates would increase the effective duration of a callable bond, and decrease the effective duration of a puttable bond.



MODULE QUIZ 25.5

1. Ron Hyatt has been asked to do a presentation on how effective duration (ED) and effective convexity (EC) are calculated with a binomial model. His presentation includes the following formulas:

$$\text{effective duration} = \text{ED} = \frac{BV_{-\Delta y} - BV_{+\Delta y}}{2 \times BV_0 \times \Delta y}$$

$$\text{effective convexity} = \text{EC} = \frac{BV_{-\Delta y} + BV_{+\Delta y} - (2 \times BV_0)}{2 \times BV_0 \times \Delta y^2}$$

where:

Δy = change in required yield, in decimal form

$BV_{-\Delta y}$ = estimated price if yield decreases by Δy

$BV_{+\Delta y}$ = estimated price if yield increases by Δy

BV_0 = initial observed bond price

Are Hyatt's formulas for effective duration and effective convexity correctly presented?

- A. The formulas are both correct.
- B. One formula is correct, the other incorrect.
- C. Both formulas are incorrect.

MODULE 25.6: KEY RATE DURATION



Video covering this content is available online.

LOS 25.k: Describe the use of one-sided durations and key rate durations to evaluate the interest rate sensitivity of bonds with embedded options.

One-Sided Durations

The computation of effective duration discussed earlier relied on computing the value of the bond for equal parallel shifts of the yield curve up and down (by the same amount). This metric captures interest rate risk reasonably well for small changes in the yield curve and for option-free bonds.

For a callable bond, when the call option is at or near the money, the change in price for a decrease in yield will be less than the change in price for an equal amount of increase in yield. The value of a callable bond is capped by its call price: the bond's value will not increase beyond the call price regardless of how low interest rates fall. Similarly, the value of a puttable bond is more sensitive to downward movements in yield curve versus upward movements.

For bonds with embedded options, **one-sided durations**—durations that apply only when interest rates rise (or, alternatively, only when rates fall)—are better at capturing interest rate sensitivity than simple effective duration. When the underlying option is at-the-money (or near-the-money), callable bonds will have lower one-sided down-duration than one-sided up-duration: the price change of a callable when rates fall is smaller than the price change for an equal increase in rates. Conversely, a near-the-money puttable bond will have larger one-sided down-duration than one-sided up-duration.

Key Rate Duration

Key rate durations or **partial durations** capture the interest rate sensitivity of a bond to changes in yields (par rates) of specific benchmark maturities. Key rate duration is used to identify the interest rate risk from changes in the shape of the yield curve (shaping risk).

The process of computing key rate duration is similar to the process of computing effective duration described earlier, except that instead of shifting the entire benchmark yield curve, only one specific par rate (key rate) is shifted before the price impact is measured.

Figure 25.2 shows the key rate durations for several 15-year *option-free* bonds, each with a different coupon rate and YTM = 3%.

Figure 25.2: Key Rate Durations of Various 15-Year Option-Free Bonds With Different Coupon Rates

Coupon	Price	Key Rate Durations					
		Total	2-year	3-year	5-year	10-year	15-year
1%	\$76.12	13.41	−0.05	−0.07	−0.22	−0.45	14.20
2%	\$88.06	12.58	−0.03	−0.05	−0.15	−0.27	13.08
3%	\$100.00	11.94	0	0	0	0	11.94
5%	\$123.88	11.03	0.02	0.1	0.15	0.32	10.44
8%	\$159.69	10.18	0.09	0.15	0.32	0.88	8.74



PROFESSOR'S NOTE

In these figures, the market interest rate is not changing; rather we are considering a number of different bonds that have different coupons.

Figure 25.3 shows the key rate durations for several 15-year European-style *callable* bonds, each with a different coupon rate (callable in 10 years at par).

Figure 25.3: Key Rate Durations of Various 15-Year Callable Bonds With Different Coupon Rates

Coupon	Price	Key Rate Durations					
		Total	2-year	3-year	5-year	10-year	15-year
1%	\$75.01	13.22	−0.03	−0.01	−0.45	−2.22	15.93
2%	\$86.55	12.33	−0.01	−0.03	−0.15	5.67	6.85
3%	\$95.66	11.45	0.00	0.00	0.00	6.40	5.05
5%	\$112.87	9.22	0.02	0.10	0.15	6.67	2.28
8%	\$139.08	8.89	0.09	0.15	0.32	7.20	1.13

Figure 25.4 shows the key rate durations for several 15-year European-style *putable* bonds, each with a different coupon rate (putable in 10 years at par).

Figure 25.4: Key Rate Durations of Various 15-Year Putable Bonds With Different Coupon Rates

Coupon	Price	Key Rate Durations					
		Total	2-year	3-year	5-year	10-year	15-year
1%	\$77.24	9.22	-0.03	-0.01	-0.45	8.66	1.05
2%	\$89.82	9.90	-0.01	-0.03	-0.15	7.23	2.86
3%	\$95.66	10.50	0.00	0.00	0.00	5.12	5.38
5%	\$123.88	10.70	0.02	0.10	0.15	2.89	7.54
8%	\$159.69	10.08	0.09	0.15	0.32	0.45	9.07

The following generalizations can be made about key rates:

1. If an option-free bond is trading at par, the bond's maturity-matched rate is the only rate that affects the bond's value. Its maturity key rate duration is the same as its effective duration, and all other rate durations are zero. In Figure 25.2, the 3% bond's (i.e., the par bond) 15-year key rate duration is same as the bond's effective duration, and all other rate durations are zero.
2. For an option-free bond *not* trading at par, the maturity-matched rate is still the most important rate. In Figure 25.2, the 15-year key rate duration is highest.
3. A bond with a low (or zero) coupon rate may have negative key rate durations for horizons other than the bond's maturity. This is evidenced by some negative key rate durations for 1% and 2% coupon bonds in all three figures.
4. A callable bond with a low coupon rate is unlikely to be called; hence, the bond's maturity-matched rate is the most critical rate (i.e., the highest key rate duration corresponds to the bond's maturity). For the 1% bond in Figure 25.3, the 15-year key rate duration exceeds all other key rate durations.
5. All else equal, higher coupon bonds are more likely to be called, and therefore the time-to-exercise rate will tend to dominate the time-to-maturity rate. For the 8% coupon bond in Figure 25.3, the 10-year key rate duration is highest.
6. Putable bonds with high coupon rates are unlikely to be put, and thus are most sensitive to their maturity-matched rates. For the 8% bond in Figure 25.4, the 15-year key rate duration is the highest.
7. All else equal, lower coupon bonds are more likely to be put, and therefore the time-to-exercise rate will tend to dominate the time-to-maturity rate. For the 1% coupon bond in Figure 25.4, the 10-year key rate duration is highest.

LOS 25.I: Compare effective convexities of callable, putable, and straight bonds.

Straight bonds have positive effective convexity: the increase in the value of an option-free bond is higher when rates fall than the decrease in value when rates increase by an equal amount. When rates are high, callable bonds are unlikely to be called and will exhibit positive convexity. When the underlying call option is near the money, its effective convexity turns negative; the upside potential of the bond's price is limited due to the call

(while the downside is not protected). Putable bonds exhibit positive convexity throughout.

MODULE 25.7: CAPPED AND FLOORED FLOATERS



Video covering
this content is
available online.

LOS 25.m: Calculate the value of a capped or floored floating-rate bond.

A floating-rate bond (“floater”) pays a coupon that adjusts every period based on an underlying reference rate. The coupon is typically paid in arrears, meaning the coupon rate is *determined* at the beginning of a period but is *paid* at the end of that period.

A **capped floater** effectively contains an issuer option that prevents the coupon rate from rising above a specified maximum rate known as the **cap**.

$$\begin{aligned} \text{value of a capped floater} \\ &= \text{value of a “straight” floater} - \text{value of the embedded cap} \end{aligned}$$

A related floating rate bond is the **floored floater**, which has a coupon rate that will not fall below a specified minimum rate known as the **floor**. In this instance, the embedded option belongs to the investor and offers protection from falling interest rates.

$$\begin{aligned} \text{value of a floored floater} \\ &= \text{value of a “straight” floater} + \text{value of the embedded floor} \end{aligned}$$

We can use the standard backward induction methodology in a binomial interest rate tree to value a capped or floored floater. As with the valuation of a bond with embedded options, we must adjust the value of the floater at each node to reflect the exercise of an in-the-money option (in this case, a cap or a floor).

EXAMPLE: Value of a capped and floored floating-rate bond

Susane Albright works as a fixed income analyst with Zedone Banks, NA. She has been asked to value a \$100 par, two-year, floating-rate note that pays MRR (set in arrears). The underlying bond has the same credit quality as reflected in the swap curve. Albright has constructed the following two-year binomial MRR tree:

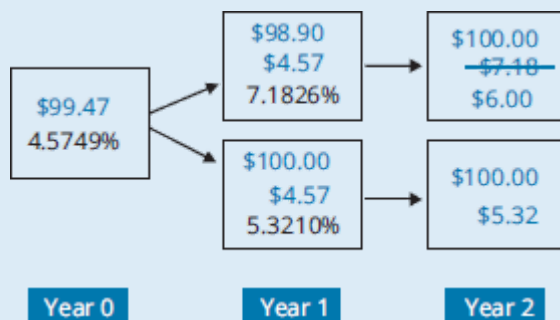
One-Period Forward Rate	
Year 0	Year 1
4.5749%	7.1826%
	5.3210%

How would we compute the following?

1. The value of the floater, assuming that it is an option-free bond.
2. The value of the floater, assuming that it is capped at a rate of 6%. Also compute the value of the embedded cap.
3. The value of the floater, assuming that it is floored at a rate of 5%. Also compute the value of the embedded floor.

Answer:

1. An option-free bond with a coupon rate equal to the required rate of return will be worth par value. Hence, the straight value of the floater is \$100.
2. The value of the capped floater is \$99.47, as shown here:



The upper node in Year 2 shows the exercise of the cap (the coupon is capped at \$6.00 instead of rising to \$7.18).

Note that when the option is not in the money, the floater is valued at par.

$$V_{1,U} = (\$100 + \$6) / (1 + 0.071826) = \$98.90$$

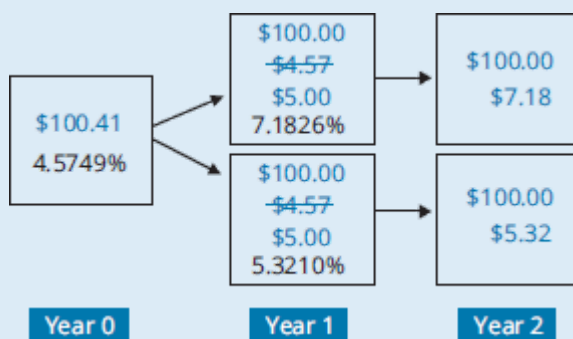
$$V_{1,L} = (100 + 5.3210) / (1 + 0.05321) = \$100$$

The Year 0 value is the average of the Year 1 values (including their adjusted coupons) discounted for one period. In this case, the Year 1 coupons require no adjustment, as the coupon rate is below the cap rate.

$$V_0 = \frac{[(98.90 + 4.57) + (100 + 4.57)] / 2}{(1.045749)} = \$99.47$$

Thus the value of the embedded cap = \$100 – \$99.47 = 0.53.

3. The value of the floored floater is \$100.41, as shown here:



The nodes for Year 2 show the coupons for that period (none of the rates are below the floor, and hence the floor is not exercised). Strikethroughs for both nodes in Year 1 indicate that the floor was in the money; we replace the MRR = market reference rate. Having another rate after it is redundant coupon with a coupon based on the floor strike rate of 5%.

The Year 0 value is the average of the Year 1 values (including their adjusted coupons) discounted for one period:

$$V_0 = \frac{[(100 + 5) + (100 + 5)] / 2}{(1.045749)} = \$100.41$$

Thus the value of the embedded floor = \$100.41 – \$100 = \$0.41.



MODULE QUIZ 25.6, 25.7

Use the following information to answer Questions 1 through 8.

Vincent Osagae, CFA, is the fixed income portfolio manager for Alpha Specialists, an institutional money manager. Vanessa Alwan, an intern, has suggested a list of bonds for Osagae's consideration as shown in Figure 1. The benchmark yield curve is currently upward sloping.

Figure 1: Selected Bonds for Consideration

- Bond X, a 3%, 15-year option-free bond.
- Bond Y, a 3%, 15-year callable bond.
- Bond Z, a 3%, 15-year putable bond.

Osagae then turns his attention to a newly issued 4%, 15-year bond issued by Suni Corp. The bond has a Bermudan-style call option exercisable annually at any time after Year 4. Osagae computes an OAS of 145 bps for this bond using a binomial interest rate tree. The tree was generated with U.S. Treasury rates and an assumed volatility of 15%, which was consistent with historical data. Option markets are currently pricing interest rate options at an implied volatility of 19%.

Every week, Alwan meets with Osagae to discuss what she had learned over the week. At one of their weekly meetings, Alwan makes the following statements about key rate duration:

Statement 1: The highest key rate duration of a low-coupon callable bond corresponds to the bond's time-to-exercise.

Statement 2: The highest key rate duration of a high-coupon putable bond corresponds to the bond's time-to-maturity.

Beth Grange, a senior analyst reporting to Osagae has generated the following two-year MRR tree using a 15% volatility assumption:

Figure 2: MRR Tree

0	1
2.0000%	3.4637%
	2.5660%

Grange has also compiled OAS for two very similar callable corporate bonds as shown in Figure 3.

Figure 3: OAS at 10% Volatility

1. In Figure 1, which bond's embedded option is *most likely* to increase in value if the yield curve flattens?
 - A. Bond Y.
 - B. Bond Z.
 - C. Neither Bond Y nor Bond Z.
2. If Osagae had used implied volatility from interest rate options to compute the OAS for Suni Corp bond, the estimated OAS would *most likely* have been:
 - A. lower than 145 bps.
 - B. higher than 145 bps.
 - C. equal to 145 bps.
3. Which bond in Figure 1 is *most likely* to have the highest effective duration?
 - A. Bond X.
 - B. Bond Y.
 - C. Bond Z.
4. Regarding Alwan's statements about key rate durations:
 - A. only one statement is correct.

- B. both statements are correct.
 - C. neither statement is correct.
5. Which bond in Figure 1 is *least likely* to experience the highest increase in value, given a parallel downward shift of 150 bps in the yield curve?
- A. Bond X.
 - B. Bond Y.
 - C. Bond Z.
6. Which bond in Figure 1 is *most likely* to experience an increase in effective duration due to an increase in interest rates?
- A. Bond X.
 - B. Bond Y.
 - C. Bond Z.
7. In Figure 3, relative to Bond A, Bond B is *most likely*:
- A. overpriced.
 - B. underpriced.
 - C. fairly priced.
8. Using the data in Figure 2, the value of a \$100 par, two-year, 3% capped floater is *closest* to:
- A. \$0.31.
 - B. \$98.67.
 - C. \$99.78.

MODULE 25.8: CONVERTIBLE BONDS



Video covering
this content is
available online.

LOS 25.n: Describe defining features of a convertible bond.

The owner of a **convertible bond** has the right to convert the bond into a fixed number of common shares of the issuer during a specified timeframe (**conversion period**) and at a fixed amount of money (**conversion price**). Convertibles allow investors to enjoy the upside on the issuer's stock, although this comes at a cost of lower yield. The issuer of a convertible bond benefits from a lower borrowing cost, but existing shareholders may face dilution if the conversion option is exercised.

The **conversion ratio** is the number of common shares for which a convertible bond can be exchanged. For example, a convertible bond issued at par with an initial conversion ratio of 10 allows its holder to convert one \$1,000 par bond into 10 shares of common stock. Equivalently, the **conversion price** of the bond is $\$1,000 / 10 \text{ shares} = \100 . For bonds not issued at par, the conversion price is the issue price divided by the conversion ratio. Offer documents will indicate how the initial conversion ratio would be modified to account for corporate actions such as stock splits or stock dividends.

Offer documents may also provide a contingent put option in the event of any change-of-control events such as mergers. Such a contingent put option can be exercised for a specific period of time after the change of control. Alternatively, a lower conversion price may be specified in the event of a change of control. The conversion ratio may also be adjusted upward if the company pays a dividend in excess of a specified **threshold dividend**. This adjustment protects the convertible bondholders in the event that the company pays out an unusually large dividend (which will cause the ex-dividend price of the stock to decline). Other put options exercisable during specific periods may also be embedded with a

convertible. These put options can be hard puts (i.e., redeemable for cash) or soft puts (i.e., the issuer decides whether to redeem the bond for cash, stock, subordinated debentures, or a combination of the three).

LOS 25.o: Calculate and interpret the components of a convertible bond's value.

The **conversion value** of a convertible bond is the value of the common stock into which the bond can be converted. The conversion ratio is the number of shares the holder receives from conversion for each bond. Conversion value is calculated as:

$$\text{conversion value} = \text{market price of stock} \times \text{conversion ratio}$$

The **straight value**, or investment value, of a convertible bond is the value of the bond if it were not convertible—the present value of the bond's cash flows discounted at the return required on a comparable option-free issue.

The **minimum value of a convertible bond** is the greater of its conversion value or its straight value. This must be the case, or arbitrage opportunities would be possible. For example, if a convertible bond were to sell for less than its conversion value, it could be purchased, immediately converted into common stock, and the stock could be sold for more than the cost of the bond.

$$\text{minimum value of a convertible bond} = \max(\text{straight value}, \text{conversion value})$$

EXAMPLE: Calculating the minimum value of a convertible bond

Business Supply Company, Inc. operates retail office equipment stores in the United States and Canada. Consider a BSC convertible bond with a 7% coupon that is currently selling at \$985 with a conversion ratio of 25 and a straight value of \$950. Suppose that the value of BSC's common stock is currently \$35 per share, and that it pays \$1 per share in dividends annually. What is this bond's minimum value?

Answer:

The conversion value of this bond is $25 \times \$35 = \875 . Since the straight value of \$950 is greater than the conversion value of \$875, the bond is worth at least \$950.

The market conversion price, or conversion parity price, is the price that the convertible bondholder would effectively pay for the stock if she bought the bond and immediately converted it. The market conversion price is given as:

$$\text{market conversion price} = \frac{\text{market price of convertible bond}}{\text{conversion ratio}}$$

EXAMPLE: Calculating market conversion price

Compute and interpret the market conversion price of the BSC bond.

Answer:

The market conversion price is: $\$985 / 25 = \39.40 . This can be viewed as the stock price at which an investor is indifferent between selling the bond and converting it.

The market conversion premium per share is the difference between the market conversion price and the stock's current market price:

$$\begin{aligned} \text{market conversion premium per share} \\ = \text{market conversion price} - \text{stock's market price} \end{aligned}$$

EXAMPLE: Calculating market conversion premium per share

Compute and interpret the market conversion premium per share of the BSC bond.

Answer:

Since BSC is selling for \$35 per share, the market conversion premium per share for the BSC bond is: $\$39.40 - \$35 = \$4.40$. This can be interpreted as the premium that investors are willing to pay for the opportunity to profit should the market price of the stock rise above the market conversion price. This is done with the assurance that even if the stock price declines, the value of the convertible bond will not fall below its straight value.

Market conversion premium per share is usually expressed as a ratio, appropriately called the **market conversion premium ratio**. Its formula is:

$$\text{market conversion premium ratio} = \frac{\text{market conversion premium per share}}{\text{market price of common stock}}$$

EXAMPLE: Calculating market conversion premium ratio

Compute the market conversion premium ratio of the BSC bond.

Answer:

The BSC bond market conversion premium ratio is:

$$\frac{\$4.40}{\$35} = 12.57\%$$

The convertible bond investor's downside risk is limited by the bond's underlying straight value because the price of a convertible bond will not fall below this value regardless of what happens to the price of the issuer's common stock.

This downside risk is measured by the **premium over straight value**, which is calculated as:

$$\text{premium over straight value} = \left(\frac{\text{market price of convertible bond}}{\text{straight value}} \right) - 1$$

EXAMPLE: Calculating premium over straight value

Compute and interpret the BSC bond's premium over straight value.

Answer:

The premium over straight value for the BSC bond is:

$$\left(\frac{\$985}{\$950}\right) - 1 = 3.68\%$$

All else equal, the greater the premium over straight value, the less attractive the convertible bond.

We need to recognize an obvious flaw with the premium over straight value metric—the straight value is not constant. It varies with changes in interest rate and with the credit spread of the bond.

LOS 25.p: Describe how a convertible bond is valued in an arbitrage-free framework.

Investing in a noncallable/nonputable convertible bond is equivalent to buying:

- an option-free bond, and
- a call option on an amount of the common stock equal to the conversion ratio.

The value of a noncallable/nonputable convertible bond can be expressed as:

$$\text{convertible, noncallable bond value} = \text{straight value} \\ + \text{value of call option on stock}$$

Most convertible bonds are callable, giving the issuer the right to call the issue prior to maturity. Incorporating this feature into the valuation of a convertible bond results in the following expression:

$$\text{callable convertible bond value} = \text{straight value of bond} \\ + \text{value of call option on stock} \\ - \text{value of call option on bond}$$

To further complicate the situation (just for fun), consider a convertible bond that is both callable and putable. The expression for value then becomes:

$$\text{callable and putable convertible bond value} = \text{straight value of bond} \\ + \text{value of call option on stock} \\ - \text{value of call option on bond} \\ + \text{value of put option on bond}$$

LOS 25.q: Compare the risk–return characteristics of a convertible bond with the risk–return characteristics of a straight bond and of the underlying common stock.

Buying convertible bonds instead of stocks limits downside risk; the price floor set by the straight bond value provides this downside protection. The cost of the downside protection is reduced upside potential due to the conversion premium. Keep in mind though, that just like investing in nonconvertible bonds, convertible bond investors must be concerned with credit risk, call risk, interest rate risk, and liquidity risk.

Consider the following two examples based on our previous BSC example.

EXAMPLE: Risk and return of a convertible bond, part 1

Calculate the return on the convertible bond and the common stock if the market price of BSC common stock increases to \$45 per share.

Answer:

The return from investing in the convertible bond is:

$$\left(\frac{\$45.00}{\$39.40}\right) - 1 = 14.2\%$$

The return from investing directly in the stock is:

$$\left(\frac{\$45.00}{\$35.00}\right) - 1 = 0.2857 = 28.6\%$$

The lower return from the convertible bond investment is attributable to the fact that the investor effectively bought the stock at the market conversion price of \$39.40 per share.

EXAMPLE: Risk and return of a convertible bond, part 2

Calculate the return on the convertible bond and on the common stock if the market price of BSC common stock falls to \$30 per share.

Answer:

Recall that the bond will trade at the greater of its straight value or its conversion value. The conversion value in this scenario is $25 \times \$30 = \750 . Assuming the straight value of the bond does not change, the bond will trade at \$950. So, the return from investing in the convertible bond is:

$$\left(\frac{\$950}{\$985}\right) - 1 = -3.55\%$$

The return from investing directly in the stock is:

$$\left(\frac{\$30}{\$35}\right) - 1 = -14.29\%$$

The loss is less for the convertible bond investment because we assumed that the straight value of the bond did not change. Even if it had changed, the loss would probably still be less than the loss on the straight stock investment, thus emphasizing how the straight value serves as a floor to cushion a decline, even if it is a moving floor.

The following comparisons can be made between ownership of the underlying stock and the risk-return characteristics of the convertible bond:

- When the stock's price falls, the returns on convertible bonds exceed those of the stock, because the convertible bond's price has a floor equal to its straight bond value. As the

stock's price approaches zero, the convertible's value will move toward the present value of estimated recovery.

- When the stock's price rises, the bond will underperform because of the conversion premium. This is the main drawback of investing in convertible bonds versus investing directly in the stock.
- If the stock's price remains stable, the return on a convertible bond may exceed the stock return due to the coupon payments received from the bond, assuming no change in interest rates or credit risk of the issuer.

Sometimes the price of the common stock associated with a convertible issue is so low that it has little or no effect on the convertible's market price, and the bond trades as though it is a straight bond. When this happens, the convertible security is referred to as a fixed-income equivalent or a **busted convertible**.

Other times, the price of the stock may be high enough that the convertible behaves as though it is an equity security. When this happens, the convertible issue is referred to as a common stock equivalent. Most of the time, however, the convertible behaves as a *hybrid security* with the characteristics of both equity and a fixed-income security.



MODULE QUIZ 25.8

1. An analyst has gathered the following information on a convertible bond and the common equity of the issuer:

- Market price of bond: \$925.00
- Annual coupon: 7.5%
- Conversion ratio: 30
- Market price of stock: \$28.50
- Annual stock dividend: \$2.15 per share

The market conversion premium ratio for the convertible bond is *closest* to:

- A. 7.56%.
 - B. 7.77%.
 - C. 8.18%.
2. Which of the following statements concerning a comparison between the risk and return of convertible bond investing versus common stock investing is *least accurate*, assuming interest rates are stable?
 - A. When stock prices fall, the returns on convertible bonds are likely to exceed those of the stock because the convertible bond's price has a floor equal to the straight bond value.
 - B. The main drawback of investing in convertible bonds versus direct stock purchases is that when stock prices rise, the convertible bond will likely underperform the stock due to the conversion premium.
 - C. Buying convertible bonds instead of direct stock investing limits upside potential to that of buying a straight bond, at the cost of increased downside risk due to the conversion premium.
 3. Data on two convertible bonds are shown in the following table.

	Convertible Bond ABC	Convertible Bond XYZ
Conversion price	\$40	\$50
Current stock price	\$123	\$8

Which factors are *most likely* to influence the market prices of ABC and XYZ: factors that affect equity prices, or factors that affect option-free bond prices?

- A. Both will be more affected by equity factors.
 - B. One will be more affected by equity factors, the other by bond factors.
 - C. Both will be more affected by bond factors.
4. The difference between the value of a callable convertible bond and the value of an otherwise comparable option-free bond is *closest* to the value of the:
- A. call option on the stock minus value of the call option on the bond.
 - B. put option on the stock plus value of the call option on the bond.
 - C. call option on the stock plus value of call option on the bond.
5. With respect to the value of a callable convertible bond, the *most likely* effects of a decrease in interest rate volatility or a decrease in the underlying stock price volatility are:
- A. both will result in an increase in value.
 - B. one will result in an increase in value, the other in a decrease.
 - C. both will result in a decrease in value.

KEY CONCEPTS

LOS 25.a

Bonds with embedded options allow issuers to manage their interest rate risk or issue bonds at attractive coupon rates. The embedded options can be simple call or put options, or more complex options such as provisions for sinking fund, estate puts, et cetera.

LOS 25.b

Value of option embedded in a callable or puttable bond:

$$V_{\text{call}} = V_{\text{straight}} - V_{\text{callable}}$$

$$V_{\text{put}} = V_{\text{puttable}} - V_{\text{straight}}$$

LOS 25.c

To value a callable or a puttable bond, the backward induction process and a binomial interest rate tree framework is used. The benchmark binomial interest rate tree is calibrated to ensure that it values benchmark bonds correctly (i.e., that it generates prices equal to their market prices).

LOS 25.d

When interest rate volatility increases, the value of both call and put options on bonds increase. As volatility increases, the value of a callable bond decreases (remember that the investor is short the call option) and the value of a puttable bond increases (remember that the investor is long the put option).

LOS 25.e

The short call in a callable bond limits the investor's upside when rates decrease, while the long put in a puttable bond hedges the investor against rate increases.

The value of the call option will be lower in an environment with an upward-sloping yield curve as the probability of the option going in the money is low. A call option gains value when the upward-sloping yield curve flattens. A put option will have a higher probability of going in the money when the yield curve is upward sloping; the option loses value if the upward-sloping yield curve flattens.

LOS 25.f

A backwards induction process is used in a binomial interest rate tree framework for valuing a callable (or puttable) bond. In the binomial tree, we use one-period forward rates for each period. For valuing a callable (puttable) bond, the value used at any node corresponding to a call (put) date must be either the price at which the issuer will call (investor will put) the bond, or the computed value if the bond is not called (put)—whichever is lower (higher).

LOS 25.g

The option adjusted spread (OAS) is the constant spread added to each forward rate in a benchmark binomial interest rate tree, such that the sum of the present values of a credit risky bond's cash flows equals its market price.

LOS 25.h

Binomial trees generated under an assumption of high volatility will lead to higher values for a call option and a corresponding lower value for a callable bond. Under a high volatility assumption, we would already have a lower computed value for the callable bond, and hence, the additional spread (i.e., the OAS) needed to force the discounted value to equal the market price will be lower.

When an analyst uses a lower-than-actual (higher-than-actual) level of volatility, the computed OAS for a callable bond will be too high (low) and the bond will be erroneously classified as underpriced (overpriced).

Similarly, when the analyst uses a lower-than-actual (higher-than-actual) level of volatility, the computed OAS for a puttable bond will be too low (high) and the bond will be erroneously classified as overpriced (underpriced).

LOS 25.i

$$\text{effective duration} = ED = \frac{BV_{-\Delta y} - BV_{+\Delta y}}{2 \times BV_0 \times \Delta y}$$

LOS 25.j

effective duration (callable) $\leq$ effective duration (straight)

effective duration (puttable) $\leq$ effective duration (straight)

effective duration (zero-coupon) $\approx$ maturity of the bond

effective duration of fixed-rate coupon bond $<$ maturity of the bond

effective duration of floater $\approx$ time (in years) to next reset

LOS 25.k

For bonds with embedded options, one-sided durations—durations when interest rates rise versus when they fall—are better at capturing interest rate sensitivity than the more common effective duration. When the underlying option is at (or near) the money, callable (putable) bonds will have lower (higher) one-sided down-duration than one-sided up-duration.

Callable bonds with low coupon rates will most likely not be called and hence their maturity matched rate is their most critical rate (and has the highest key rate duration). As the coupon rate increases, a callable bond is more likely to be called and the time-to-exercise rate will start dominating the time-to-maturity rate.

Putable bonds with high coupon rates are unlikely to be put and are most sensitive to its maturity-matched rate. As the coupon rate decreases, a putable bond is more likely to be put and the time-to-exercise rate will start dominating the time-to-maturity rate.

LOS 25.l

Straight and putable bonds exhibit positive convexity throughout. Callable bonds also exhibit positive convexity when rates are high. However, at lower rates, callable bonds exhibit negative convexity.

LOS 25.m

A capped floater contains an issuer option that prevents the coupon rate on a floater from rising above a specified maximum (i.e., the cap) rate.

$$\begin{aligned} \text{value of a capped floater} \\ &= \text{value of a "straight" floater} - \text{value of the embedded cap} \end{aligned}$$

A related floating-rate bond is the floored floater where the coupon rate will not fall below a specified minimum (i.e., the floor).

$$\begin{aligned} \text{value of a floored floater} \\ &= \text{value of a "straight" floater} + \text{value of the embedded floor} \end{aligned}$$

LOS 25.n

The owner of a convertible bond can exchange the bond for the common shares of the issuer. A convertible bond includes an embedded call option giving the bondholder the right to buy the common stock of the issuer.

LOS 25.o

The conversion ratio is the number of common shares for which a convertible bond can be exchanged.

$$\text{conversion value} = \text{market price of stock} \times \text{conversion ratio}$$

$$\text{market conversion price} = \text{market price of convertible bond} / \text{conversion ratio}$$

$$\text{market conversion premium per share} = \text{market conversion price} - \text{market price}$$

The minimum value at which a convertible bond trades is its straight value or its conversion value, whichever is greater.

LOS 25.p

The value of a bond with embedded options is determined as the value of the straight bond plus (minus) the value of options that the investor is long (short).

$$\begin{aligned}\text{callable and putable convertible bond value} &= \text{straight value of bond} \\ &+ \text{value of call option on stock} \\ &- \text{value of call option on bond} \\ &+ \text{value of put option on bond}\end{aligned}$$

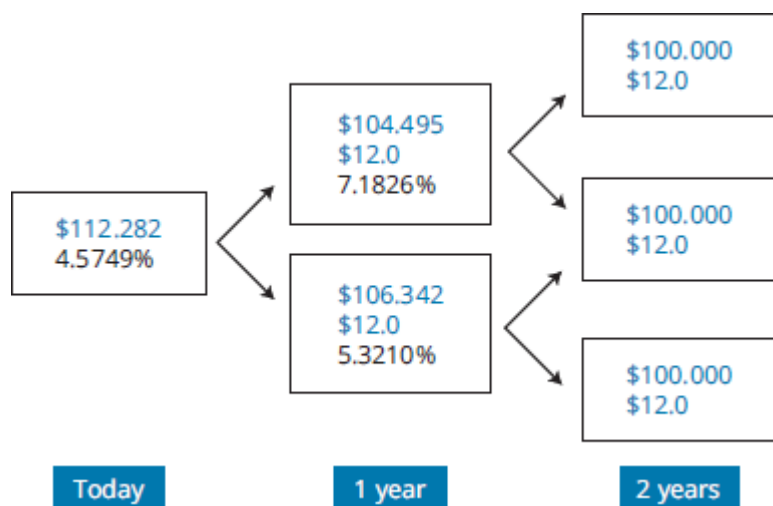
LOS 25.q

- The major benefit from investing in convertible bonds is the price appreciation resulting from an increase in the value of the common stock.
- The main drawback of investing in a convertible bond versus investing directly in the stock is that when the stock price rises, the bond will underperform the stock because of the conversion premium of the bond.
- If the stock price remains stable, the return on the bond may exceed the stock returns due to the coupon payments received from the bond.
- If the stock price falls, the straight value of the bond limits downside risk (assuming bond yields remain stable).

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 25.1, 25.2

1. **B** The value at any given node in a binomial tree is the average of the present values of the cash flows at the two possible states immediately to the right of the given node, discounted at the one-period rate at the node under examination. (Module 25.2, LOS 25.c)
2. **C** The tree should look like this:



Consider the value of the bond at the *upper* node for Period 1, $V_{1,U}$:

$$V_{1,U} = \frac{1}{2} \times \left[\frac{\$100 + \$12}{1.071826} + \frac{\$100 + \$12}{1.071826} \right] = \$104.495$$

Similarly, the value of the bond at the *lower* node for Period 1, $V_{1,L}$ is:

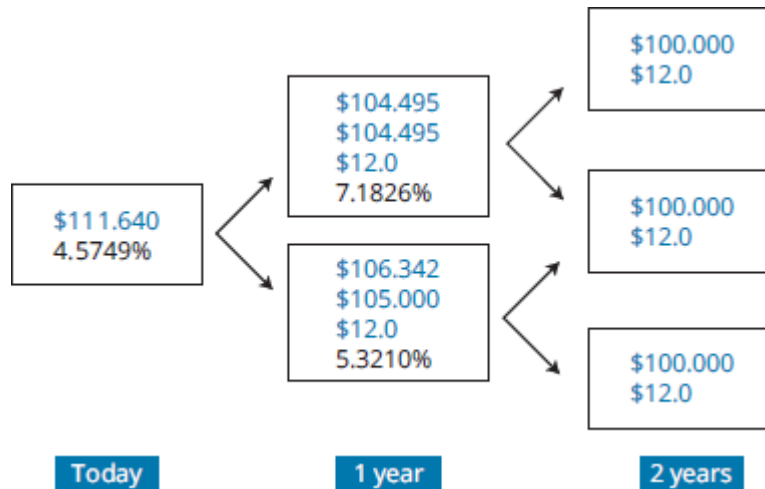
$$V_{1,L} = \frac{1}{2} \times \left[\frac{\$100 + \$12}{1.053210} + \frac{\$100 + \$12}{1.053210} \right] = \$106.342$$

Now calculate V_0 , the current value of the bond at Node 0:

$$V_0 = \frac{1}{2} \times \left[\frac{\$104.495 + \$12}{1.045749} + \frac{\$106.342 + \$12}{1.045749} \right] = \$112.282$$

(Module 25.2, LOS 25.f)

3. **C** The tree should look like this:



Consider the value of the bond at the *upper* node for Period 1, $V_{1,U}$:

$$V_{1,U} = \frac{1}{2} \times \left[\frac{\$100 + \$12}{1.071826} + \frac{\$100 + \$12}{1.071826} \right] = \$104.495$$

Similarly, the value of the bond at the *lower* node for Period 1, $V_{1,L}$ is:

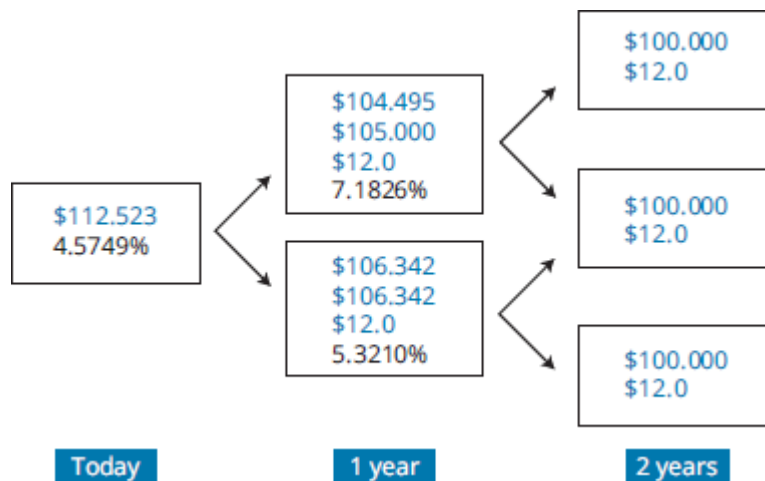
$$V_{1,L} = \frac{1}{2} \times \left[\frac{\$100 + \$12}{1.053210} + \frac{\$100 + \$12}{1.053210} \right] = \$106.342$$

Now calculate V_0 , the current value of the bond at Node 0:

$$V_0 = \frac{1}{2} \times \left[\frac{\$104.495 + \$12}{1.045749} + \frac{\$105.00 + \$12}{1.045749} \right] = \$111.640$$

The value of the embedded call option is $\$112.282 - \$111.640 = \$0.642$. (Module 25.2, LOS 25.f)

4. **A** The tree should look like this:



Consider the value of the bond at the *upper* node for Period 1, $V_{1,U}$:

$$V_{1,U} = \frac{1}{2} \times \left[\frac{\$100 + \$12}{1.071826} + \frac{\$100 + \$12}{1.071826} \right] = \$104.495$$

Similarly, the value of the bond at the *lower* node for Period 1, $V_{1,L}$, is:

$$V_{1,L} = \frac{1}{2} \times \left[\frac{\$100 + \$12}{1.053210} + \frac{\$100 + \$12}{1.053210} \right] = \$106.342$$

Now calculate V_0 , the current value of the bond at Node 0:

$$V_0 = \frac{1}{2} \times \left[\frac{\$105.000 + \$12}{1.045749} + \frac{\$106.342 + \$12}{1.045749} \right] = \$112.523$$

The value of the embedded put option is $\$112.523 - \$112.282 = \$0.241$. (Module 25.2, LOS 25.f)

Module Quiz 25.3, 25.4

1. **C** Let's construct a table of the risk differences between the issuer's callable bond and on-the-run Treasuries to help us answer this question.

Type of Risk	Equal?
Credit	No
Option	Removed by OAS

Therefore, the OAS reflects the credit risk of the corporate callable bond over Treasuries since option risk has been removed. (Module 25.4, LOS 25.g)

2. **B** Like ordinary options, the value of an embedded option increases as volatility increases. Furthermore, the arbitrage-free value of an option-free bond ($V_{\text{option-free}}$) is independent of the assumed volatility. This implies that the arbitrage-free value of a callable bond (V_{callable}) decreases as volatility increases the value of the embedded call option (V_{call}). This can be seen from the expression for the value of a callable bond:

$$\downarrow V_{\text{callable}} = V_{\text{option-free}} - \uparrow V_{\text{call}}$$

The value of the puttable bond (V_{puttable}) increases as the assumed volatility increases the value of the embedded put option (V_{put}).

$$\uparrow V_{\text{puttable}} = V_{\text{option-free}} + \uparrow V_{\text{put}}$$

(Module 25.3, LOS 25.d)

Module Quiz 25.5

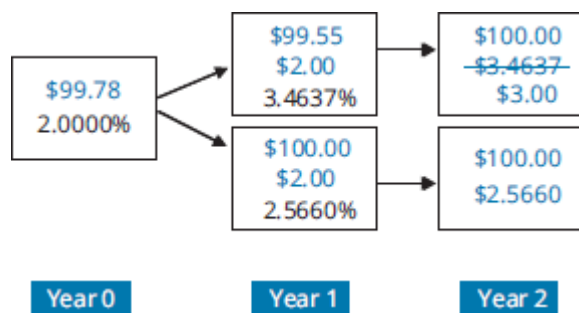
1. **B** The duration formula is presented correctly. The convexity formula is presented incorrectly; the "2" should not appear in the denominator of the convexity formula.

$$\text{effective convexity} = EC = \frac{BV_{-\Delta y} + BV_{+\Delta y} - (2 \times BV_0)}{BV_0 \times \Delta y^2}$$

(LOS 25.i)

Module Quiz 25.6, 25.7

1. **A** When an upward sloping yield curve flattens, call options increase in value while put options decrease in value. (Module 25.3, LOS 25.e)
2. **A** When the assumed volatility in a binomial tree increases, the computed value of OAS will decrease. When an analyst uses a lower-than-actual level of volatility like the 15% volatility assumed here, the computed OAS for a callable bond will be too high. Using the 19% implied volatility instead would have resulted in an estimated OAS lower than 145 bps. (Module 25.4, LOS 25.h)
3. **A** Straight bonds generally have higher effective durations than bonds with embedded options. Both call and put options have the potential to reduce the life of a bond, so the duration of callable and putable bonds will be less than or equal to that of their straight counterparts. (Module 25.5, LOS 25.j)
4. **A** Statement 1 is incorrect. Low-coupon callable bonds are unlikely to be called; hence, their highest key rate duration corresponds to their time-to-maturity. Statement 2 is correct. High coupon putable bonds are unlikely to be put and hence, their highest key rate duration corresponds to their time-to-maturity. (Module 25.6, LOS 25.k)
5. **B** Straight and putable bonds exhibit positive convexity at all interest rate levels. The price appreciation for a callable bond due to decline in interest rate is limited due to the call feature; callables exhibit negative convexity at low rates. Hence a decline in rates is least likely to result in best price performance for a callable bond. (Module 25.6, LOS 25.l)
6. **B** When interest rates increase, a callable bond becomes less likely to be called (its duration will increase). The put option in a putable bond would be more likely to be exercised in a rising interest rate scenario and hence, the duration of a putable bond would decrease. Duration of an option-free bond would also decrease as interest rate increases (but not as significantly). (Module 25.6, LOS 25.k)
7. **A** Relative to bond A, bond B has lower OAS. Given that the two bonds have similar credit risk, bond B offers a lower OAS for the same level of risk as bond A and thus would be considered overpriced. Alternatively, bond A is more attractive (underpriced) relative to bond B. (Module 25.4, LOS 25.g)
8. **C** The value of the capped floater is \$99.78, as shown here:



The upper node at Year 2 is subject to the 3% cap, and the coupon is adjusted accordingly.

$$V_{1,U} = \$103 / (1.03467) = \$99.55$$

$$V_{1,L} = 100 \text{ (option is not exercised)}$$

$$V_0 = \frac{[(100 + 2) + (99.55 + 2)] / 2}{(1.02)} = 99.78$$

(Module 25.7, LOS 25.m)

Module Quiz 25.8

1. **C** The market conversion premium per share is the market conversion price per share minus the market price per share. The market conversion price per share is

$$\frac{925.00}{30} = \$30.833, \text{ so the conversion premium per share is } \$30.833 - \$28.50 = \$2.333.$$

market conversion premium ratio

$$= \frac{\text{market conversion premium per share}}{\text{market price of common stock}}$$

$$= \frac{2.33}{28.50} = 8.18\%$$

(LOS 25.o)

2. **C** Buying convertible bonds in lieu of direct stock investing limits downside risk to that of straight bond investing, at the cost of reduced upside potential due to the conversion premium. (Note that this analysis assumes that interest rates remain stable. Otherwise, the interest rate risk associated with the straight bond investing must be considered.) When stock prices fall, the returns on convertible bonds are likely to exceed those of the stock, because the convertible bond's price has a floor equal to the straight bond value. The main drawback of investing in convertible bonds versus direct stock purchases is that when stock prices rise, the convertible bond is likely to underperform due to the conversion premium. If the stock price remains stable, the return on the bond may exceed the stock's return if the bond's coupon payment exceeds the dividend income of the stock. (LOS 25.q)

3. **B** ABC has a conversion price much less than the current stock price, so the conversion option is deep in the money. Bond ABC effectively trades like equity and is more likely to be influenced by the same factors that affect equity prices, in general, than the factors that affect bond prices.

A busted convertible like XYZ, with a stock price significantly less than the conversion price, trades like a bond (that's why a busted convertible is also called a fixed-income equivalent) and is therefore more likely to be influenced by the factors that affect bond prices. (LOS 25.q)

4. **A** A bond that is both callable and convertible contains two embedded options: (1) a call option on the stock and (2) a call option on the bond. The investor has a *short* position in the call option on the bond (the issuer has the right to call the bond) and a *long* position in the call option on the stock (the investor has the right to convert the bond into shares of stock). Therefore, the difference in value between the callable convertible bond and the value of the comparable option-free bond to the

investor is equal to the value of the call option on the stock minus the value of the call option on the bond. (LOS 25.p)

5. **B** A decrease in interest rate volatility will decrease the value of the embedded short call on the bond (but have no effect on the value of the embedded call on the stock) and increase the value of the convertible bond.

A decrease in stock price volatility will decrease the value of the embedded call on the stock (but have no effect on the embedded call on the bond) and decrease the value of the convertible bond. (LOS 25.p)

READING 26

CREDIT ANALYSIS MODELS

EXAM FOCUS

This topic review augments credit analysis covered at Level I with newer models of credit analysis including structural models and reduced form models. Be able to differentiate between the two and know the merits and drawbacks of both. Be able to compute the credit valuation adjustment and expected change in a bond's price given credit migration, and credit spread given CVA. Finally, understand how credit analysis of ABS differs from credit analysis of corporate debt.

This topic review provides a broad overview of evaluation of credit risk. Do note that while credit risk and default risk are often used interchangeably, credit risk is a broader term. Credit risk *includes* the risk of default, as well as the risk of worsening credit quality (even if default does not occur).

MODULE 26.1: CREDIT RISK MEASURES



Video covering
this content is
available online.

LOS 26.a: Explain expected exposure, the loss given default, the probability of default, and the credit valuation adjustment.

Expected exposure is the amount of money a bond investor in a credit risky bond stands to lose at a point in time *before* any recovery is factored in. The expected exposure is equal to the present value (based on the risk-free rate) of the bond's remaining cash flows. A bond's expected exposure changes over time. **Recovery rate** is the percentage recovered in the event of a default. Recovery rate is the opposite of loss severity; given a loss severity of 40%, the recovery rate would be 60%. **Loss given default (LGD)** is equal to loss severity multiplied by exposure.

Probability of default is the likelihood of default occurring in a given year. The **hazard rate** is the conditional probability of default given that default has previously not occurred.

Probability of survival is $1 -$ the cumulative probability of default. If we assume a constant hazard rate, then the probability of survival for time period t is given as:

$$PS_t = (1 - \text{hazard rate})^t$$

From this expression, it can be seen that the probability of survival decreases over time. PD, the probability of default, depends on the probability of survival from the prior period.

$$PD_t = \text{hazard rate} \times PS_{(t-1)}$$

In the first year, the probability of default = hazard rate (because PS = 1 at inception). In subsequent years, PD < hazard rate. The **expected loss** for any period is the loss given default (LGD) for that period multiplied by the PD for that period.

Credit valuation adjustment (CVA) is the sum of the present value of the expected loss for each period. CVA is the monetary value of the credit risk in present value terms; it is the difference in value between a risk-free bond and an otherwise identical risky bond:

$$\text{CVA} = \text{price of risk-free bond} - \text{price of risky bond}.$$

We illustrate the computation of CVA in the following example.

EXAMPLE: Credit valuation adjustment (CVA), part 1

A 3-year, \$100 par, zero-coupon corporate bond has a hazard rate of 2% per year. Its recovery rate is 60% and the benchmark rate curve is flat at 3%. Calculate the expected exposure, probability of survival, probability of default, loss given default, CVA, and the credit spread on the bond.

Answer:

Year	Exposure	LGD	PS	PD	Expected Loss	DF	PV of Expected Loss
1	94.260	37.704	98.000%	2.000%	0.754077	0.970874	0.732113
2	97.087	38.835	96.040%	1.9600%	0.761165	0.942596	0.717471
3	100.000	40.000	94.119%	1.9208%	0.76832	0.915142	0.703122
CVA							2.152706

Exposure at the end of three years is the face value of the bond (\$100). Exposure at the end of Year 2 is the present value of \$100 discounted for 1 period at 3% ($= \$100 / 1.03$). Exposure at the end of Year 1 is the present value discounted for 2 periods ($= \$100 / 1.03^2$).

Loss given default (LGD) is $(1 - \text{recovery rate})$ or 40% of the exposure.

Probabilities of survival (PS) for Years 1, 2, and 3 are calculated as $(1 - 0.02)^1 = 0.98$, $(1 - 0.02)^2 = 0.9604$, and $(1 - 0.02)^3 = 0.9412$.

Probability of default is 2% in the first year, and decreases with the PS. PD for Year 2 = 2% of PS for Year 1 $= 0.02 \times 0.98 = 0.0196$ or 1.96%. Similarly, the PD for Year 3 $= 0.02 \times 0.9604 = 0.019208$ or 1.9208%.

expected loss for each year = LGD $\times$ PD.

expected loss for Year 3 $= 0.019208 \times 40 = 0.7683$.

DF is the present value of \$1. Year 2 DF $= 1 / (1.03)^2 = 0.942596$.

PV of expected loss = DF $\times$ expected loss

CVA = sum of the three PV of expected loss = \$2.15

An identical benchmark bond should trade at $\$100 / 1.03^3 = \91.51 . Therefore, the value of credit risky bond = $91.50 - \text{CVA} = 91.51 - 2.15 = \89.36 .

Credit spread calculation:

Step 1: YTM on risk-free bond (given): 3%

Step 2: YTM on risky bond:

$N = 3$, $PV = -89.36$, $PMT = 0$, $FV = 100$, $\text{CPT I/Y} = 3.82\%$

Step 3: Credit spread = YTM risky – YTM risk-free = $3.82\% - 3\% = 0.82\%$

EXAMPLE: Credit valuation adjustment (CVA), part 2

Continuing the previous example: now suppose that the bond pays a 4% annual coupon and everything else is the same.

Calculate the expected exposure, probability of survival, probability of default, loss given default, and CVA.

Answer:

Year	Exposure	LGD	PS	PD	Expected Loss	DF	PV of Expected Loss
1	105.913	42.365	98.000%	2.000%	0.847308	0.970874	0.822629
2	104.971	41.988	96.040%	1.960%	0.822972	0.942596	0.775730
3	104.000	41.600	94.119%	1.921%	0.799053	0.915142	0.731247
CVA							2.329605

The probability of default and discount factors will remain the same as before, but the expected exposure changes due to the coupon.

Exposure at the end of three years is the face value of the bond plus the final coupon (\$104). Exposure at the end of Year 2 is the present value of \$104 discounted at 3% for 1 period ($\$104 / 1.03 = \100.971), plus the \$4 Year 2 coupon ($\$100.971 + \$4.00 = \104.971). Exposure at the end of Year 1 is the present value of the Year 3 cash flows discounted for 2 periods, plus the present value of the Year 2 coupon, plus the Year 1 coupon: ($\$104/1.03^2 + 4/1.03 + 4 = \105.913).

Other calculations follow the same process.

Risk Neutral Probability of Default

In the previous example, we used a probability of default that was based on the expected likelihood of default in any given year. However, in practice, we use the *risk neutral probability* of default, which is the probability of default *implied* in the current market price.

While the computation of risk-neutral probability of default is not needed for the exam, consider a 1-year, zero-coupon, \$100 par bond trading at \$95. The benchmark 1-year rate is 3% and the recovery rate is assumed to be 60%. At the end of the year, there are two possible outcomes: either the bond does not default (pays the promised par value of \$100) or defaults (the cash flow is recovery rate $\times$ 100 = \$60).

Assuming a probability of default of p , the expected year-end cash flow = $60p + 100(1 - p) = 100 - 40p$. Using the risk-free (benchmark) rate, the present value of this expected cash flow = $100 - 40p / (1.03)$.

Setting this value to be equal to the bond's market price:

$$95 = \frac{100 - 40p}{1.03}$$

implies $p = 5.38\%$.

While calculating the risk neutral probability of default, we assumed a recovery rate of 60%. Using similar mechanics, we can also calculate the implied recovery rate in the market price *given* the probability of default. If we had assumed a probability of default greater than 5.38%, then the implied recovery rate would have been higher (keeping the market price constant).

In general, given the market price (and hence the credit spread), the estimated risk neutral probabilities of default and recovery rates are positively correlated.



PROFESSOR'S NOTE

This is an important statement. We need to recognize that there are several inputs in the bond valuation model—some are known (such as the bond's coupon rate, maturity, and the benchmark term structure), while two of the inputs are estimates (probability of default and recovery rate). We have to assume one to calculate the value of the other implied in the current market price.

ESG Considerations

Analysts should also consider environmental, social, and governance factors when evaluating the default risk of the company. For example, polluters may violate environmental regulations, resulting in fines or business curtailments. Companies with poor labor practices may experience loss of reputation, customers, and profitability. Similarly, companies with poor governance systems may resort to fraudulent accounting to mask debt service problems. On the flip side, some countries provide tax incentives for investing in **green bonds** (i.e., bonds that fund green projects). The WHO issued pandemic bonds in 2017 that offered a high interest rate, but in the event of a pandemic, the principal would be diverted as aid to poor countries. Due to the COVID-19 pandemic, by July 2020, the entire principal of these bonds was wiped out.



MODULE QUIZ 26.1

1. Manny Zeld is evaluating a 5%, annual pay, \$100 par, 5-year Barry Corp. bond and has calculated the CVA as \$12.67. Benchmark rates are flat at 2%. Zeld would *most appropriately* value Barry Corp. bonds at:

- A. \$101.47.
 - B. \$110.22.
 - C. \$114.76.
2. If the annual hazard rate for a bond is 1.25%, the probability of default (PD) in Year 1 is:
 - A. less than 1.25%.
 - B. equal to 1.25%.
 - C. greater than 1.25%.
 3. If the annual hazard rate for a bond is 1.25%, the probability that the bond does not default over the next three years is *closest* to:
 - A. 94.32%.
 - B. 95.20%.
 - C. 96.30%.
 4. For a risky bond, the expected loss for a specific year is *most appropriately* calculated as:
 - A. exposure multiplied by the recovery rate multiplied by the hazard rate.
 - B. loss given default multiplied by the probability of default.
 - C. exposure multiplied by the probability of default.
 5. The CVA for a risky bond is *most likely* the:
 - A. price discount for a credit risky bond compared to an otherwise identical risk-free bond.
 - B. cumulative amount of expected loss.
 - C. mean cumulative present value of expected loss.
 6. Joel Abramson, CFA, is the fixed income portfolio manager of VZ Bank. Based on his analysis of bonds issued by Tinta Corp, including an evaluation of Tinta's balance sheet, Joel estimates the hazard rate for the bond. Abramson would *most accurately* consider the bond to be an attractive purchase if the:
 - A. estimated hazard rate is less than the bond's risk-neutral probability of default.
 - B. estimated hazard rate is less than 2%.
 - C. estimated hazard rate is greater than the risk-neutral probability of default.
 7. Given a risky bond's market price, its risk-neutral probability of default is *most likely* to be:
 - A. positively correlated with the assumed recovery rate.
 - B. negatively correlated with the assumed recovery rate.
 - C. independent of the assumed recovery rate.

MODULE 26.2: ANALYSIS OF CREDIT RISK

Consider our previous example where we were evaluating a 3-year, \$100 par, zero-coupon corporate bond. Its recovery rate is 60% and benchmark rates are flat at 3%. We calculated the exposures at the end of Years 1, 2, and 3 as \$94.26, \$97.09, and \$100, respectively. CVA was calculated as \$2.15. The bond price today = value of risk-free bond – CVA = $[100 / (1.03)^3] - 2.15 = \89.36 . Assume that default only occurs at year-end.

The cash flows on this bond in the event of default and corresponding internal rate of return (IRR), as well as the IRR if the bond does not default, are shown in Figure 26.1.



Video covering this content is available online.

Figure 26.1: Expected Rates of Return

	Year 1		Year 2		Year 3	
	Cash Flow	IRR	Cash Flow	IRR	Cash Flow	IRR
Default	\$56.56	-36.71%	\$58.25	-19.26%	\$60.00	-12.43%
No Default	0	-	0	-	\$100.00	3.82%

At the end of Year 1, there are two possible outcomes: default or no-default. If default occurs, a recovery of 60% of exposure or $0.6 \times 94.26 = \$56.56$ occurs. Similarly, in Years 2 and 3, in case of default, the recovery amounts are \$58.25 and \$60, respectively.

If the bond defaults in Year 1, the investor's IRR is calculated as:

$$PV = -89.36, N = 1, FV = 56.56, \text{CPT I/Y} = -36.71\%$$

Similarly, the IRR in case default occurs in Years 2 and 3 are:

$$\text{Year 2: } PV = -89.36, N = 2, FV = 58.25, \text{CPT I/Y} = -19.26\%$$

$$\text{Year 3: } PV = -89.36, N = 3, FV = 60.0, \text{CPT I/Y} = -12.43\%$$

If the bond does not default over its life, the investor would earn an IRR of:

$$PV = -89.36, N = 3, FV = 100.0, \text{CPT I/Y} = 3.82\%$$

Relative Credit Risk Analysis

While comparing the credit risk of several bonds, the metric that combines the probability of default as well as loss severity is the **expected loss**. Everything else constant, for a given period, the higher the expected loss, the higher the credit risk.

EXAMPLE: Relative risk evaluation

Elsa Jaitley is comparing three corporate bonds for inclusion in her fixed-income portfolio. For the next year, Jaitley has collected the following information on the three bonds:

Bond	Exposure (per \$100 par)	Recovery (per \$100 par)	Probability of Default
X	102	40	1.25%
Y	88	45	1.30%
Z	92	32	1.65%

On a relative basis, which bond has the highest risk? Based on this information, what would be an appropriate trading strategy?

Answer:

Credit risk can be evaluated based on the expected loss.

Bond	Exposure (per \$100 par)	Recovery (per \$100 par)	Probability of Default	LGD (per \$100 par)	Expected Loss
X	102	40	1.25%	62	0.775
Y	88	45	1.30%	43	0.559
Z	92	32	1.65%	60	0.99

Since the exposure and recovery amounts are given per \$100 par,

loss given default (LGD) per \$100 par = exposure – recovery

For example, for bond Y, LGD = 88 – 45 = \$43.

expected loss = LGD × probability of default

Based on expected loss, Bond Z is the most risky while Bond Y has the least credit risk.

Because we are not given the market price of the three bonds, no trading strategy can be recommended based on only this information.



MODULE QUIZ 26.2

1. A 3-year, zero-coupon corporate bond trades at a price of \$89.49. The benchmark yield curve is flat at 3%. Given a recovery rate of 45%, what is the IRR on the bond if it defaults in Year 2?
A. -24.66%.
B. -27.43%.
C. -30.13%.
2. Given the following information, which bond has the *least* amount of credit risk?

Bond	Exposure (per \$100 par)	Recovery (per \$100 par)	Probability of Default
P	98	50	2.50%
Q	94	56	3.00%
R	89	49	4.65%

- A. Bond P.
 - B. Bond Q.
 - C. Bond R.
3. A bond trader observes a 5% annual-pay, 3-year, corporate bond trading at \$103. Benchmark rates are flat at 2.50%. The trader has collected the following information on the bond:

Year	Exposure	LGD	PD	PS	Expected Loss	DF	PV of Expected Loss
1	109.82	43.927	2.00%	98.000%	0.879	0.9756	0.86
2	107.44	42.976	1.96%	96.040%	0.842	0.9518	0.80
3	105.00	42.000	1.92%	94.119%	0.807	0.9286	0.75

Based on the trader's analysis the bond is *most likely*:

- A. correctly priced.
- B. overvalued.

MODULE 26.3: CREDIT SCORES AND CREDIT RATINGS



Video covering
this content is
available online.

LOS 26.b: Explain credit scores and credit ratings.

Credit scoring is used for small businesses and individuals. A higher **credit score** indicates better credit quality. FICO is a well-known example of a credit scoring model used in the United States. FICO scores are higher for those with (a) longer credit histories (age of oldest account), (b) absence of delinquencies, (c) lower utilization (outstanding balance divided by available credit line), (d) fewer credit inquiries, and (e) a wider variety of types of credit used.

Credit ratings are issued for corporate debt, asset-backed securities, and government and quasi-government debt. Similar to credit scores, credit ratings are ordinal ratings (higher = better). Three major global credit rating agencies are Moody's Investor Service, Standard & Poor's, and Fitch Ratings. The issuer rating for a company is typically for its senior unsecured debt. Ratings on other classes of debt by the same issuer may be notched. Notching is the practice of lowering the rating by one or more levels for more subordinate debt of the issuer. Notching accounts for LGD differences between different classes of debt by the same issuer (higher LGD for issues with lower seniority).

In addition to a letter grade, rating agencies provide an outlook (positive, negative, or stable). Higher-rated bonds trade at lower spreads relative to their benchmark rates.

LOS 26.c: Calculate the expected return on a bond given transition in its credit rating.

Bond portfolio managers often want to evaluate the performance of a bond in the event of **credit migration** (i.e., change in rating). A change in a credit rating generally reflects a change in the bond's credit risk. The change in the price of the bond depends on the modified duration of the bond and the change in spread resulting from the change in credit risk as reflected by the credit migration.

$$\Delta\%P = -(\text{modified duration of the bond}) \times (\Delta \text{ spread})$$

EXAMPLE: Credit migration

Suppose a bond with a modified duration of 6.32 gets downgraded from AAA to AA. The typical AAA credit spread is 60 bps, while the typical AA credit spread is 87 bps. Calculate the percentage change in price of the bond assuming that the bond is priced at typical spreads.

Answer:

$$\text{change in spread} = (87 - 60) = 27 \text{ bps} = 0.0027$$

$$\Delta\%P = -(\text{modified duration of the bond}) \times (\Delta \text{ spread}) = -6.32 \times 0.0027 = -0.0171 \text{ or } -1.71\%$$



MODULE QUIZ 26.3

- Credit ratings incorporate:
 - default probabilities only.
 - loss given default only.
 - both the default probability and the loss given default.
- Credit scores and credit ratings are:
 - ordinal.
 - cardinal.
 - optimized.
- In credit rating, the practice of notching accounts for differences in:
 - probability of default.
 - business cycle impact.
 - loss given default.
- Bjørn Johansen, portfolio manager for Agnes Advisors, is concerned about one of the holdings in Agnes's fixed income portfolio. Beta, Inc., bonds are currently rated A, but with a negative outlook. Beta bonds have a modified duration of 9.20. Johansen collects the following information about average spreads by ratings class.

Rating	AAA	AA	A	BBB	BB	B	CCC/ CC/C
Avg. Spread	0.24%	0.32%	0.49%	0.60%	0.77%	0.95%	1.22%

Assuming that the spread on Beta bonds is equal to the average spread in its rating class, what is the expected change in price of Beta, Inc., bonds if Beta gets downgraded to BBB?

- 1.01%.
- 0.82%.
- 0.92%.

MODULE 26.4: STRUCTURAL AND REDUCED FORM MODELS



Video covering this content is available online.

LOS 26.d: Explain structural and reduced-form models of corporate credit risk, including assumptions, strengths, and weaknesses.

Structural Models

Structural models of corporate credit risk are based on the structure of a company's balance sheet and rely on insights provided by option pricing theory.

Option Analogy

Consider a hypothetical company with assets that are financed by equity and a single issue of zero-coupon debt. The value of the assets at any point in time is the sum of the value of equity and the value of debt.

Due to the limited liability nature of corporate equity, the shareholders effectively have a European call option on the company's assets with a strike price equal to the face value of debt. If at the maturity of the debt, the value of the company's assets is higher than the face value of debt, shareholders will exercise their call option to acquire the assets (and then pay off the debt and keep the residual). On the other hand, if the value of the company's assets is less than the face value of debt, the shareholders will let the option expire worthless (i.e., default on the debt), leaving the company's assets to the debt investors.

Hence, at time T , (corresponding to the maturity of debt):

$$\text{value of equity}_T = \max(0, A_T - K)$$

$$\text{value of debt} = A_T - \text{value of equity} = A_T - \max(0, A_T - K)$$

which means that $\text{value of debt}_T = \min(A_T, K)$

where:

A_T = value of company's assets at time T (i.e., at maturity of debt)

K = face value of debt

An alternate, but related interpretation considers equity investors as long the net assets of the company (with a time T value of $A_T - K$) and long a European put option, allowing them to sell the assets at an exercise price of K . Default is then synonymous to exercising the put option.

$$\text{value of the put option} = \max(0, K - A_T)$$

Under the put option analogy, the investors in risky debt can be construed to have a long position in risk-free debt and a short position in that put option.

$$\text{value of risky debt} = \text{value of risk-free debt} - \text{value of put option}$$

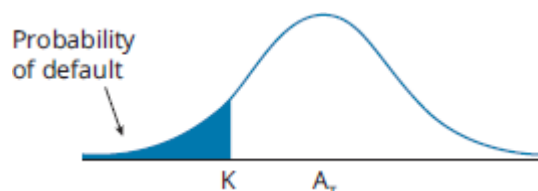
From our earlier discussion on CVA, recall that:

$$\text{value of risky debt} = \text{value of risk-free debt} - \text{CVA}$$

Therefore, the value of the put option = CVA.

Figure 26.2 shows the distribution of asset values at time T . If the value of assets falls below the default barrier K , the company defaults. The probability of default is indicated by the region in the left tail below the default barrier of K .

Figure 26.2: Distribution of Asset Value at Time T



Advantages of structural models:

1. Structural models provide an economic rationale for default (i.e., $A_T < K$) and explain why default occurs.
2. Structural models utilize option pricing models to value risky debt.

Disadvantages of structural models:

1. Because structural models assume a simple balance sheet structure, complex balance sheets cannot be modeled. Additionally, when companies have off-balance sheet debt, the default barrier under structural models (K) would be inaccurate and hence the estimated outputs of the model will be inaccurate.
2. One of the key assumptions of the structural model is that the assets of the company are traded in the market. This restrictive assumption makes the structural model impractical.

Reduced Form Models

Reduced form (RF) models do not rely on the structure of a company's balance sheet and therefore do not assume that the assets of the company trade. Unlike the structural model, reduced form models do not explain *why* default occurs. Instead, they statistically model *when* default occurs. Default under the RF model is a randomly occurring *exogenous* variable. As such, the RF model imposes assumptions on the output of the structural model (asset values, recovery rates, etc.).

A key input into the RF model is the **default intensity**, which is the probability of default over the next (small) time period. Default intensity can be estimated using regression models. These regression models employ several independent variables including company specific variables (e.g., leverage, beta, interest coverage ratio) as well as macro-economic variables.

Advantages of reduced form models:

1. RF models do not assume that the assets of a company trade.
2. Default intensity is allowed to vary as company fundamentals change, as well as when the state of the economy changes.

Disadvantages of reduced form models:

1. RF models do not explain why default occurs.
2. Under RF models, default is treated as a random event (i.e., a surprise), but in reality, default is rarely a surprise (it is often preceded by several downgrades).



MODULE QUIZ 26.4

1. The probability of default under the structural model is *most likely*:
 - A. endogenous.
 - B. exogenous.
 - C. estimated using a regression model.
2. Under the structural model, risky debt can be thought of as equivalent to a portfolio comprising a long position in a risk-free bond and:
 - A. a short put option on assets of the company, with a strike price equal to the face value of debt.

- B. a long put option on the assets of the company, with a strike price equal to the face value of debt.
 - C. short put option on assets of the company, with a strike price equal to the present value of debt.
3. Which one of the following is *least likely* a disadvantage of structural models?
- A. Structural models are not appropriate in the presence of off-balance sheet financing.
 - B. Structural models assume that company assets trade.
 - C. Structural models do not explain why default occurs.
4. Default intensity is *least likely* to be:
- A. estimated using company specific and macroeconomic variables.
 - B. the probability of default over the next time period.
 - C. constant over the life of the risky bond.

MODULE 26.5: CREDIT SPREAD ANALYSIS



Video covering
this content is
available online.

LOS 26.e: Calculate the value of a bond and its credit spread, given assumptions about the credit risk parameters.

credit spread on a risky bond = YTM of risky bond – YTM of benchmark

The value of a risky bond, assuming it does not default, is its **value given no default (VND)**. VND is calculated using the risk-free rate to value the risky bond.

EXAMPLE: Credit spread

Jack Gordon, a fixed income analyst for Omega Bank PLC, is evaluating an AA corporate bond for inclusion in the bank's portfolio. The \$100 par, 3.50%, annual-pay, 5-year bond is currently priced with a credit spread of 135 bps over the benchmark par rate of 2%. Calculate the bond's CVA implied in its market price.

Answer:

VND for the bond = present value of bond's cash flows using benchmark YTM.

$N = 5$, $PMT = 3.50$, $I/Y = 2$, $FV = 100$, CPT PV = 107.07

value of the risky bond using credit spread = 2% benchmark rate + 1.35% credit spread = 3.35% as YTM:

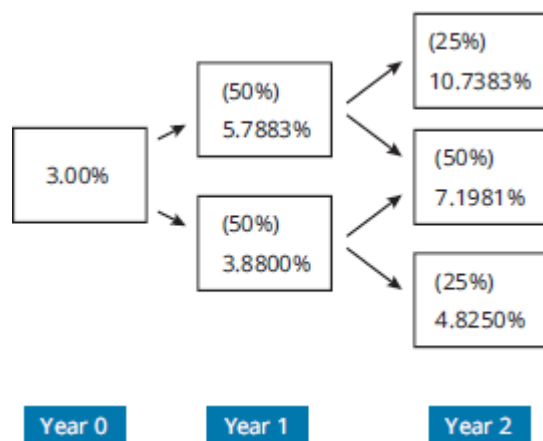
$FV = 100$, $N = 5$, $PMT = 3.5$, $I/Y = 3.35$, CPT PV = 100.68

$CVA = VND - \text{value of risky debt} = 107.07 - 100.68 = \6.39

In our earlier discussion on the computation of CVA, we assumed zero volatility for benchmark rates. We now introduce volatility in future 1-period benchmark rates using the binomial tree framework.

Figure 26.3 shows a three-year benchmark interest rate tree. Numbers in parentheses in each node indicate the probability for that node (note that probabilities add up to 100% for each period).

Figure 26.3: Benchmark Interest Rate Tree



We use our standard backward induction process to value a risky bond given the benchmark rate tree. Recall that for a binomial interest rate tree, the probability of up = probability of down = 0.5. Because the rates in the tree are benchmark rates, the value obtained is the bond's VND.

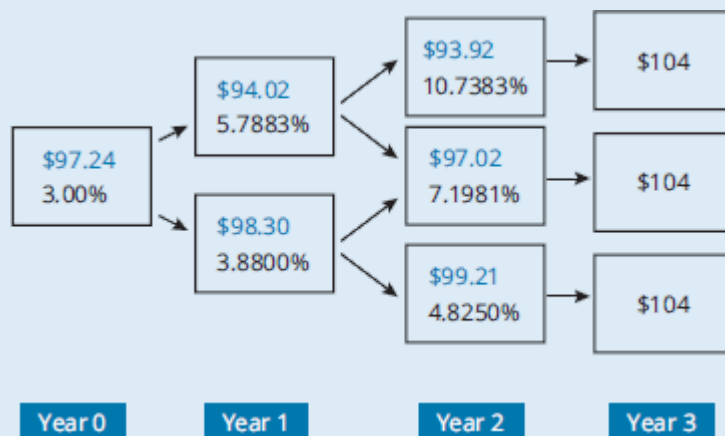
We illustrate the computation of VND using the following example.

EXAMPLE: Calculation of VND

For a 3-year, annual pay, 4% coupon, \$100 par corporate bond using the interest rate tree in Figure 26.3, calculate the VND for the bond and the expected exposure for each year.

Answer:

The completed tree is shown as follows:



Given a 4% coupon, the cash flow at the end of three years is 104. The value at the bottom node in Year 2 = $104 / 1.04825 = \$99.21$. The value of bottom node in Year 1 is calculated as the present value of the average of the two values in Year 2 plus the coupon.

$$[(99.21 + 97.02) / 2] + 4 / (1.0388) = 98.30.$$

Using the same procedure for Year 0, the VND for the bond is \$97.24.

expected exposure for Year t

$$= \sum (\text{value in node } t \text{ at time } t \times \text{node probability}) + \text{coupon for Year } t$$

expected exposure for Year 1 = $(0.5)(98.30) + (0.5)(94.02) + 4 = \100.16

expected exposure for Year 2 = $(0.25)(93.92) + (0.5)(97.02) + (0.25)(99.21) + 4 = \100.79

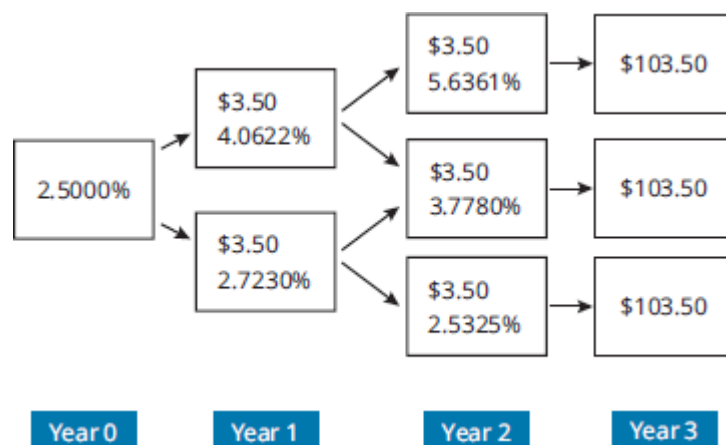
expected exposure for Year 3 = \$104 (no need to calculate!)

Once the expected exposure for each period is calculated, given an estimated unconditional probability of default and a recovery rate, we use the same method discussed earlier to calculate the CVA for the bond.



MODULE QUIZ 26.5

1. An analyst has completed the following interest rate tree:



The VND for a 3-year, annual-pay, 3.50% corporate bond is *closest* to:

- A. \$99.78.
B. \$100.70.
C. \$101.54.
2. Jill Smith, CFA, is evaluating an annual pay 4%, 3-year corporate bond and has compiled the following information. Smith has assumed a recovery rate of 60% and a flat benchmark rate of 2.50%. Unfortunately, some of the information is missing. Complete the missing information to answer the question.

Year	Exposure	LGD	PD	PS	Expected Loss	DF	PV of Expected Loss
1	106.89	42.756	1.00%	99.000%	0.428	0.9756	0.42
2			0.99%	98.010%		0.9518	
3	104.00	41.600	0.98%	97.030%	0.408	0.9286	0.38

The CVA on the bond is *closest* to:

- A. \$1.20.
B. \$2.58.
C. \$2.82.
3. Jill Smith, CFA, is evaluating a 4%, annual-pay, 3-year corporate bond with a CVA of \$1.88. The bond is trading at \$103.53. The benchmark par curve is given here.

Year	1	2	3	4	5
Par Rate	2%	2.25%	2.50%	2.75%	3.00%

Smith is *most likely* to conclude that the corporate bond is:

- A. overvalued.
- B. undervalued.
- C. correctly valued.

MODULE 26.6: CREDIT SPREAD



Video covering
this content is
available online.

LOS 26.f: Interpret changes in a credit spread.

A benchmark yield should be equal to the real risk-free rate plus expected inflation, as well as a risk-premium for uncertainty in future inflation. **Credit spreads** include compensation for default, liquidity, and taxation risks relative to the benchmark. Adjustment to the price for all these risk factors together is known as the XVA. In this reading, we focus only on the default risk component (i.e., CVA) as it is the most important and most-commonly-used in practice.

Credit spreads change as investor perceptions about the future probability of default and recovery rates change. These perceptions depend on expectations about the state of the economy. Expectations of impending recessions lead to expectations of higher defaults and lower recovery rates.

EXAMPLE: Changes in probability of default and recovery rates

Joan De Silva, a junior analyst, works for a regional bank. Currently, De Silva is evaluating a 3-year, annual pay, 3% XYZ corporate bond priced at \$102. The benchmark yield curve is flat at 1.75%.

De Silva prepares Table 1 assuming a hazard rate of 1.25% and a recovery rate of 70%.

Table 1: Credit Analysis of 3%, 3-Year Bond With 1.25% PD and 70% RR

Year	Exposure	LGD	PD	PS	Expected Loss	DF	PV of Expected Loss
1	105.44	31.63	1.25%	98.750%	0.395	0.983	0.389
2	104.23	31.27	1.23%	97.516%	0.386	0.966	0.373
3	103.00	30.90	1.22%	96.297%	0.377	0.949	0.358
						CVA	1.12

Expected exposure is the amount of money a bond investor in a credit risky bond stands to lose at a point in time before any recovery is factored in.

The expected exposure for each period is calculated given the coupon of \$3 and risk-free rate of 1.75%:

expected exposure for Year 1 = $(104.23 / 1.0175) + 3 = \105.44

expected exposure for Year 2 = $(103 / 1.0175) + 3 = \$104.23$

expected exposure for Year 3 = \$103 (no need to calculate)

De Silva prepares a report including Table 1 for Susan Collins, Chief Investment Officer. Collins states that based on the expectations of a slowdown in the economy, a 1.50% hazard rate and a recovery rate of 60% would be more appropriate for XYZ.

De Silva updates those estimates and prepares Table 2.

Table 2: Credit Analysis of 3%, 3-Year Bond With 1.50% PD and 60% RR

Year	Exposure	LGD	PD	PS	Expected Loss	DF	PV of Expected Loss
1	105.44	42.174	1.50%	98.500%	0.633	0.983	0.622
2	104.23	41.691	1.48%	97.023%	0.616	0.966	0.595
3	103.00	41.20	1.46%	95.567%	0.600	0.949	0.569
						CVA	1.79

1. Using De Silva's estimates of hazard rate and recovery rate, XYZ bond is currently *most likely*:
 - A. undervalued.
 - B. overvalued.
 - C. fairly priced.
2. Using the market price of the bond, the credit spread on XYZ bond is *closest* to:
 - A. 0.44%.
 - B. 0.49%.
 - C. 0.55%.
3. Assuming the market price changes to reflect Collins's expectations of PD and recovery rate, the new credit spread would be *closest* to:
 - A. 0.52%.
 - B. 0.61%.
 - C. 0.79%.

Answer:

1. **A** Using the benchmark rate, XYZ bond's VND is calculated as:

$$N = 3, PMT = 3, I/Y = 1.75, FV = 100, CPT PV = 103.62$$

$$\text{value of risky bond} = VND - CVA = 103.62 - 1.12 = 102.50$$

The market price of \$102 for the bond implies that the bond is undervalued.

2. **C** YTM for 3-year risk-free bond = 1.75% (given)

$$\text{YTM for XYZ bond: } PV = -102, N = 3, PMT = 3, FV = 100, CPT I/Y = 2.30\%$$

$$\text{credit spread} = 2.30 - 1.75 = 0.55\%$$

3. B Based on the revised PD and recovery rate, $CVA = 1.79$.

value of risky bond = $VND - CVA = 103.62 - 1.79 = 101.83$

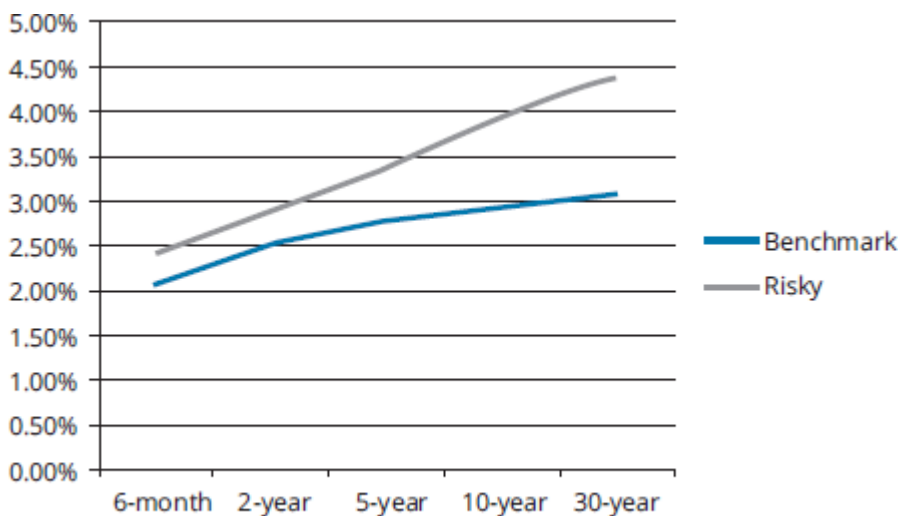
YTM for XYZ bond: $PV = -101.83$, $N = 3$, $PMT = 3$, $FV = 100$, $CPT I/Y = 2.36\%$

credit spread = $2.36 - 1.75 = 0.61\%$

LOS 26.g: Explain the determinants of the term structure of credit spreads and interpret a term structure of credit spreads.

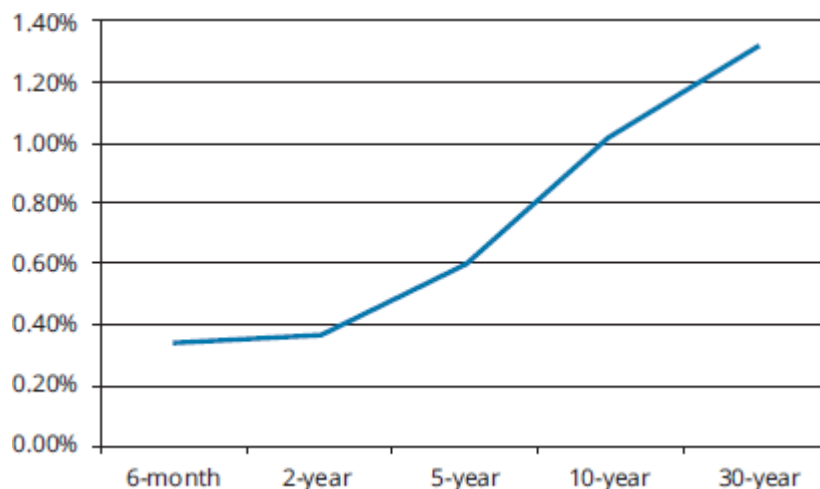
The **term structure of credit spreads** shows the relationship between credit spreads and maturity. Term structures can be for different bonds issued by the same issuer (with varying maturities) or different bonds within a sector (e.g., AA term structure). Term structures can be useful for investors to price a new issue, as well as to determine the relative valuation of an existing issue. A **credit spread curve** displays this relationship graphically. Figure 26.4 shows yields on U.S. Treasury bonds and high-quality (AAA) corporate bonds. Figure 26.5 shows the resulting HQM spread curve.

Figure 26.4: U.S. Treasury and High-Quality Corporate Bond Yields



Source: St. Louis Federal Reserve Bank, May 2018.

Figure 26.5: Credit Spread Curve: High-Quality Corporates



The credit spread is inversely related to the recovery rate and positively related to the probability of default. Recall that credit spread is calculated as the difference between the yields of a risky security and a benchmark security with the same maturity. However, oftentimes a maturity-matched, liquid (i.e., frequently traded) benchmark security is not available. In such cases, interpolation from available benchmark securities is used. When such data is not easily available, or when benchmark bonds are thinly traded, a spread relative to a swap fixed rate (corresponding to the maturity of the bond) may be used.

To create a spread curve, all of the bonds whose spreads are used should have similar credit characteristics. Differences in seniority, first/second lien provisions, and embedded options distort the computed credit spread. A spread curve using these distorted spreads would not be an accurate reflection of the term structure.

Key determinants of the shape of the credit spread curve include expectations about future recovery rates and default probabilities. If default probabilities are expected to be higher (or recovery rates lower) in the future, the credit curve would be expected to be positively sloped. Flat credit curves indicate stable expectations over time.

Determinants of Term Structure of Credit Spreads

1. *Credit quality* is an important factor driving term structure; AAA term structures tend to be flat or slightly upward sloping. Lower-rated sectors tend to have steeper spread curves, reflecting greater uncertainty as well as greater sensitivity to the business cycle.
2. *Financial conditions* affect the credit spread curve. Spreads narrow during economic expansions and widen during cyclical downturns. During boom times, benchmark yields tend to be higher while credit spreads tend to be narrower.
3. *Market demand and supply* influence the shape of the spread curve. Recall that a credit spread includes a premium for lack of liquidity. Hence, less liquid maturities would show higher spreads (even if the expectations for that time period are stable). Due to low liquidity in most corporate issues, the credit curves are most heavily influenced by more heavily traded bonds. Because newly issued bonds are generally more liquid, when an issuer refinances a near-dated bond with a longer-term bond, the spread may appear to narrow for the longer maturity (possibly leading to an inverted credit spread curve).
4. *Equity market volatility*: Company-value models (structural models discussed earlier) employ a company's stock price volatility and balance sheet structure in determining the probability of default. Increases in equity volatility therefore tend to widen spreads and influence the shape of the credit spread curve.



MODULE QUIZ 26.6

1. Mossimo Gulzar is compiling data on spreads for AAA corporate bonds. Mossimo has compiled the following information from the firm's senior economist.

Year	1	3	5	10
Benchmark Par Rate	0.75%	1.25%	1.75%	2.25%
AAA CVA	\$1.79	\$2.12	\$3.24	\$5.84

Using a typical 4% coupon AAA bond for each maturity category, Mossimo would *most likely* conclude that the credit spread curve is:

- A. upward sloping.

- B. downward sloping.
 - C. flat.
2. Credit spreads are *most likely*:
- A. positively related to probability of default and loss severity.
 - B. positively related to probability of default and recovery rate.
 - C. negatively related to probability of default and recovery rate.
3. Credit spread curves of the highest rated bond sectors tend to be:
- A. flat or downward sloping.
 - B. flat or slightly upward sloping.
 - C. steeply upward sloping.
4. Flat credit curves are *most likely* indicative of:
- A. expectations of economic expansion.
 - B. expectations of a recession.
 - C. stable expectations about future recovery rates and default probabilities.
5. The shape of the credit curve is *least likely* to be affected by:
- A. demand and supply forces in the market.
 - B. sector quality.
 - C. the swap rate curve.

MODULE 26.7: CREDIT ANALYSIS OF SECURITIZED DEBT



Video covering
this content is
available online.

LOS 26.h: Compare the credit analysis required for securitized debt to the credit analysis of corporate debt.

Securitized debt entails financing of specific assets (e.g., auto loans, credit card receivables, and mortgages) without financing the entire balance sheet (which would be the case with the general obligation nature of corporate and sovereign bonds). Secured debt is usually financed via a bankruptcy-remote SPE. This isolation of securitized assets allows for higher leverage and lower cost to the issuer. Investors also benefit from greater diversification, more stable cash flows and a higher risk premium relative to similarly-rated general obligation bonds (due to higher complexity associated with collateralized debt).

Components of Credit Analysis of Secured Debt

1. **Collateral pool:** Credit analysis of structured, securitized debt begins with the collateral pool. Homogeneity of a pool refers to the similarity of the assets within the collateral pool. Granularity refers to the transparency of assets within the pool. A highly granular pool would have hundreds of clearly defined loans, allowing for use of summary statistics as opposed to investigating each borrower. A more-discrete pool of a few loans would warrant examination of each obligation separately.

Short-term granular and homogenous structured finance vehicles are evaluated using a statistical-based approach. Medium-term granular and homogenous obligations are evaluated using a portfolio-based approach because the portfolio composition varies over time. Discrete and non-granular portfolios have to be evaluated at the individual loan level.

2. **Servicer quality** is important to evaluate the ability of the servicer to manage the origination and servicing of the collateral pool. After origination, investors in secured debt face the operational and counterparty risk of the servicer. A servicer's past history is often used as an indication of servicer quality.
3. **Structure** determines the tranching or other management of credit and other risks in a collateral pool. One key structural element is credit enhancement, which may be internal or external.

Examples of internal credit enhancements include tranching of credit risk among classes with differing seniority (i.e., the distribution waterfall), overcollateralization, and excess servicing spread (whereby such excess collateral or spread becomes the first line of defense against credit losses). Third party guarantees (e.g., bank, insurance companies, or loan originators) are an example of external credit enhancements.

A special structure is the case of a **covered bond**. Issued by a financial institution, covered bonds are senior, secured bonds backed by a collateral pool *as well as* by the issuer (i.e., covered bond investors have recourse rights). Originated in Germany, covered bonds have spread to the rest of Europe, Asia, and Australia. While the collateral (known as the cover pool) types vary by jurisdiction, common forms are commercial and residential mortgages and public sector loans. The cover pool is dynamic; the sponsor has to replace any prepaid or nonperforming asset in the pool so that the pool generates sufficient cash flow to service the bonds.

In the event the financial institution sponsor misses a payment, a soft or a hard bullet may be triggered. In the case of soft-bullet covered bonds, the maturity date of the bond is extended up to one year and all payments are not immediately due. In the hard-bullet covered bonds, the payments are accelerated.



MODULE QUIZ 26.7

1. Short-term granular and homogenous structured finance vehicles are *most appropriately* evaluated using a:
 - A. statistical-based approach.
 - B. portfolio-based approach.
 - C. loan level analysis.
2. Medium-term granular and homogenous structured finance vehicles are *most appropriately* evaluated using a:
 - A. statistical-based approach.
 - B. portfolio-based approach.
 - C. loan level analysis.
3. With respect to the servicer in a secured debt, investors *most likely* face:
 - A. operational and financial risk only.
 - B. financial and counterparty risk only.
 - C. operational and counterparty risk only.
4. Covered bonds are *most likely* issued by a financial institution and:
 - A. are backed by the government.
 - B. are backed by the collateral pool only.

C. have recourse rights as well as backing of the collateral pool.

KEY CONCEPTS

LOS 26.a

Expected exposure is the amount of money a bond investor in a credit risky bond stands to lose at a point in time *before* any recovery is factored in. Loss given default is equal to loss severity multiplied by exposure. Credit valuation adjustment (CVA) is the sum of the present values of expected losses for each period.

LOS 26.b

Credit scoring is used for small businesses and individuals. Credit ratings are issued for corporate debt, asset-backed securities, and government and quasi-government debt. Credit scores and ratings are ordinal rankings (higher = better). Notching different issues of the same issuer based on seniority ensures that credit ratings incorporate the probability of default as well as loss given default.

LOS 26.c

The change in the price of a bond resulting from credit migration depends on the modified duration of the bond and the change in spread.

$$\Delta\%P = -(\text{modified duration of the bond}) \times (\Delta \text{ spread})$$

LOS 26.d

Structural models of corporate credit risk are based on the structure of a company's balance sheet and rely on insights provided by option pricing theory. Structural models consider equity as a call option on company assets. Alternately, an investment in a risky bond can be viewed as equivalent to purchasing a risk-free bond and writing a put option on company assets.

Unlike the structural model, reduced-form models do not explain why default occurs. Instead, they statistically model when default occurs. Default under a reduced-form model is a randomly occurring exogenous variable.

LOS 26.e

$$\text{credit spread on a risky bond} = \text{YTM of risky bond} - \text{YTM of benchmark}$$

The value of a risky bond, assuming it does not default, is its value given no default (VND). VND is calculated using the risk-free rate to value the risky bond.

A backward induction procedure using a risky bond's cash flows and a benchmark interest-rate tree can also be used to calculate the bond's VND.

LOS 26.f

Credit spreads change as investors' perceptions change about the future probability of default and recovery rates. Expectations of impending recessions lead to expectations of higher default and lower recovery rates.

LOS 26.g

The term structure of credit spreads shows the relationship between credit spreads and maturity. Term structure depends on credit quality, financial conditions, demand and supply in the bond market, and expected volatility in the equity markets.

LOS 26.h

Credit analysis of ABS involves the analysis of the collateral pool, servicer quality, and the structure of secured debt (i.e., distribution waterfall).

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 26.1

1. **A** Value of benchmark 5% annual pay, 5-year bond (using benchmark rate of 2%): $N = 5$, $PMT = 5$, $FV = 100$, $I/Y = 2$, CPT $PV = 114.14$
Value of Barry Corp. bonds would be lower by its CVA = $114.14 - 12.67 = \$101.47$. (LOS 26.a)
2. **B** The hazard rate (the conditional probability of default) will be equal to the probability of default in the first year. The probability of default will be less than the hazard rate in all years after the first. (LOS 26.a)
3. **C** Probability of survival is the probability that the bond does not default. For Year 3, the probability of survival is $PS = (1 - 0.0125)^3 = 96.30\%$. (LOS 26.a)
4. **B** $\text{expected loss} = \text{exposure} \times (1 - \text{recovery rate}) \times PD$
 $\text{exposure} \times (1 - \text{recovery rate}) = \text{loss given default (LGD)}$
Hence, $\text{expected loss} = LGD \times PD$. (LOS 26.a)
5. **A** $CVA = \text{price of risk-free bond} - \text{price of risky bond}$
CVA is the sum (not the mean) of the present value of expected loss. (LOS 26.a)
6. **A** Risk-neutral probability of default is the probability of default that is priced by the market. If Abramson's estimated hazard rate is less than this, the market has priced-in a higher credit risk in the bond (i.e., the bond is undervalued). (LOS 26.a)
7. **A** Given the market price, if the assumed recovery rate is increased, risk-neutral probability of default (that is implied in the current market price) has to be higher to offset the higher recovery rate. As such, the assumed recovery rate is positively correlated with the risk-neutral probability of default. (LOS 26.a)

Module Quiz 26.2

1. **C** $\text{exposure in Year 2} = 100 / 1.03 = 97.09$
 $\text{recovery cash flow} = \text{exposure} \times \text{recovery rate} = 97.09 \times 0.45 = 43.69$
 $PV = -89.49$, $N = 2$, $FV = 43.69$, CPT $I/Y = -30.13\%$
(LOS 26.a)
2. **B** Bond Q has the lowest expected loss and hence the lowest credit risk.

Bond	Exposure (per \$100 par)	Recovery (per \$100 par)	Probability of Default	LGD (per \$100 par)	Expected Loss
P	98	50	2.50%	48	1.20
Q	94	56	3.00%	38	1.14
R	89	49	4.65%	40	1.86

(LOS 26.a)

3. **C** The value of a comparable risk-free bond (using the benchmark rate of 2.50%) can be calculated to be \$107.41 as follows:

$N = 3$, $PMT = 5$, $FV = 100$, $I/Y = 2.50$, $CPT PV = 107.14$.

The sum of the PV of expected loss column is the CVA = 2.41.

Hence, the risky bond value = $107.41 - 2.41 = \$104.73$, which is greater than the market price of \$103. The bond is undervalued. (LOS 26.a)

Module Quiz 26.3

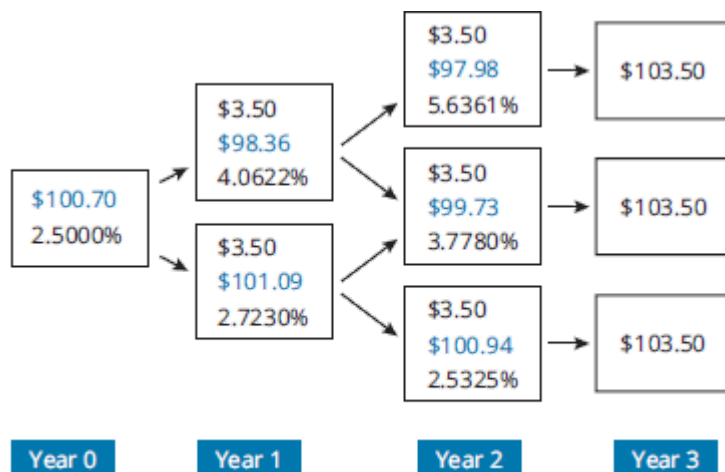
1. **C** Due to the practice of notching (lower ratings for subordinated debt as compared to senior debt of the same issuer), the loss given default is incorporated into credit ratings along with the probability of default. (LOS 26.b)
2. **A** Credit scores and credit ratings are ordinal measures; a higher rating implies better credit risk, but the difference in scores or ratings is not proportional to the difference in risk. (LOS 26.b)
3. **C** Notching accounts for LGD differences between different classes of debt by the same issuer (higher LGD for issues with lower seniority). (LOS 26.b)
4. **A** The change in spread from A to BBB is $0.60\% - 0.49\% = +0.11\%$. The percent change in price = $-9.20 \times 0.11\% = -1.01\%$. (LOS 26.c)

Module Quiz 26.4

1. **A** Probability of default under the structural model is endogenous and hence is not estimated using a regression model. It is the probability that the uncertain future value of the company assets is below the default barrier. (LOS 26.d)
2. **A** Because of limited liability of equity investors, risky debt investors are effectively short a put option on company assets with a strike price equal to the face value of debt. Hence, a risky debt investment is equivalent to an investment in risk-free debt as well as a short put option on company assets. (LOS 26.d)
3. **C** Disadvantages of structural models include (1) SMs assume that assets of the company trade and (2) SMs are inappropriate when the balance sheet is complex or there are off-balance sheet liabilities. Structural models explain why default occurs; reduced form models do not. (LOS 26.d)
4. **C** Default intensity is the probability of default over the next time increment. Default intensity varies over the life of the risky bond as company circumstances and the state of the economy change. (LOS 26.d)

Module Quiz 26.5

1. **B** The completed tree is shown here.



$$\text{VND} = [(101.09 + 98.36) / 2 + 3.50] / 1.025 = \$100.70$$

(LOS 26.e)

2. **A** We need to calculate the expected exposure, LGD, expected loss, and PV of expected loss for Year 2. The coupon of \$4 and risk-free rate of 2.5% give expected exposure in Year 2 = $(104 / 1.025) + 4 = \$105.46$.

$$\text{LGD in Year 2} = \text{exposure} \times (1 - \text{recovery rate}) = 105.46 \times 0.4 = 42.19$$

$$\text{expected loss} = \text{LGD} \times \text{PD} = 42.19 \times 0.0099 = \$0.418$$

$$\text{PV of expected loss} = \text{expected loss} \times \text{DF} = 0.42 \times 0.9518 = \$0.40$$

$$\text{CVA} = \text{sum of PV of expected loss} = 0.42 + 0.40 + 0.38 = \$1.20$$

(LOS 26.e)

3. **A** Using the 3-year par rate, we can calculate the VND for the corporate bond as:

$$\text{FV} = 100, \text{N} = 3, \text{PMT} = 4, \text{I/Y} = 2.50, \text{CPT PV} = 104.28$$

$$\text{value of the corporate bond} = \text{VND} - \text{CVA} = 104.28 - 1.88 = \$102.40$$

Since the bond is trading at \$103.53, it is overvalued. (LOS 26.e)

Module Quiz 26.6

1. **B** The coupon rate is given as 4%. The following table shows the results:

Year	1	3	5	10
Par Rate	0.75%	1.25%	1.75%	2.25%
AAA CVA	\$1.79	\$2.12	\$3.24	\$5.84
VND	\$103.23	\$108.05	\$110.68	\$115.52
V (Risky)	\$101.44	\$105.93	\$107.44	\$109.68
Yield (Risky)	2.53%	1.95%	2.40%	2.88%
Credit Spread	1.78%	0.70%	0.65%	0.63%

VND is calculated using the benchmark par rate. For example, VND for 5-year bond: $\text{N} = 5, \text{FV} = 100, \text{PMT} = 4, \text{I/Y} = 1.75, \text{CPT PV} = 110.68$

$$\text{value (risky)} = \text{VND} - \text{CVA}$$

Yield (risky) is computed using the value of risky bond as its price.

Yield (risky) for 3-year bond: $N = 3$, $PV = -105.93$, $PMT = 4$, $FV = 100$, CPT I/Y = 1.95%

credit spread = yield (risky) – par rate

Credit spreads appear to decline with maturity, resulting in a downward sloping credit spread curve. (LOS 26.f)

2. **A** Credit spreads are positively related to probability of default and loss severity. Loss severity = $1 - \text{recovery rate}$. (LOS 26.g)
3. **B** The highest rated bond sectors tend to have a flat or slightly upward sloping credit spread curve. (LOS 26.g)
4. **C** When expectations about probabilities of default and recovery rates are stable, credit spread curves tend to be flat. (LOS 26.g)
5. **C** The shape of the credit spread curve is determined by sector quality, market demand and supply forces, company-value models, and financial conditions in the economy. (LOS 26.g)

Module Quiz 26.7

1. **A** Short-term granular and homogenous structured finance vehicles are evaluated using a statistical-based approach. (LOS 26.h)
2. **B** Medium-term granular and homogenous obligations are evaluated using a portfolio-based approach because the portfolio composition varies over time. (LOS 26.h)
3. **C** After origination, investors in a secured debt face operational and counterparty risk of the servicer. (LOS 26.h)
4. **C** Covered bonds are senior, secured bonds issued by a financial institution. Covered bonds are backed by a collateral pool *as well as* by the issuer (i.e., covered bond investors have recourse rights). (LOS 26.h)

READING 27

CREDIT DEFAULT SWAPS

EXAM FOCUS

A credit default swap (CDS) is a contract between two parties in which one party purchases protection from another party against losses from the default of a borrower. For the exam, you should be able to describe CDS, as well as related securities like index CDS. You should know what a credit event is and how the different protocols for settlement work. You should be familiar with the principles and factors that drive market pricing of CDS. Be able to describe how CDS are used to manage credit exposure, and how they can be used to profit from anticipated changes in the credit curve. You should understand how CDS are used for arbitrage to take advantage of relative mispricings of different risky securities.

MODULE 27.1: CDS FEATURES AND TERMS



Video covering
this content is
available online.

Credit Default Swaps

A **credit default swap (CDS)** is essentially an insurance contract. If a credit event occurs, the *credit protection buyer* gets compensated by the *credit protection seller*. To obtain this coverage, the protection buyer pays the seller a premium called the **CDS spread**. The protection seller is assuming (i.e., long) credit risk, while the protection buyer is short credit risk. Note that the CDS does not provide protection against market-wide interest rate risk, only against credit risk. The contract is written on a face value of protection called the **notional principal** (or “notional”).



PROFESSOR'S NOTE

The terminology related to CDS is counterintuitive: the protection **buyer** is the short party (short the credit risk of the reference asset), while the **seller** is the long party (long the credit risk of the reference asset).

Even though the CDS spread should be based on the underlying credit risk of the reference obligation, standardization in the market has led to a fixed **coupon** on CDS products: 1% for investment-grade securities and 5% for high-yield securities. Hence, the coupon rate on the CDS and the actual credit spread may be different. The present value of the difference between the standardized coupon rate and the credit spread on the reference obligation is paid upfront by one of the parties to the contract. For example, a CDS on an investment-grade bond with a credit spread of 75 basis points (bps) would require a premium payment of 100bps (CDS coupon rate) by the protection buyer. To compensate the protection buyer (who pays a higher-than-market premium), the protection seller would then pay upfront to the buyer the present value of 25bps of the notional principal.

For a protection buyer, a CDS has some of the characteristics of a put option—when the underlying performs poorly, the holder of the put option has a right to exercise the option.

The **International Swaps and Derivatives Association (ISDA)**, the unofficial governing body of the industry, publishes standardized contract terms and conventions in the **ISDA Master Agreement** to facilitate smooth functioning of the CDS market.

LOS 27.a: Describe credit default swaps (CDS), single-name and index CDS, and the parameters that define a given CDS product.

Single-Name CDS

In the case of a single-name CDS, the **reference obligation** is the fixed-income security on which the swap is written, usually a senior unsecured obligation (in the case of a **senior CDS**). The issuer of the reference obligation is called the **reference entity**. The CDS pays off not only when the reference entity defaults on the reference obligation but also when the reference entity defaults on any other issue that is ranked *pari passu* (i.e., same rank) or higher. The CDS payoff is based on the market value of the **cheapest-to-deliver (CTD)** bond that has the same seniority as the reference obligation.



PROFESSOR'S NOTE

The cheapest-to-deliver bond is the debt instrument with the same seniority as the reference obligation but that can be purchased and delivered at the lowest cost.

EXAMPLE: Cheapest-to-deliver

Party X is a protection buyer in a \$10 million notional principal senior CDS of Alpha, Inc. There is a credit event (i.e., Alpha defaults) and the market prices of Alpha's bonds after the credit event are as follows:

- Bond P, a subordinated unsecured debenture, is trading at 15% of par.
- Bond Q, a five-year senior unsecured debenture, is trading at 25% of par.
- Bond R, a three-year senior unsecured debenture, is trading at 30% of par.

What will be the payoff on the CDS?

Answer:

The cheapest-to-deliver senior unsecured debenture (i.e., same seniority as the senior CDS) is bond Q. The payoff will be the difference between the notional principal and market value of the CTD.

$$\text{payoff} = \$10 \text{ million} - (0.25)(\$10 \text{ million}) = \$7.5 \text{ million}$$

Index CDS

An *index CDS* covers multiple issuers, allowing market participants to take on an exposure to the credit risk of several companies simultaneously in the same way that stock indexes allow investors to take on an equity exposure to several companies at once. In this case, the protection for each issuer is equal (i.e., equally weighted) and the total notional principal is the sum of the protection on all the issuers.

EXAMPLE: Index CDS

Party X is a protection buyer in a five-year, \$100 million notional principal CDS for CDX-IG, which contains 125 entities. One of the index constituents, company A, defaults and its bonds trade at 30% of par after default.

1. What will be the payoff on the CDS?
2. What will be the notional principal of the CDS after default?

Answer:

1. The notional principal attributable to entity A is $\$100 \text{ million} / 125 = \0.8 million .
Party X should receive payment of $\$0.8 \text{ million} - (0.3)(\$0.8 \text{ million}) = \$560,000$.
2. Post the default event, the remainder of the CDS continues with a notional principal of \$99.2 million.

The pricing of an index CDS is dependent on the correlation of default (credit correlation) among the entities in the index. The higher the correlation of default among index constituents, the higher the spread on the index CDS.

LOS 27.b: Describe credit events and settlement protocols with respect to CDS.

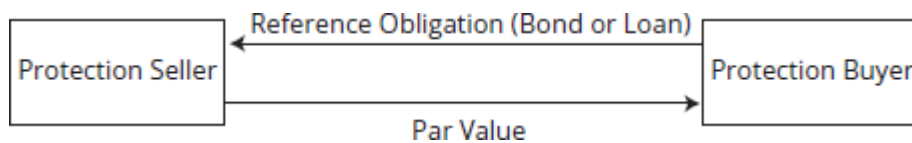
A default is defined as the occurrence of a credit event. Common types of credit events specified in CDS agreements include bankruptcy, failure to pay, and restructuring.

- *Bankruptcy.* A bankruptcy protection filing allows the defaulting party to work with creditors under the supervision of the court so as to avoid full liquidation.
- *Failure to pay.* Occurs when the issuer misses a scheduled coupon or principal payment without filing for formal bankruptcy.
- *Restructuring.* Occurs when the issuer forces its creditors to accept terms that are different than those specified in the original issue. Restructuring is less common in the United States as issuers prefer to go the bankruptcy protection route.

A 15-member group of the ISDA called the **Determinations Committee (DC)** declares when a credit event has occurred. A supermajority vote (at least 12 members) is required for a credit event to be declared.

When there is a credit event, the swap will be settled in cash or by physical delivery. With physical delivery, the protection seller receives the reference obligation (i.e., the bond or loan) and pays the protection buyer the notional amount, as shown in Figure 27.1.

Figure 27.1: Physical Settlement on Credit Default Swap After a Credit Event



In the case of a cash settlement, the payout amount is the payout ratio times the notional principal. The payout ratio depends on the recovery rate (i.e., the proportion of par that the bond trades at after default) as shown in Figure 27.2.

$$\text{payout amount} = \text{payout ratio} \times \text{notional principal}$$

where:

$$\text{payout ratio} = 1 - \text{recovery rate (\%)}$$

Figure 27.2: Cash Settlement on Credit Default Swap After a Credit Event



MODULE 27.2: FACTORS AFFECTING CDS PRICING

LOS 27.c: Explain the principles underlying and factors that influence the market's pricing of CDS.



Video covering this content is available online.

The factors that influence the pricing of CDS (i.e., CDS spread) include the probability of default, the loss given default, and the coupon rate on the swap. The CDS spread is higher for higher probability of default and for higher loss given default. For this discussion, we will focus on single-name CDS. However, the principles apply to index CDS as well.

Earlier, we showed the calculation of CVA, which is the present value of expected loss. CVA, therefore, would be the best estimate of the value of the hedging instrument to cover the risk of the protection seller.

Probability of default (POD) is the likelihood of default by the reference entity in a given year. However, because the CDS typically covers a multi-year horizon, the probability of default is not constant; the probability of default usually increases over time. Hence, the probability of default in any given year assumes that no default has occurred in the preceding years. We call the probability of default *given that it has not already occurred* the **conditional probability of default** or **hazard rate**. The credit risk of a reference obligation and hence the cost of protection is proportional to the hazard rate. The POD in any year is equal to the probability of survival at the end of the prior year multiplied by the hazard rate of that year.

$$\text{POD}_t = \text{PS}_{(t-1)} \times \text{hazard rate}_t$$

EXAMPLE: Hazard rate

Consider a five-year senior CDS on Xeon Corp. Xeon's hazard rate is 2% and increases by 1% per year.

Compute the survival rate in five years.

Answer:

The hazard rates for the five years are: 2%, 3%, 4%, 5%, and 6%.

$$\begin{aligned} \text{survival rate in five years} \\ = (1 - 0.02)(1 - 0.03)(1 - 0.04)(1 - 0.05)(1 - 0.06) = 0.815 \text{ or } 81.5\% \end{aligned}$$

For a single-period CDS, ignoring the time value of money, we can estimate the CDS premium as:

$$\text{CDS spread} \approx (1 - \text{RR}) \times \text{POD}$$

For example, if the recovery rate is 50% and POD is 3%, CDS spread = $0.5 \times 0.03 = 0.015$ or 150 bps.

The cash payments made by the protection buyer on the CDS (i.e., the coupon payments) cease when there is a default (i.e., when the CDS terminates). Hence, the expected value of the coupon payments also depends on the hazard rate.

The payments made by the protection buyer to the seller are the **premium leg**. On the other side of the contract, the protection seller must make a payment to the protection buyer in case of a default; these contingent payments make up the **protection leg**.

The difference between the present value of the premium leg and the present value of the protection leg determines the upfront payment.

$$\text{upfront payment (by protection buyer)} = \text{PV}(\text{protection leg}) - \text{PV}(\text{premium leg})$$

We can approximate the upfront premium as the difference between the CDS spread and the CDS coupon rate, multiplied by the duration of the CDS. Again, the CDS spread is the compensation for bearing the credit risk of the reference obligation.



PROFESSOR'S NOTE

Be careful here! The formula uses duration of the CDS and not the duration of the reference obligation.

$$\text{upfront premium \%} \approx (\text{CDS spread} - \text{CDS coupon}) \times \text{duration}$$

which means that:

$$\text{CDS spread} = \frac{\text{upfront premium \%}}{\text{duration}} + \text{CDS coupon}$$

We can also quote the **CDS price** as:

$$\text{price of CDS (per \$100 notional)} \approx \$100 - \text{upfront premium (\%)}$$

EXAMPLE: Upfront premium and price of CDS

Aki Mutaro, bond portfolio manager for a regional bank, is considering buying protection on one of the bank's high-yield holdings: Alpha, Inc., bonds. Ten-year CDS on Alpha bonds have a coupon rate of 5% while the 10-year Alpha CDS spread is 3.5%. The duration of the CDS is 7.

Calculate the approximate upfront premium and price of a 10-year Alpha Inc., CDS.

Answer:

$$\begin{aligned}\text{upfront premium \%} &\approx (\text{CDS spread} - \text{CDS coupon}) \times \text{duration} \\ &= (3.5\% - 5.0\%) \times 7 = -10.5\%\end{aligned}$$

Hence, the protection seller would pay (approximately) 10.5% of the notional to the protection buyer upfront because the CDS coupon is higher than the credit spread.

$$\text{CDS price} = 100 - (-10.5) = \$110.50 \text{ per } \$100 \text{ notional}$$

Valuation After Inception of CDS

At inception of a CDS, the CDS spread (and the upfront premium) is computed based on the credit quality of the reference entity. After inception, the credit quality of the reference entity (or the credit risk premium in the overall market) may change. This will lead to the underlying CDS having a nonzero value. For example, if the credit spread declines, the protection seller, having locked in a higher credit spread at initiation, would gain.

The change in value of a CDS after inception can be approximated by the change in spread multiplied by the duration of the CDS:

$$\text{profit for protection buyer} \approx \text{change in spread} \times \text{duration} \times \text{notional principal}$$

or

$$\text{profit for protection buyer (\%)} \approx \text{change in spread (\%)} \times \text{duration}$$

Note that the protection buyer is short credit risk and hence benefits (i.e., profit is positive) when credit spreads widen.

The protection buyer (or seller) can unwind an existing CDS exposure (prior to expiration or default) by entering into an offsetting transaction. For example, a protection buyer can remove his exposure to the CDS by selling protection with the same terms as the original CDS and maturity equal to the remaining maturity on the existing CDS. The difference between the upfront premium paid and received should be (approximately) equal to the profit for the protection buyer. This process of capturing value from an in-the-money CDS exposure is called **monetizing** the gain.

MODULE 27.3: CDS USAGE

LOS 27.d: Describe the use of CDS to manage credit exposures and to express views regarding changes in the shape and/or level of the credit curve.



Video covering this content is available online.

Credit Curve

The **credit curve** is the relationship between credit spreads for different bonds issued by an entity, and the bonds' maturities. The credit curve is similar to the term structure of interest rates. If the longer maturity bonds have a higher credit spread compared to

shorter maturity bonds, the credit curve will be upward sloping. However, if the hazard rate is constant, the credit curve will be flat.

CDS can be used to manage credit exposures of a bond portfolio. For example, in anticipation of declining (increasing) credit spreads, a portfolio manager may increase (decrease) credit exposure in the portfolio by being a protection seller (buyer).

In a **naked CDS**, an investor with no underlying exposure purchases (or sells) protection in the CDS market. In a **long/short trade**, an investor purchases protection on one reference entity while simultaneously selling protection on another (often related) reference entity. The investor is betting that the difference in credit spreads between the two reference entities will change to the investor's advantage. This is similar to going long (protection seller exposure) in one reference entity bond and simultaneously going short (protection buyer exposure) in the other reference entity bond.

A **curve trade** is a type of long/short trade where the investor is buying and selling protection on the *same* reference entity but with a different maturity. If the investor expects that an upward-sloping credit curve on a specific corporate issuer will flatten, she may take the position of protection buyer in a short maturity CDS and the position of protection seller in a long maturity CDS.

An investor concerned about the credit risk of an issuer in the near term while being more confident of the long-term prospects of the issuer might buy protection in the short-term CDS and offset the premium cost by selling protection in the long-term CDS. An investor who believes that the short-term outlook for the reference entity is better than the long-term outlook can use a curve-steepening trade; buying protection in a long-term CDS and selling protection in a short-term CDS. The investor will profit if the credit curve steepens; that is, if long-term credit risk increases relative to short-term credit risk. Conversely, an investor who is bearish about the reference entity's prospects in the short term will enter into a curve-flattening trade.

LOS 27.e: Describe the use of CDS to take advantage of valuation disparities among separate markets, such as bonds, loans, equities, and equity-linked instruments.

Uses of CDS

Earning arbitrage profits is another motivation for trading in the CDS market. Differences in pricing between asset and derivative markets, or differences in pricing of different products in the market, may offer potential arbitrage profits.

A **basis trade** is an attempt to exploit the difference in credit spreads between bond markets and the CDS market. Basis trades rely on the idea that such mispricing will be temporary and that disparity should eventually disappear after it is recognized. For example, if a specific bond is trading at a credit spread of 4% over MRR in the bond market but the CDS spread on the same bond is 3%, a trader can profit by buying the bond and taking the protection buyer position in the CDS market. If the expected convergence occurs, the trader will make a profit.

Another arbitrage transaction involves buying and selling debt instruments issued by the same entity based on which instruments the CDS market suggests to be undervalued or overvalued.

In a leveraged buyout (LBO), the firm will issue a great amount of debt in order to repurchase all of the company's publicly traded equity. This additional debt will increase the CDS spread because default is now more likely. An investor who anticipates an LBO might purchase both the stock and CDS protection, both of which will increase in value when the LBO eventually occurs.

In the case of an index CDS, the value of the index should be equal to the sum of the values of the index components. An arbitrage transaction is possible if the credit risk of the index constituents is priced differently than the index CDS spread.

Collateralized debt obligations (CDO) are claims against a portfolio of debt securities. A synthetic CDO has similar credit risk exposure to that of a cash CDO but is assembled using CDS rather than debt securities. If the synthetic CDO can be created at a cost lower than that of the cash CDO, investors can buy the synthetic CDO and sell the cash CDO, engaging in a profitable arbitrage.



MODULE QUIZ 27.1, 27.2, 27.3

Use the following information to answer Questions 1 through 6.

Jamshed Banaji, CFA, manages a \$400 million bond portfolio for a large public pension fund. Banaji is concerned about volatility in the credit markets and expects credit spreads to widen in the short-term but revert back to current levels over the long-term.

Banaji has flagged two of his holdings for further scrutiny: IDG Corp. and Zeta Corp.

The portfolio currently has \$10 million par value of 6% 10-year senior unsecured IDG Corp. bonds. Because he is concerned about IDG's credit risk, Banaji enters into a credit default swap as a protection buyer. Banaji selects a five-year senior CDS for IDG with a coupon rate of 5% and a duration of 4. IDG bonds have a yield-to-maturity of 6.5%. The MRR yield curve is flat at 2%.

Banaji is also concerned about the Zeta Corp. bonds that he holds. Zeta's management is planning to pursue a recapitalization plan that involves a large stock buyback program financed by new debt.

1. The *most* appropriate strategy for Banaji, given his expectation about changing credit spreads, is a:
 - A. credit curve trade; selling protection in the short-term and purchasing protection in the long-term.
 - B. credit curve trade; buying protection in the short-term and selling protection in the long-term.
 - C. CDS trade; buying protection in the short-term only.
2. At inception of the CDS for IDG bonds, Banaji is *most likely* to:
 - A. receive a premium of \$200,000.
 - B. pay a premium of \$300,000.
 - C. receive a premium of \$400,000.
3. For this question only, suppose that six months after the inception of the swap, IDG declares bankruptcy. Figure 1 shows the market prices of IDG bonds after the company files for

bankruptcy.

Exhibit 1: Market Price of IDG Bonds Post Bankruptcy Filing

Description	Market Price (%) of Par
9.5-year 6% senior unsecured	45% of par
5-year 5% senior unsecured	40% of par
5-year 6% subordinated unsecured	30% of par

If Banaji has a choice of settlement procedure, he is *most likely* to choose:

- A. physical settlement.
 - B. cash settlement and the payoff would be \$6 million.
 - C. cash settlement and the payoff would be \$7 million.
4. Which of the following statements about hazard rate is *most accurate*? Hazard rate:
- A. is the probability of default given that default has already occurred in a previous period.
 - B. affects both the premium leg as well as the protection leg in a CDS.
 - C. is higher for higher loss given default.
5. The *most appropriate* strategy for Banaji to follow in regard to Zeta Corp. would be to buy Zeta Corp.:
- A. stock and buy CDS protection on Zeta Corp. bonds.
 - B. bonds and sell CDS protection on Zeta Corp. bonds.
 - C. stock and sell CDS protection on Zeta Corp. bonds.
6. The statement “credit spreads are positively related to loss given default and to hazard rate” is:
- A. correct.
 - B. correct regarding loss given default but incorrect regarding hazard rate.
 - C. correct regarding hazard rate but incorrect regarding loss given default.

KEY CONCEPTS

LOS 27.a

A credit default swap (CDS) is essentially an insurance contract wherein upon occurrence of a credit event, the credit protection buyer gets compensated by the credit protection seller. To obtain this coverage, the protection buyer pays the seller a premium called the CDS spread. In the case of a single-name CDS, the reference obligation is the fixed income security on which the swap is written. An index CDS covers an equally-weighted combination of borrowers.

LOS 27.b

A default is defined as occurrence of a credit event. Common types of credit events specified in CDS agreements include bankruptcy, failure to pay, and restructuring.

When there is a credit event, the swap will be settled in cash or by physical delivery.

LOS 27.c

The factors that influence the pricing of CDS (i.e., CDS spread) include the probability of default, the loss given default, and the coupon rate on the swap. The CDS spread is higher for a higher probability of default and for a higher loss given default. The conditional

probability of default (i.e., the probability of default given that default has not already occurred) is called the hazard rate.

$$(\text{expected loss})_t = (\text{probability of default})_t \times (\text{loss given default})_t$$

The upfront premium on a CDS can be computed as:

$$\text{upfront payment (by protection buyer)} = \text{PV}(\text{protection leg}) - \text{PV}(\text{premium leg})$$

Or approximately:

$$\text{upfront premium} \approx (\text{CDS spread} - \text{CDS coupon}) \times \text{duration}$$

The change in value for a CDS after inception can be approximated by the change in spread multiplied by the duration of the CDS.

$$\text{profit for protection buyer} \approx \text{change in spread} \times \text{duration} \times \text{notional principal}$$

$$\text{profit for protection buyer (\%)} \approx \text{change in spread (\%)} \times \text{duration}$$

LOS 27.d

In a naked CDS, an investor with no exposure to the underlying purchases protection in the CDS market.

In a long/short trade, an investor purchases protection on one reference entity while selling protection on another reference entity.

A curve trade is a type of long/short trade where the investor is buying and selling protection on the same reference entity but with different maturities. An investor who believes the short-term outlook for the reference entity is better than the long-term outlook can use a curve-steepening trade (buying protection in a long-term CDS and selling protection in a short-term CDS) to profit if the credit curve steepens. Conversely, an investor who is bearish about the reference entity's prospects in the short term will enter into a curve-flattening trade.

LOS 27.e

A basis trade is an attempt to exploit the difference in credit spreads between bond markets and the CDS market. Basis trades rely on the idea that such mispricings will be temporary and that disparity should eventually disappear after it is recognized.

If a synthetic CDO can be created at a cost lower than that of the equivalent cash CDO, investors can buy the synthetic CDO and sell the cash CDO, producing a profitable arbitrage.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 27.1, 27.2, 27.3

1. **B** Banaji expects credit spreads to widen in the short-term; therefore, the appropriate strategy is to buy short-term CDS protection. Similarly, long-term credit spreads are expected to revert back to current levels (narrow) and hence Banaji can sell protection in the long-term CDS. Buying protection only would cost more money (the protection buyer premium is not offset by premium income from selling protection)

and does not use Banaji's entire information set and, therefore, is not most appropriate. (Module 27.2, LOS 27.c)

2. **A** credit spread on IDG bonds = yield – MRR = 6.5% – 2% = 4.5%

upfront premium (paid by protection buyer) $\approx$ (CDS spread – CDS coupon) $\times$ duration $\times$ notional principal

= (0.045 – 0.05) $\times$ 4 $\times$ \$10 million = –\$200,000

Because the computed value is negative, \$200,000 would be received by Banaji as the protection buyer. (Module 27.2, LOS 27.c)

1. **B** The CDS in the question is a senior CDS, hence the reference obligation is a senior unsecured bond. The payoff on the CDS is based on the CTD with same seniority as reference obligation. From the three choices given, the five-year 5% senior unsecured is the cheapest to deliver. Hence, the payoff will be notional principal – market value of the CTD = \$10 million – \$4 million = \$6 million.

Note that physical settlement would not be advantageous to Banaji; Figure 1 indicates that the IDG bonds that Banaji is currently holding have a market value of \$4.5 million, so the implied payoff of physically delivering these bonds in exchange for \$10 million would be only \$5.5 million (\$10 million – \$4.5 million). (Module 27.2, LOS 27.c)

1. **B** Hazard rate is the conditional probability of default given that default has not occurred in previous periods. The hazard rate affects the protection leg: the higher the hazard rate, the higher the expected value of payoffs made by the protection seller upon default. Hazard rate also affects the premium leg because once default occurs, the CDS ceases to exist and premium income would also cease. Loss given default depends on the recovery rate and not on hazard rate (probability of default). (Module 27.2, LOS 27.c)

2. **A** Due to leveraged recapitalization of Zeta Corp., it can be expected that the credit spread on Zeta bonds would widen leading to increased value for CDS protection buyer. Additionally, the increase in stock buyback would be expected to increase the value of Zeta stock. Banaji should purchase both the stock and CDS protection, both of which will increase in value when the LBO occurs. (Module 27.3, LOS 27.d)

3. **A** Credit spreads are positively related to hazard rates and loss given default, and negatively related to recovery rates. (Module 27.3, LOS 27.d)

Topic Quiz: Fixed Income

You have now finished the Fixed Income topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow three minutes per question.